Astrodynamics
Comprehensive Derivation of Governing Equations

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A personal project motivated by my interest in orbital mechanics

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Part I

The General Two Body Problem

Chapter 1

Foundational Dynamics

Introduction

The fundamental tools that govern the motion of celestial bodies lay with the elegant field of dynamics. Many of the incredible relationships that define orbital motion come out as a direct consequence of solving the dynamics of the system. Like all dynamical systems, the quantities that govern the system come from a set of rules that must be adhered to in order to be accurate. This chapter will focus on rigorously ensuring that these dynamical quantities are defined, derived, and obtained while adhering to the rules demanded by the field of mechanics.

1.1 Cylindrical Coordinates

Cylindrical coordinates may seem like an arbitrary coordinate system, but turns out to be the most convenient coordinate system to work with in the context of general astrodynamics. In essence, the cylindrical coordinate system is just a polar coordinate system with a vertical axis to add the third dimension. The position vector has a radial and vertical component.

$$\boxed{\underline{r} = r\,\hat{r} + z\,\hat{z}} \tag{1.1}$$

The velocity vector has two parts. The cylindrical coordinate system is non-inertial. To get the inertial velocity, transport theorem must be taken into account [1]. The cylindrical coordinate system rotates with respect to the inertial reference frame about the z-axis by:

$$\underline{\omega}^B = \dot{\theta}\,\hat{z} \tag{1.2}$$

Then by the transport theorem, the inertial velocity is:

$$\dot{\underline{r}} = \frac{{}^B d}{dt}\left(\underline{r}\right) + \underline{\omega}^B \times \underline{r} = \left(\dot{r}\,\hat{r} + \dot{z}\,\hat{z}\right) + \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ 0 & 0 & \dot{\theta} \\ r & 0 & z \end{vmatrix} \tag{1.3}$$

Which results in:

$$\boxed{\dot{\underline{r}} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta} + \dot{z}\,\hat{z}} \tag{1.4}$$

The acceleration vector, is determined with a time derivative of the inertial velocity vector. Since the cylindrical coordinate system has been established as non-inertial, transport theorem must be used in order to obtain the inertial acceleration.

$$\underline{\ddot{r}} = \frac{{}^{B}d}{dt}\left(\underline{\dot{r}}\right) + \underline{\omega}^{B} \times \underline{\dot{r}} = \left(\ddot{r}\,\hat{r} + \dot{r}\dot{\theta}\,\hat{\theta} + r\ddot{\theta}\,\hat{\theta} + \dot{z}\,\hat{z}\right) + \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ 0 & 0 & \dot{\theta} \\ \dot{r} & r\dot{\theta} & \dot{z} \end{vmatrix} \tag{1.5}$$

$$\underline{\ddot{r}} = \ddot{r}\,\hat{r} + \dot{r}\dot{\theta}\,\hat{\theta} + r\ddot{\theta}\,\hat{\theta} + \dot{z}\,\hat{z} - r\dot{\theta}^{2}\,\hat{r} + \dot{r}\dot{\theta}\,\hat{\theta} \tag{1.6}$$

Which results in:

$$\boxed{\underline{\ddot{r}} = \left(\ddot{r} - r\dot{\theta}^{2}\right)\,\hat{r} + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\,\hat{\theta} + \ddot{z}\,\hat{z}} \tag{1.7}$$

The time rate of change of the position vector and velocity vector with respect to the inertial frame is one of the core requirements in determining key dynamical quantities as will be discussed in greater detail later on. The choice of the cylindrical frame will also become apparent the more time that is spent detailing the dynamical quantities involved in the solutions to the Kepler two body problem.

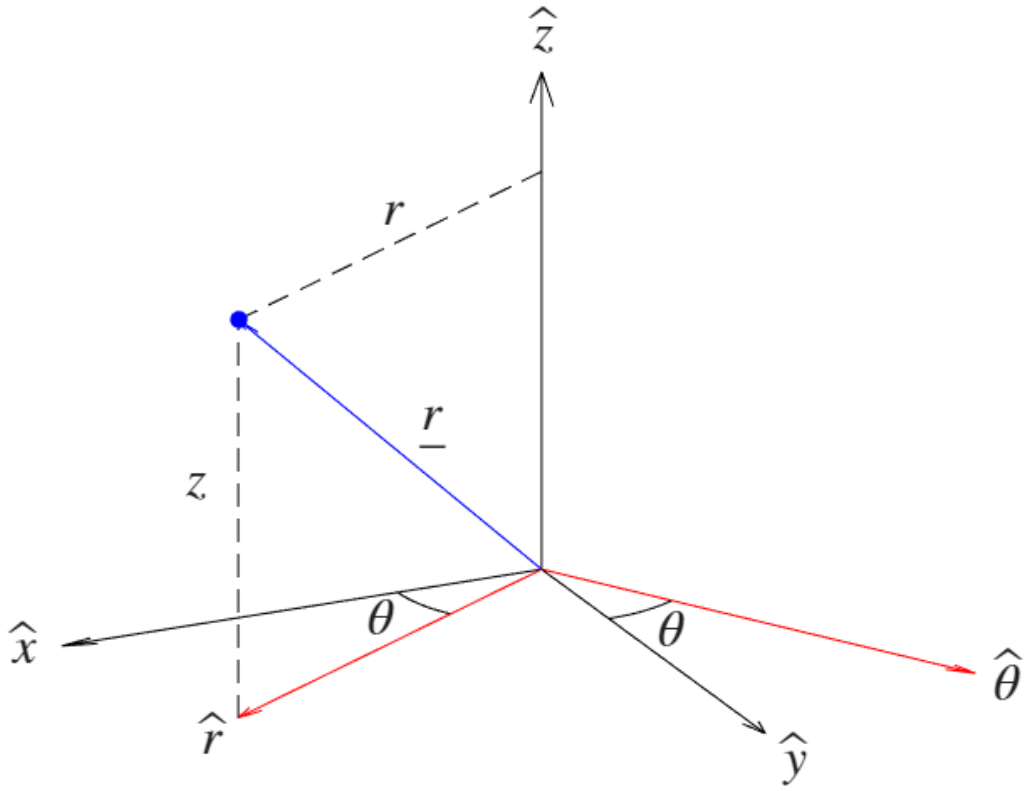


Figure 1.1: The cylindrical coordinate reference frame.

1.2 The Force of Gravity

In 1687, Sir Isaac Newton published his universal law of gravitation, who used his already known laws of motion to interpret the motion of celestial bodies. From the prior works of Johannes Kepler, Newton new that the moon and other celestial bodies in the solar system orbit in near circular paths, which he then related to the laws of circular motion.

Circular motion is governed by what is called a centripetal or "center-seeking" force. This force, alongside Newton's second law of motion yields the relationship:

$$F = ma = m\frac{v^2}{r} \tag{1.8}$$

Another important result of circular motion is period of rotation. The period of rotation is the amount of time a particle takes to make a full circular rotation.

$$P = \frac{2\pi r}{v} \tag{1.9}$$

Newton also knew from the works of Johannes Kepler that the square of the orbital period was proportional to the cube of the mean orbital radius [2]. Now taking the analysis to proportionality, Newton related Kepler's orbital period relation to the period of circular motion. Using proportions is a straightforward way to determine how important variables relate to one another, and a constant of proportionality can always be determined later.

$$P^2 \propto r^3 \propto \frac{r^2}{v^2} \quad \longrightarrow \quad v^2 \propto \frac{1}{r} \tag{1.10}$$

When paired with the centripetal force from circular motion yields the following for the force of gravity.

$$F_g \propto \frac{m}{r^2} \tag{1.11}$$

Newton's third law of motion states that for every force there is an equal and opposite force. So for a planet in near circular orbit around the sun, the same force that acts as centripetal force on the planet also acts on the sun as a result of the planet. This symmetry led Newton to conclude that the force of gravity between two objects must be dependent on the masses of *both* objects.

$$F_g \propto \frac{m_1 m_2}{r^2} \tag{1.12}$$

Finally, Newton removes the proportionality symbol in favor of an equal sign by introducing the constant of proportionality. The direction of centripetal force is opposite the direction from the center of the circular motion to the mass. Gravity is a vector quantity, and thus the full definition of the force of gravity includes magnitude and direction.

$$\boxed{\underline{F}_g = -\frac{Gm_1 m_2}{r^2}\,\hat{r}} \tag{1.13}$$

The numerical value of the constant of proportionality is $G = 6.67430 \cdot 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-1}}$.

1.3 Relative Position & Center of Mass

As discussed in the previous section, the force of gravity is dependent on the relative position between the two masses in question. Without a formal definition for this relative position, the force of gravity is just a neat formula. In this text, the definition of the relative position will be the position of mass 2 with respect to mass 1.

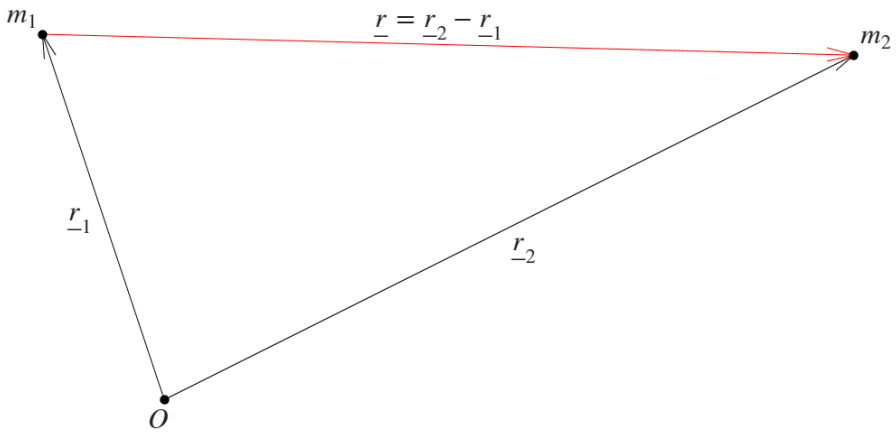


Figure 1.2: Position of mass 1 and mass 2 in relation to the inertial reference frame, as well as the relative position of mass 2 with respect to mass 1.

Notice that figure 1.2 is a plane. Huge simplifications can be made to the coming analysis by defining the reference frame such that the components of $\underline{r}_1$ and $\underline{r}_2$ do not contain a z component. As such the plane shown by the relative positions of mass 1 and mass 2 reside in the $r - \theta$ plane of the non-inertial cylindrical coordinate system.

The relative position vector $\underline{r}$ in figure 1.2 is defined by the following relation:

$$\underline{r} = \underline{r}_2 - \underline{r}_1 \tag{1.14}$$

Taking inertial derivatives of equation 1.14 results in expressions for the relative velocity and relative acceleration vectors.

$$\dot{\underline{r}} = \dot{\underline{r}}_2 - \dot{\underline{r}}_1 \tag{1.15}$$

$$\ddot{\underline{r}} = \ddot{\underline{r}}_2 - \ddot{\underline{r}}_1 \tag{1.16}$$

The center of mass of the system is always of great importance no matter the dynamical problem. In the case of the two body problem, it is useful for determining the inertial positions of the masses. For a two mass particle system, the center of mass is defined as:

$$\underline{R}_{CM} = \frac{m_1\underline{r}_1 + m_2\underline{r}_2}{m_1 + m_2} \tag{1.17}$$

Rearranging equation 1.14 and inserting it into equation 1.17 will result in expressions for the inertial position of the two masses in terms of only the system center of mass and the relative position of the two masses.

For mass 1:

$$\underline{R}_{CM} = \frac{m_1\underline{r}_1 + m_2(\underline{r}_1 + \underline{r})}{m_1 + m_2} = \frac{(m_1 + m_2)\underline{r}_1 + m_2\underline{r}}{m_1 + m_2} \tag{1.18}$$

$$\underline{r}_1 = \frac{(m_1 + m_2)\underline{R}_{CM} - m_2\underline{r}}{m_1 + m_2} \tag{1.19}$$

$$\boxed{\underline{r}_1 = \underline{R}_{CM} - \frac{m_2}{m_1 + m_2}\underline{r}} \tag{1.20}$$

For mass 2:

$$\underline{R}_{CM} = \frac{m_1(\underline{r}_2 - \underline{r}) + m_2\underline{r}_2}{m_1 + m_2} = \frac{(m_1 + m_2)\underline{r}_2 - m_1\underline{r}}{m_1 + m_2} \tag{1.21}$$

$$\underline{r}_2 = \frac{(m_1 + m_2)\underline{R}_{CM} + m_1\underline{r}}{m_1 + m_2} \tag{1.22}$$

$$\boxed{\underline{r}_2 = \underline{R}_{CM} + \frac{m_1}{m_1 + m_2}\underline{r}} \tag{1.23}$$

The position of a mass particle can be expressed as the position of the center of mass added to the position of the mass with respect to the center of mass. In the case of mass 1:

$$\underline{r}_1 = \underline{R}_{CM} - \frac{m_2}{m_1 + m_2}\underline{r} = \underline{R}_{CM} + \underline{r}_{1/CM} \tag{1.24}$$

$$\underline{r}_{1/CM} = -\frac{m_2}{m_1 + m_2}\underline{r} \tag{1.25}$$

In the case of mass 2:

$$\underline{r}_2 = \underline{R}_{CM} + \frac{m_1}{m_1 + m_2}\underline{r} = \underline{R}_{CM} + \underline{r}_{2/CM} \tag{1.26}$$

$$\underline{r}_{2/CM} = \frac{m_1}{m_1 + m_2}\underline{r} \tag{1.27}$$

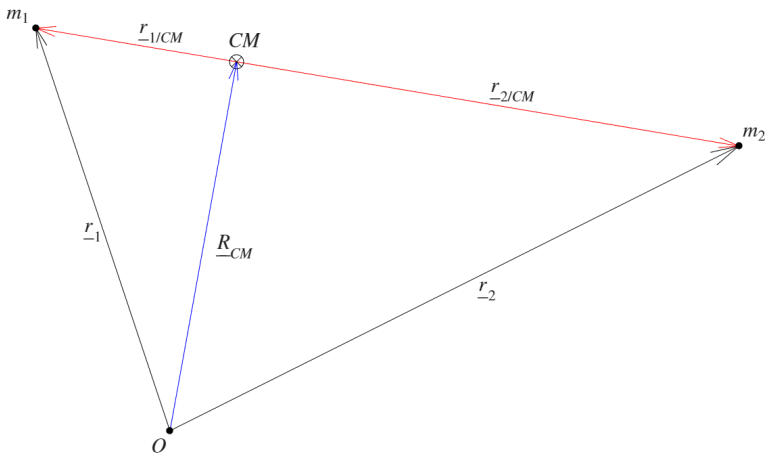


Figure 1.3: Mass 1 and mass 2 as they relate to the system's center of mass.

1.4 Two Body Kinetic Energy

The total kinetic energy of the two body system is:

$$T = T_1 + T_2 = \frac{1}{2} m_1 \dot{\underline{r}}_1 \cdot \dot{\underline{r}}_1 + \frac{1}{2} m_2 \dot{\underline{r}}_2 \cdot \dot{\underline{r}}_2 \tag{1.28}$$

The inertial velocities are determined via time derivatives of equations 1.20 and 1.23.

$$\dot{\underline{r}}_1 = \dot{\underline{R}}_{CM} - \frac{m_2}{m_1 + m_2} \dot{\underline{r}} \tag{1.29}$$

$$\dot{\underline{r}}_2 = \dot{\underline{R}}_{CM} + \frac{m_1}{m_1 + m_2} \dot{\underline{r}} \tag{1.30}$$

The kinetic energy can now be reevaluated using equation 1.29 and equation 1.30 alongside the associative property of vector dot products. For the kinetic energy of mass 1:

$$T_1 = \frac{1}{2} m_1 \left(\dot{\underline{R}}_{CM} - \frac{m_2}{m_1 + m_2} \dot{\underline{r}} \right) \cdot \left(\dot{\underline{R}}_{CM} - \frac{m_2}{m_1 + m_2} \dot{\underline{r}} \right) \tag{1.31}$$

$$T_1 = \frac{1}{2} m_1 \left(\dot{\underline{R}}_{CM} \cdot \dot{\underline{R}}_{CM} - 2 \frac{m_2}{m_1 + m_2} \dot{\underline{R}}_{CM} \cdot \dot{\underline{r}} + \frac{m_2^2}{(m_1 + m_2)^2} \dot{\underline{r}} \cdot \dot{\underline{r}} \right) \tag{1.32}$$

For the kinetic energy of mass 2:

$$T_2 = \frac{1}{2} m_2 \left(\dot{\underline{R}}_{CM} + \frac{m_1}{m_1 + m_2} \dot{\underline{r}} \right) \cdot \left(\dot{\underline{R}}_{CM} + \frac{m_1}{m_1 + m_2} \dot{\underline{r}} \right) \tag{1.33}$$

$$T_2 = \frac{1}{2} m_2 \left(\dot{\underline{R}}_{CM} \cdot \dot{\underline{R}}_{CM} + 2 \frac{m_1}{m_1 + m_2} \dot{\underline{R}}_{CM} \cdot \dot{\underline{r}} + \frac{m_1^2}{(m_1 + m_2)^2} \dot{\underline{r}} \cdot \dot{\underline{r}} \right) \tag{1.34}$$

Adding the two kinetic energy terms together results in the total system kinetic energy.

$$T = \frac{1}{2}(m_1 + m_2) \dot{\underline{R}}_{CM} \cdot \dot{\underline{R}}_{CM} + \frac{m_1 m_2 (1 - 1)}{m_1 + m_2} \dot{\underline{R}}_{CM} \cdot \dot{\underline{r}} + \frac{1}{2} \frac{m_1 m_2 (m_1 + m_2)}{(m_1 + m_2)^2} \dot{\underline{r}} \cdot \dot{\underline{r}} \tag{1.35}$$

$$T = \frac{1}{2}(m_1 + m_2) \dot{\underline{R}}_{CM} \cdot \dot{\underline{R}}_{CM} + \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \dot{\underline{r}} \cdot \dot{\underline{r}} \tag{1.36}$$

$\dot{\underline{R}}_{CM}$ can be given arbitrarily due to it not having any relation to the force of gravity between the two masses. If the center of mass is given in cartesian coordinates, in general, it can have components X, Y, and Z for each of the basis directions. $\dot{\underline{R}}_{CM}$ is given as:

$$\dot{\underline{R}}_{CM} = \dot{X} \underline{e}_1 + \dot{Y} \underline{e}_2 + \dot{Z} \underline{e}_3 \tag{1.37}$$

Then the square of the magnitude of the inertial velocity of the system's center of mass is:

$$\dot{\underline{R}}_{CM} \cdot \dot{\underline{R}}_{CM} = \dot{X}^2 + \dot{Y}^2 + \dot{Z}^2 \tag{1.38}$$

The expression for $\dot{\underline{r}}$ was given by equation 1.4. The difference being that the component of the position and velocity in the $\hat{z}$ direction is zero due to the way $\underline{r} = \underline{r}_2 - \underline{r}_1$ has been defined. The square of the magnitude of the inertial velocity of the relative position is then:

$$\dot{\underline{r}} \cdot \dot{\underline{r}} = \dot{r}^2 + r^2 \dot{\theta}^2 \tag{1.39}$$

The total two body system's kinetic energy is then:

$$\boxed{T = \frac{1}{2}(m_1 + m_2) \left[\dot{X}^2 + \dot{Y}^2 + \dot{Z}^2 \right] + \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right)} \tag{1.40}$$

1.5 Gravitational Potential Energy

The work done due to a force is defined as:

$$W = \int_a^b \underline{F} \cdot d\underline{r} \tag{1.41}$$

A force is defined as conservative if the work done between two points is always the same regardless of the path. Therefore, if the force is conservative, then the work done around a closed path (the start point is the same as the end point) is always zero because the *net* displacement is zero.

$$\oint_C \underline{F} \cdot d\underline{r} = \iint_S (\nabla \times \underline{F}) \cdot d\underline{S} = 0 \tag{1.42}$$

Stokes' theorem is a useful mathematical principle that relates a closed path integral to the curl of the vector integrated over a surface. Stokes' theorem is useful for conservative forces because the closed path integral is forced to zero when the force is conservative. Since this result holds for **every** closed curve C and every corresponding surface S, the curl of the force must vanish.

$$\nabla \times \underline{F} = \underline{0} \tag{1.43}$$

Equation 1.43 can be used to check if a force is conservative or not. The question can then asked: is Newton's law of universal gravitation a conservative force?

$$\nabla \times \underline{F}_g = \begin{pmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r}\frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} -\frac{Gm_1m_2}{r^2} \\ 0 \\ 0 \end{pmatrix} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{1}{r}\frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ -\frac{Gm_1m_2}{r^2} & 0 & 0 \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \underline{0} \tag{1.44}$$

The answer: Newton's law of universal gravitation is a conservative force.

The total mechanical energy of a system is the sum of kinetic and potential energy. If no non-conservative forces act on the system, then the total mechanical energy remains constant. Differentiating the total system energy:

$$T + U = \text{constant} \quad \longrightarrow \quad dT + dU = 0 \quad \longrightarrow \quad dU = -dT \tag{1.45}$$

From the work-energy theorem in differential form:

$$dU = -dT = -dW = -\underline{F} \cdot d\underline{r} \tag{1.46}$$

The gravitational force is inversely proportional to the square of the distance of the masses apart. Therefore, if the masses are infinitely far apart, then there would be zero force, zero work, and consequently zero potential energy.

$$\int_U^0 dU = -\int_r^\infty \underline{F}_g \cdot d\underline{r} = -\int_r^\infty \begin{pmatrix} -\frac{Gm_1m_2}{r^2} \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} dr \\ r\,d\theta \\ dz \end{pmatrix} \tag{1.47}$$

$$0 - U = \int_r^\infty \frac{Gm_1m_2}{r^2} dr = \left[-\frac{Gm_1m_2}{r}\right]_r^\infty = \left[-\frac{Gm_1m_2}{\infty} + \frac{Gm_1m_2}{r}\right] \tag{1.48}$$

$$\boxed{U = -\frac{Gm_1m_2}{r}} \tag{1.49}$$

1.6 Orbital Angular Momentum

The angular momentum of most dynamical systems tend to give insight into the equations of motion relating to changes in angle, and occasionally identifies a conserved quantity within the dynamics. The orbital angular momentum is the sum of all moments of linear momentum for each mass particle. In the case of the two body gravitational system, there are two mass particles that must be evaluated.

$$\underline{H} = \underline{H_1} + \underline{H_2} = \underline{r}_1 \times m_1 \dot{\underline{r}}_1 + \underline{r}_2 \times m_2 \dot{\underline{r}}_2 \tag{1.50}$$

This expression is rewritten using the form of $\underline{r}_1$, $\underline{r}_2$, $\dot{\underline{r}}_1$, and $\dot{\underline{r}}_2$ using equations 1.20, 1.23, 1.29, and 1.30 respectively. This evaluation puts the angular momentum in terms of the center of mass and relative position, which reduces the overall number of unknown variables for the final expression. For the mass 1 term:

$$\underline{H_1} = \left(\underline{R}_{CM} - \frac{m_2}{m_1 + m_2}\underline{r}\right) \times m_1 \left(\dot{\underline{R}}_{CM} - \frac{m_2}{m_1 + m_2}\dot{\underline{r}}\right) \tag{1.51}$$

$$\underline{H_1} = m_1 \, \underline{R}_{CM} \times \dot{\underline{R}}_{CM} - \frac{m_1 m_2}{m_1 + m_2}\underline{R}_{CM} \times \dot{\underline{r}} - \frac{m_1 m_2}{m_1 + m_2}\underline{r} \times \dot{\underline{R}}_{CM} + \frac{m_1 m_2^2}{(m_1 + m_2)^2}\underline{r} \times \dot{\underline{r}} \tag{1.52}$$

For the mass 2 term:

$$\underline{H_2} = \left(\underline{R}_{CM} + \frac{m_1}{m_1 + m_2}\underline{r}\right) \times m_2 \left(\dot{\underline{R}}_{CM} + \frac{m_1}{m_1 + m_2}\dot{\underline{r}}\right) \tag{1.53}$$

$$\underline{H_2} = m_2 \, \underline{R}_{CM} \times \dot{\underline{R}}_{CM} + \frac{m_1 m_2}{m_1 + m_2}\underline{R}_{CM} \times \dot{\underline{r}} + \frac{m_1 m_2}{m_1 + m_2}\underline{r} \times \dot{\underline{R}}_{CM} + \frac{m_1^2 m_2}{(m_1 + m_2)^2}\underline{r} \times \dot{\underline{r}} \tag{1.54}$$

Adding the two terms together results in the orbital angular momentum of the two body gravitational system.

$$\underline{H} = \underline{H_1} + \underline{H_2} = (m_1 + m_2) \, \underline{R}_{CM} \times \dot{\underline{R}}_{CM} + \frac{m_1 m_2 (m_1 + m_2)}{(m_1 + m_2)^2}\underline{r} \times \dot{\underline{r}} \tag{1.55}$$

$$\boxed{\underline{H} = (m_1 + m_2) \, \underline{R}_{CM} \times \dot{\underline{R}}_{CM} + \frac{m_1 m_2}{m_1 + m_2}\underline{r} \times \dot{\underline{r}}} \tag{1.56}$$

With the completion of the orbital angular momentum given in terms of the center of mass and relative position, the next step is to determine whether or not the orbital angular momentum is conserved. The time rate of change of the angular momentum is equal to the sum of moments on the two body gravitational system. This is known as Euler's second law of rotational motion: $\sum \underline{M} = \frac{d}{dt}(\underline{H})$.

$$\sum \underline{M} = \underline{r}_1 \times \frac{G m_1 m_2}{r^2}\hat{r}_{2/1} + \underline{r}_2 \times \frac{G m_1 m_2}{r^2}\hat{r}_{1/2} = \frac{G m_1 m_2}{r^3}\left(\underline{r}_1 \times \underline{r} - \underline{r}_2 \times \underline{r}\right) \tag{1.57}$$

Factor using the distributive property of cross products.

$$\sum \underline{M} = \frac{G m_1 m_2}{r^3}\left(\underline{r}_1 - \underline{r}_2\right) \times \underline{r} = -\frac{G m_1 m_2}{r^3}\left(\underline{r}_2 - \underline{r}_1\right) \times \underline{r} = -\frac{G m_1 m_2}{r^3}\underline{r} \times \underline{r} \tag{1.58}$$

$$\boxed{\frac{d}{dt}\left(\underline{H}\right) = \sum \underline{M} = \underline{0}} \tag{1.59}$$

Orbital angular momentum is conserved in the two body gravitational system. Also note that orbital angular momentum H is not to be confused with the system Hamiltonian $\mathcal{H}$.

1.7 The Equations of Motion

A triumph of mathematical physics from the calculus of variations comes in the form of the Euler-Lagrange equations.

$$\frac{\partial \mathscr{L}}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{q}_i}\right) = 0 \tag{1.60}$$

The equations of motion for each of the generalized coordinates are the result of applying the Euler-Lagrange equations to a dynamical system. The general Lagrangian $\mathscr{L}$ is:

$$\mathscr{L} = T - U \tag{1.61}$$

For the two body gravitational system, using equations 1.40 and 1.49, the Lagrangian is:

$$\mathscr{L} = \frac{1}{2}(m_1 + m_2)\left[\dot{X}^2 + \dot{Y}^2 + \dot{Z}^2\right] + \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{G m_1 m_2}{r} \tag{1.62}$$

The equation of motion for X:

$$\frac{\partial \mathscr{L}}{\partial X} - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{X}}\right) = (0) - \frac{d}{dt}\Big((m_1 + m_2)\dot{X}\Big) = (m_1 + m_2)\ddot{X} = 0 \quad \longrightarrow \quad \boxed{\ddot{X} = 0} \tag{1.63}$$

The equation of motion for Y:

$$\frac{\partial \mathscr{L}}{\partial Y} - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{Y}}\right) = (0) - \frac{d}{dt}\Big((m_1 + m_2)\dot{Y}\Big) = (m_1 + m_2)\ddot{Y} = 0 \quad \longrightarrow \quad \boxed{\ddot{Y} = 0} \tag{1.64}$$

The equation of motion for Z:

$$\frac{\partial \mathscr{L}}{\partial Z} - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{Z}}\right) = (0) - \frac{d}{dt}\Big((m_1 + m_2)\dot{Z}\Big) = (m_1 + m_2)\ddot{Z} = 0 \quad \longrightarrow \quad \boxed{\ddot{Z} = 0} \tag{1.65}$$

The equation of motion for r:

$$\frac{\partial \mathscr{L}}{\partial r} - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{r}}\right) = \left(\frac{m_1 m_2}{m_1 + m_2} r\dot{\theta}^2 - \frac{G m_1 m_2}{r^2}\right) - \frac{d}{dt}\left(\frac{m_1 m_2}{m_1 + m_2}\dot{r}\right) = 0 \tag{1.66}$$

$$\frac{\partial \mathscr{L}}{\partial r} - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{r}}\right) = r\dot{\theta}^2 - \frac{G(m_1 + m_2)}{r^2} - \ddot{r} = 0 \tag{1.67}$$

$$\boxed{\ddot{r} = r\dot{\theta}^2 - \frac{G(m_1 + m_2)}{r^2}} \tag{1.68}$$

The equation of motion for θ:

$$\frac{\partial \mathscr{L}}{\partial \theta} - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{\theta}}\right) = (0) - \frac{d}{dt}\Big(\frac{m_1 m_2}{m_1 + m_2} r^2\dot{\theta}\Big) = -2r\,\dot{r}\,\dot{\theta} - r^2\ddot{\theta} = 0 \tag{1.69}$$

$$\boxed{\ddot{\theta} = -\frac{2\,\dot{r}\,\dot{\theta}}{r}} \tag{1.70}$$

$\dot{X}$, $\dot{Y}$, and $\dot{Z}$ are constant in time. Recall that these generalized coordinates describe the location of the center of mass for the two body gravitational system. In other words, the motion of the center of mass is constant under the influence of gravity alone. It is a huge simplification to continue analysis within the center of mass frame, which acts as the origin to the attached non-inertial cylindrical coordinate system.

1.8 The Center of Mass Fixed Reference Frame

As stated after deriving of the equations of motion for the two body gravitational system, the analysis continues only for the center of mass (COM) fixed reference frame. The general dynamical quantities that were derived in previous sections will drop all terms relating to the COM due to the analysis pertaining only to the motion about the COM and not including the motion of the COM itself. The analysis will move forward using the following definitions for the dynamical quantities of the two body gravitational system.

Total Mechanical Energy

The total mechanical energy of the system is the sum of kinetic and potential energies. Using equation 1.40 for kinetic energy and equation 1.49 for the potential energy, the total mechanical energy is:

$$\boxed{\Sigma = \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{G m_1 m_2}{r}} \tag{1.71}$$

Σ is used here to denote the total mechanical energy because later on E will have a very different meaning that has nothing to do with mechanical energy.

Orbital Angular Momentum

The orbital angular momentum from the center of mass fixed reference frame uses equation 1.56 but drops all terms relating to the center of mass. The orbital angular momentum from this frame involves only the moments of linear momentum about the center of mass only.

$$\underline{H} = \frac{m_1 m_2}{m_1 + m_2}\,\underline{r} \times \underline{\dot{r}} = \frac{m_1 m_2}{m_1 + m_2}\begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ r & 0 & 0 \\ \dot{r} & r\dot{\theta} & 0 \end{vmatrix} \tag{1.72}$$

$$\boxed{\underline{H} = \frac{m_1 m_2}{m_1 + m_2}\,\underline{r} \times \underline{\dot{r}} = \frac{m_1 m_2}{m_1 + m_2}\,r^2\dot{\theta}\,\hat{z}} \tag{1.73}$$

The magnitude of the orbital angular momentum and $\dot{\theta}$ is therefore:

$$H = \frac{m_1 m_2}{m_1 + m_2}\,r^2\dot{\theta} \quad \longleftrightarrow \quad \boxed{\dot{\theta} = \frac{m_1 + m_2}{m_1 m_2}\frac{H}{r^2}} \tag{1.74}$$

Equations of Motion

Equations 1.68 and equation 1.69 can be used alongside Newton's second law of motion and equation 1.7 to obtain the acceleration of the inertial position vector. The generalized coordinates X, Y, and Z are unnecessary.

$$\underline{\ddot{r}} = \left(\ddot{r} - r\dot{\theta}^2 = -\frac{G(m_1 + m_2)}{r^2}\right)\hat{r} + \left(2\dot{r}\dot{\theta} + r^2\ddot{\theta} = 0\right)\hat{\theta} + 0\,\hat{z} \tag{1.75}$$

$$\boxed{\underline{\ddot{r}} = -\frac{G(m_1 + m_2)}{r^2}\,\hat{r}} \tag{1.76}$$

1.9 Gravitational Parameter and Reduced Mass

Now that the dynamics of the system has been established, two new quantities can be identified to simplify the expressions. Neither of these quantities introduce new physics, they only establish a name and shorthand for quantities that have already been determined. The first quantity is called the gravitational parameter:

$$\mu \equiv G(m_1 + m_2) \tag{1.77}$$

The gravitational parameter can be thought of as a quantification the gravitational strength of the two body system. The second quantity is called the reduced mass:

$$\mathscr{M} = \frac{m_1 m_2}{m_1 + m_2} \tag{1.78}$$

The reduced mass can be thought of as the effective inertial mass that resists a change in motion for two interacting bodies. The reduced mass also allows one to analyze a two body system as a single object with reduced mass moving relative to the system center of mass. As such, the total mechanical energy of the gravitational two body system is also written as:

$$\Sigma = \frac{1}{2}\mathscr{M}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{Gm_1m_2}{r} \tag{1.79}$$

With the total orbital angular momentum being similarly rewritten as:

$$\underline{H} = \mathscr{M}\,\underline{r} \times \underline{\dot{r}} = \mathscr{M}\, r^2\dot{\theta}\,\hat{z} \tag{1.80}$$

These quantities also pave the way to introduce the specific mechanical energy and specific orbital angular momentum.

$$\varepsilon \equiv \frac{\Sigma}{\mathscr{M}} = \frac{1}{2}\frac{m_1 + m_2}{\mathscr{M}}\left[\dot{X}^2 + \dot{Y}^2 + \dot{Z}^2\right] + \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{G(m_1 + m_2)}{r} \tag{1.81}$$

Notice the gravitational parameter in the numerator of the potential energy term. The specific mechanical energy becomes:

$$\boxed{\varepsilon = \frac{1}{2}\frac{m_1 + m_2}{\mathscr{M}}\left[\dot{X}^2 + \dot{Y}^2 + \dot{Z}^2\right] + \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{\mu}{r}} \tag{1.82}$$

Applying the same methodology, the specific orbital angular momentum is therefore:

$$\underline{h} \equiv \frac{\underline{H}}{\mathscr{M}} \tag{1.83}$$

$$\boxed{\underline{h} = \underline{r} \times \underline{\dot{r}} = r^2\dot{\theta}\,\hat{z}} \tag{1.84}$$

Then $\dot{\theta}$ becomes a much simpler expression:

$$\boxed{\dot{\theta} = \frac{h}{r^2}} \tag{1.85}$$

Notice also that the vector equation of motion can be written as:

$$\boxed{\underline{\ddot{r}} = -\frac{\mu}{r^3}\,\underline{r}} \tag{1.86}$$

1.10 Summary

The relative position between two orbiting bodies:

$$\underline{r} = \underline{r}_2 - \underline{r}_1$$

The force of gravity:

$$\underline{F}_g = -\frac{Gm_1m_2}{r^2}\,\hat{r}$$

Simplifying quantities:

$$\mu = G(m_1 + m_2)$$

$$\mathscr{M} = \frac{m_1m_2}{m_1 + m_2}$$

Total and specific mechanical energy. $\dot{X}$, $\dot{Y}$, & $\dot{Z}$ are neglected in the COM fixed frame:

$$\Sigma = T + U = \frac{1}{2}(m_1 + m_2)\left[\dot{X}^2 + \dot{Y}^2 + \dot{Z}^2\right] + \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{Gm_1m_2}{r}$$

$$\varepsilon = \frac{\Sigma}{\mathscr{M}} = \frac{1}{2}\frac{m_1 + m_2}{\mathscr{M}}\left[\dot{X}^2 + \dot{Y}^2 + \dot{Z}^2\right] + \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{\mu}{r}$$

Total and specific orbital angular momentum:

$$\underline{H} = \frac{m_1m_2}{m_1 + m_2}\,\underline{r} \times \dot{\underline{r}} = \frac{m_1m_2}{m_1 + m_2}\,r^2\dot{\theta}\,\hat{z}$$

$$\underline{h} = \frac{\underline{H}}{\mathscr{M}} = \underline{r} \times \dot{\underline{r}} = r^2\dot{\theta}\,\hat{z}$$

The gravitational two body system's vector equation of motion:

$$\ddot{\underline{r}} = -\frac{\mu}{r^3}\,\underline{r}$$

References

[1] N. Jeremy Kasdin and Derek A. Paley. *Engineering Dynamics: A Comprehensive Introduction*. Princeton University Press, 2011.

[2] Barbara Mattson. *Newton's Law of Gravitation Derivation*. `https://imagine.gsfc.nasa.gov/observatories/learning/swift/classroom/law_grav_derivation.html`. (Visited on 07/02/2026).

Chapter 2

Orbit Geometry

Introduction

In the last chapter, the dynamical framework was laid out such that useful information can now be gathered from it. This dynamical framework holds information about orbiting satellites, binary star systems, and even the Earth-Moon system. However, a list of conserved quantities like energy and angular momentum alone do not explain much about how the system behaves. The equations of motion do give a deeper look into the behavior of a two body system.

This chapter will focus primarily on using the equations of motion to obtain information that is useful in determining the behavior of a two body system. The equations of motion will provide behavioral information, and the conservation laws will make things simpler and, at times, more elegant. Unlike many complicated dynamical systems, the Keplerian two body problem has a complete analytic solution. The geometrical properties of this solution will be detailed now.

2.1 The Orbit Equation

The equations of motion is the starting point for any dynamical system analysis. Equation 1.68 and equation 1.70 are the two useful equations of motion that come out of the two body gravitational system. This is the starting point to the solution to the Kepler problem. Recall also that orbital angular momentum is conserved in the two body gravitational system, and that $\dot{\theta}$ was rewritten in terms of h in equation 1.85. This fact will return again and again to make things easier. Equation 1.85 is now used to rewrite the equation of motion for r. Substituting the gravitational parameter:

$$\ddot{r} = r\left(\frac{h}{r^2}\right)^2 - \frac{\mu}{r^2} = \frac{h^2}{r^3} - \frac{\mu}{r^2} \tag{2.1}$$

Notice that the radial equation of motion is now expressed in terms of only r. Even with this simplification, the resulting differential equation is extremely difficult to solve.

At the moment, the goal is to determine how the system behaves. Swapping θ for time as the independent variable might make things simpler to solve, while also finding a relationship

between relative position and θ. The chain rule provides an easy route for this technique.

$$\dot{r} = \frac{dr}{dt} = \frac{d\theta}{dt}\frac{dr}{d\theta} \quad \longrightarrow \quad \dot{r} = \dot{\theta}\,\frac{dr}{d\theta} \tag{2.2}$$

$$\ddot{r} = \frac{d}{dt}\left(\dot{\theta}\frac{dr}{d\theta}\right) = \frac{d\dot{\theta}}{dt}\frac{dr}{d\theta} + \dot{\theta}\frac{d\theta}{dt}\frac{d}{d\theta}\frac{dr}{d\theta} \quad \longrightarrow \quad \ddot{r} = \ddot{\theta}\,\frac{dr}{d\theta} + \dot{\theta}^2\,\frac{d^2r}{d\theta^2} \tag{2.3}$$

Substitute the θ equation of motion (equation 1.70) into equation 2.3 for $\ddot{\theta}$.

$$\ddot{r} = \left(-\frac{2\dot{r}\dot{\theta}}{r}\right)\,\frac{dr}{d\theta} + \dot{\theta}^2\,\frac{d^2r}{d\theta^2} \tag{2.4}$$

Substitute in equation 1.85 into equation 2.4.

$$\ddot{r} = \left(-\frac{2\dot{r}\left(\frac{h}{r^2}\right)}{r}\right)\,\frac{dr}{d\theta} + \left(\frac{h}{r^2}\right)^2\,\frac{d^2r}{d\theta^2} = \frac{h^2}{r^4}\,\frac{d^2r}{d\theta^2} - 2\dot{r}\frac{h}{r^3}\frac{dr}{d\theta} \tag{2.5}$$

Substitute equation 2.2 into equation 2.5 for $\dot{r}$.

$$\ddot{r} = \frac{h^2}{r^4}\,\frac{d^2r}{d\theta^2} - 2\left(\dot{\theta}\,\frac{dr}{d\theta}\right)\frac{h}{r^3}\frac{dr}{d\theta} = \frac{h^2}{r^4}\,\frac{d^2r}{d\theta^2} - 2\dot{\theta}\frac{h}{r^3}\left(\frac{dr}{d\theta}\right)^2 \tag{2.6}$$

Substitute equation 1.85 into equation 2.6 for $\dot{\theta}$.

$$\ddot{r} = \frac{h^2}{r^4}\,\frac{d^2r}{d\theta^2} - 2\left(\frac{h}{r^2}\right)\frac{h}{r^3}\left(\frac{dr}{d\theta}\right)^2 = \frac{h^2}{r^4}\,\frac{d^2r}{d\theta^2} - 2\frac{h^2}{r^5}\left(\frac{dr}{d\theta}\right)^2 = \frac{h^2}{r^4}\left[\frac{d^2r}{d\theta^2} - \frac{2}{r}\left(\frac{dr}{d\theta}\right)^2\right] \tag{2.7}$$

Equation 2.7 goes on the left hand side of equation 2.1 for $\ddot{r}$.

$$\frac{h^2}{r^4}\left[\frac{d^2r}{d\theta^2} - \frac{2}{r}\left(\frac{dr}{d\theta}\right)^2\right] = \frac{h^2}{r^3} - \frac{\mu}{r^2} \tag{2.8}$$

$$\frac{d^2r}{d\theta^2} - \frac{2}{r}\left(\frac{dr}{d\theta}\right)^2 = r - \frac{\mu}{h^2}r^2 \tag{2.9}$$

Equation 2.9 is a second order, non-linear differential equation. This complicates things, as there aren't a great deal of simple ways to solve a differential equation such as this one. A path through the mess that is equation 2.9 is to make a change of variables from r to u, where u is defined as:

$$u = \frac{1}{r} \tag{2.10}$$

In order for this to work, a few things need to happen. All derivatives need to be with respect to u instead of r, and all r's that appear in equation 2.9 need to be replaced with $1/u$. The chain rule is used to redefine the derivatives with respect to u.

$$\frac{dr}{d\theta} = \frac{dr}{du}\frac{du}{d\theta} = \frac{d}{du}\left(\frac{1}{u}\right)\frac{du}{d\theta} \quad \longrightarrow \quad \frac{dr}{d\theta} = -\frac{1}{u^2}\frac{du}{d\theta} \tag{2.11}$$

$$\frac{d^2r}{d\theta^2} = \frac{d}{d\theta}\left(-\frac{1}{u^2}\frac{du}{d\theta}\right) = \frac{du}{d\theta}\frac{d}{du}\left(-\frac{1}{u^2}\right)\frac{du}{d\theta} - \frac{1}{u^2}\frac{d}{d\theta}\left(\frac{du}{d\theta}\right) \tag{2.12}$$

$$\frac{d^2r}{d\theta^2} = \frac{2}{u^3}\left(\frac{du}{d\theta}\right)^2 - \frac{1}{u^2}\frac{d^2u}{d\theta^2} \tag{2.13}$$

Equation 2.11 and equation 2.13 are to be substituted into equation 2.9, as well as replacing every r with the inverse of u.

$$\frac{2}{u^3}\left(\frac{du}{d\theta}\right)^2 - \frac{1}{u^2}\frac{d^2u}{d\theta^2} - 2u\left(-\frac{1}{u^2}\frac{du}{d\theta}\right)^2 = \frac{1}{u} - \frac{\mu}{h^2}\frac{1}{u^2} \tag{2.14}$$

Expanding parenthesis and multiplying through by $-u^2$:

$$\frac{d^2u}{d\theta^2} + \frac{2}{u}\left(\frac{du}{d\theta}\right)^2 - \frac{2}{u}\left(\frac{du}{d\theta}\right)^2 = \frac{\mu}{h^2} - u \tag{2.15}$$

Simplifying equation 2.15 results in a remarkably simple differential equation.

$$\boxed{\frac{d^2u}{d\theta^2} + u = \frac{\mu}{h^2}} \tag{2.16}$$

The form that equation 2.16 takes is the same as that of a forced harmonic oscillator with a constant forcing function. This equation has a very straightforward solution. The particular solution is trivial since the right hand side of equation 2.16 is constant.

$$u_p = \frac{\mu}{h^2} \tag{2.17}$$

The homogeneous solution satisfies equation 2.16 if the right hand side were equal to zero.

$$\frac{d^2u_h}{d\theta^2} + u_h = 0 \tag{2.18}$$

A common way of solving differential equations is by assuming a solution. In this case, assume the homogeneous solution takes the form of an exponential.

$$u_h = e^{x\theta} \tag{2.19}$$

Plugging the assumed homogeneous solution into equation 2.18 and solving for the unknown x will (hopefully) yield the true homogeneous solution.

$$\frac{d^2}{d\theta^2}\left(e^{x\theta}\right) + e^{x\theta} = x\frac{d}{d\theta}\left(e^{x\theta}\right) + e^{x\theta} = x^2e^{x\theta} + e^{x\theta} = (x^2+1)e^{x\theta} = 0 \tag{2.20}$$

$$x^2 + 1 = 0 \quad \longrightarrow \quad x = \pm i \tag{2.21}$$

Since x has two possible values, there are generally two basic solutions. These can be taken as a linear combination without loss of generality.

$$u_h = c_1e^{i\theta} + c_2e^{-i\theta} = c_1(\cos(\theta) + i\sin(\theta)) + c_2(\cos(\theta) - i\sin(\theta)) \tag{2.22}$$

$$u_h = (c_1 + c_2)\cos(\theta) + i(c_1 - c_2)\sin(\theta) \tag{2.23}$$

$$u_h = C_1\cos(\theta) + C_2\sin(\theta) \tag{2.24}$$

The values of C_1 and C_2 are to be found from initial conditions. The solution to equation 2.16 is the sum of the particular and homogeneous solutions.

$$u = u_p + u_h \tag{2.25}$$

$$u = \frac{\mu}{h^2} + C_1 \cos(\theta) + C_2 \sin(\theta) \tag{2.26}$$

The constants may be determined by using $u = 1/r$ and finding $du/d\theta$ for the left hand side, and using the solution that was determined by equation 2.26 for the right hand side.

$$u = \frac{1}{r} = \frac{\mu}{h^2} + C_1 \cos(\theta) + C_2 \sin(\theta) \tag{2.27}$$

$$\boxed{\frac{1}{r} - \frac{\mu}{h^2} = C_1 \cos(\theta) + C_2 \sin(\theta)} \tag{2.28}$$

$$\frac{du}{d\theta} = \frac{d}{d\theta}\left(\frac{1}{r}\right) = \frac{dt}{d\theta}\frac{dr}{dt}\frac{d}{dr}\left(\frac{1}{r}\right) = \frac{\dot{r}}{\dot{\theta}}\frac{d}{dr}\left(\frac{1}{r}\right) = -\frac{\dot{r}}{r^2\dot{\theta}} \tag{2.29}$$

Substituting in equation 1.85 for $\dot{\theta}$:

$$\frac{du}{d\theta} = -\frac{\dot{r}}{h} \tag{2.30}$$

Equation 2.30 is to be used as the left hand side of a boundary condition equation for $du/d\theta$, and the right hand side will be found by differentiating equation 2.26 with respect to θ.

$$\frac{du}{d\theta} = \frac{d}{d\theta}\left(\frac{\mu}{h^2} + C_1 \cos(\theta) + C_2 \sin(\theta)\right) = -C_1 \sin(\theta) + C_2 \sin(\theta) \tag{2.31}$$

Set equal to equation 2.30.

$$\boxed{-\frac{\dot{r}}{h} = -C_1 \sin(\theta) + C_2 \cos(\theta)} \tag{2.32}$$

The term periapsis refers to the point during two body's mutual orbit where the relative position between them is at the absolute minimum. Because of this, the relative position at periapsis r_p has an instantaneous rate of change equal to zero ($\dot{r} = 0$) due to experiencing a mathematical turning point. It is convenient to have the location of periapsis correspond to $\theta = 0$, which is the boundary condition to be imposed on equation 2.28 and equation 2.32.

$$\frac{1}{r_p} - \frac{\mu}{h^2} = C_1 \cos(0) + C_2 \sin(0) \quad \longrightarrow \quad \boxed{C_1 = \frac{1}{r_p} - \frac{\mu}{h^2}} \tag{2.33}$$

$$-\frac{(0)}{h} = 0 = -C_1 \sin(0) + C_2 \cos(0) \quad \longrightarrow \quad \boxed{C_2 = 0} \tag{2.34}$$

C_1 and C_2 were obtained by applying the periapsis condition to equations 2.28 and 2.32. They will still apply to every point ($0 \leq \theta \leq 2\pi$) because they are **constant**. Applying C_1 and C_2 to equation 2.26 finally yields a useful relationship.

$$u = \frac{1}{r} = \frac{\mu}{h^2} + \left(\frac{1}{r_p} - \frac{\mu}{h^2}\right)\cos(\theta) + (0)\sin(\theta) = \frac{\mu}{h^2}\left[1 + \left(\frac{h^2}{\mu r_p} - 1\right)\cos(\theta)\right] \tag{2.35}$$

Inverting equation 2.35 on both sides results in a relationship that yields the relative position as a function of θ. This relationship is known as the orbit equation.

$$\boxed{r = \frac{\frac{h^2}{\mu}}{1 + \left(\frac{h^2}{\mu r_p} - 1\right)\cos(\theta)}} \tag{2.36}$$

Rotational Structure

The system of equations necessary to determine the constants (equation 2.28 and equation 2.32) can be put into matrix form so that linear algebra operations are allowed.

$$\begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{r} - \frac{\mu}{h^2} \\ -\frac{\dot{r}}{h} \end{pmatrix} \tag{2.37}$$

The coefficient matrix on the left hand side is orthonormal, and its matrix inverse is simply its transpose. The two constants can be isolated to the left hand side only.

$$\boxed{\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{pmatrix} \frac{1}{r} - \frac{\mu}{h^2} \\ -\frac{\dot{r}}{h} \end{pmatrix}} \tag{2.38}$$

Equation 2.38 is identical in form to the standard transformation between the polar and cartesian components of a vector in a plane. Considering the left hand side is composed of constants, it may be interpreted as the cartesian components of a constant vector.

2.2 Orbit Geometry as Conic Sections

The climax of the previous section resulted in equation 2.36. If the form of that equation is carefully examined, it has the same form as the polar form of a conic section.

$$r = \frac{p}{1 + e\cos(\theta)} \tag{2.39}$$

The variable p is called the semi-latus rectum, and is defined as the half chord that runs through a focus of the conic section and is parallel to the directrix. "The directrix of a conic section is the line which, together with the point known as the focus, serves to define a conic section as the locus of points whose distance from the focus is proportional to the horizontal distance from the directrix" [1].

The variable e is called the eccentricity, and parametrizes how much a conic section deviates from a circle. Eccentricities exactly equal to zero are perfect circles, eccentricities between zero and one represent circles that have been stretched into ellipses, an eccentricity of exactly one represents a parabola, and eccentricities greater than one represent hyperbolas.

Conic section geometry becomes very important when dealing with the Kepler two body problem. After all, orbits take the exact shape of a conic section as evidenced by equation 2.36. Two more parameters are absolutely essential in order to characterize a conic section: the semi-major axis length and the semi-minor axis length.

The semi-major axis length (a) is half the length of a conic section's major axis. The major axis is the line that connects the turning points of the long axis in an ellipse and the two branching curves in a hyperbola. A parabola has a major axis that extends off to infinity due to its open and non-branching curve.

The semi-minor axis (b) is half the length of a conic section's minor axis. For an ellipse, the minor axis is the line that connects the turning points of the short axis. For a hyperbola, the minor axis corresponds to the vertical segment between the asymptotes at a horizontal offset equal to the semi-major axis. A parabola has no minor axis because its semi-major axis is unbounded.

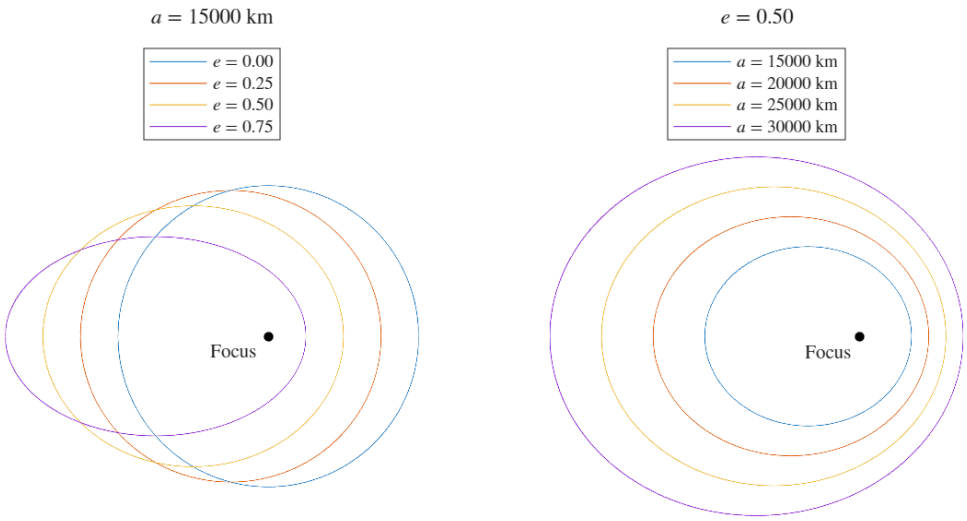


Figure 2.1: Affects of eccentricity and semi-major axis on closed orbit types.

Because the relative position of the two bodies evolves in the shape of a conic section, the dynamics of the system will dictate the type of conic section the system will correspond to. This is, obvious, but nevertheless a powerful statement. The first step to take is to equate equation 2.36 and equation 2.39 and compare terms.

$$r = \frac{p}{1 + e\cos(\theta)} = \frac{\frac{h^2}{\mu}}{1 + \left(\frac{h^2}{\mu r_p} - 1\right)\cos(\theta)} \tag{2.40}$$

By observation, the semi-latus rectum and eccentricity now have a definition that comes directly from the dynamics.

$$p = \frac{h^2}{\mu} \tag{2.41}$$

$$e = \frac{h^2}{\mu r_p} - 1 = \frac{p}{r_p} - 1 \tag{2.42}$$

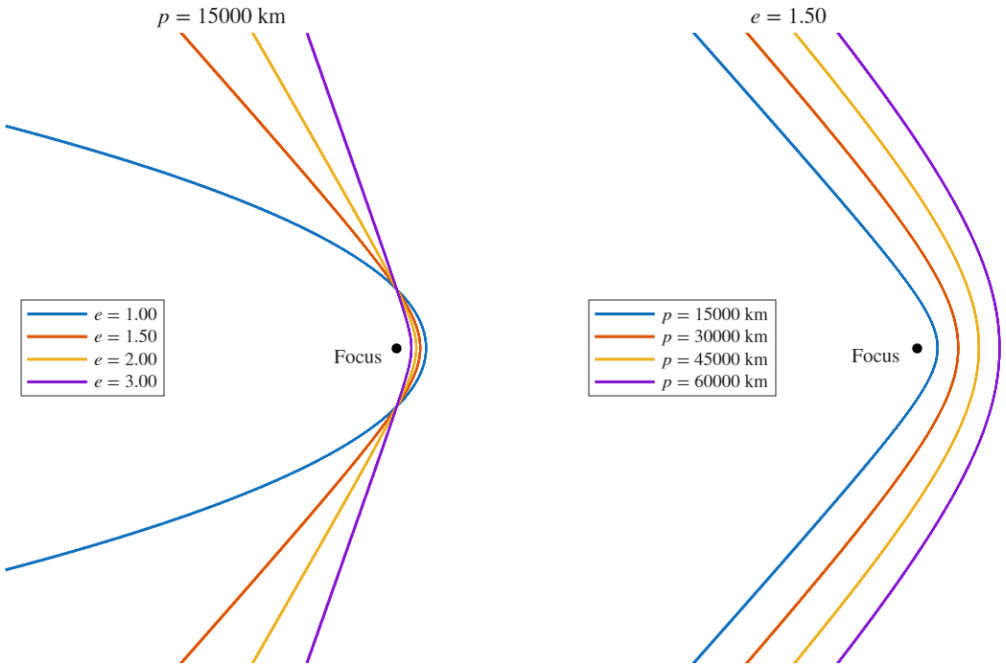


Figure 2.2: Affects of eccentricity and semi-latus rectum on open orbit types.

Defining r_p in terms of conic section parameters will require a bit of finesse. By definition, the focus of a conic section is located ae away from the center point of an ellipse or hyperbola. By definition, the semi-major axis length a equals the center-to-focus distance plus the focus-to-periapsis distance.

$$a = ae + r_p \tag{2.43}$$

Rearranging:

$$r_p = a - ae = a(1 - e) \tag{2.44}$$

Plugging this definition of r_p back into equation 2.42 results in a very useful relationship.

$$e = \frac{h^2}{\mu r_p} - 1 \tag{2.45}$$

$$e = \frac{p}{a(1 - e)} - 1 \tag{2.46}$$

$$1 + e = \frac{p}{a(1 - e)} \tag{2.47}$$

$$a(1 + e)(1 - e) = p \tag{2.48}$$

$$\boxed{p = a(1 - e^2)} \tag{2.49}$$

Example

For orbits to do no extend to infinity (bound orbits) there exists a minimum and maximum orbital radius. The radius of closest approach is called the periapsis and the radius of furthest approach is called the apoapsis. Determine the semi-major axis length of such an orbit as a function of the periapsis and apoapsis radii. Periapsis has already been established to be located at $\theta = 0$ and by the symmetry of conic sections, the apoapsis must be located at $\theta = \pi$.

$$r_p = \frac{p}{1 + e\cos(0)} = \frac{a(1-e^2)}{1+e} = a(1-e) \quad \rightarrow \quad e = 1 - \frac{r_p}{a}$$
$$r_a = \frac{p}{1 + e\cos(\pi)} = \frac{a(1-e^2)}{1-e} = a(1+e) \quad \rightarrow \quad e = \frac{r_a}{a} - 1$$

Setting $e = e$:

$$1 - \frac{r_p}{a} = \frac{r_a}{a} - 1 \quad \rightarrow \quad a - r_p = r_a - a \quad \rightarrow \quad 2a = r_p + r_a$$

$$\boxed{a = \frac{r_p + r_a}{2}} \quad , \quad \text{For Bound Orbits} \tag{2.50}$$

In closed orbits, the semi-major axis length characterizes the radius of the auxiliary circle that can be drawn around the orbit to perfectly enclose it. This auxiliary circle is a major component of closed orbits when determining time dependencies.

Some useful relationships are obtained by equating equation 2.41 and 2.49. These allow for expressing the semi-major axis length and orbital angular momentum in common terms.

$$\frac{h^2}{\mu} = a(1-e^2) \tag{2.51}$$
$$h = \sqrt{\mu a(1-e^2)} \tag{2.52}$$
$$a = \frac{h^2}{\mu(1-e^2)} \tag{2.53}$$

In the previous chapter it was found that the force of gravity is conservative, and thus the mechanical energy is conserved. As a result, the specific mechanical can be calculated at periapsis with the quantity being conserved around the whole orbit.

$$\varepsilon = \frac{1}{2}\left((0)^2 + r_p^2\left(\frac{h}{r_p^2}\right)^2\right) - \frac{\mu}{r_p} \tag{2.54}$$
$$\varepsilon = \frac{1}{2}\frac{h^2}{r_p^2} - \frac{\mu}{r_p} \tag{2.55}$$

Substitute equation 2.44 for r_p and equation 2.52 for h.

$$\varepsilon = \frac{1}{2}\frac{\left(\sqrt{\mu a(1-e^2)}\right)^2}{\left(a(1-e)\right)^2} - \frac{\mu}{a(1-e)} \tag{2.56}$$

$$\varepsilon = \frac{\mu a(1-e^2) - 2\mu a(1-e)}{2a^2(1-e)^2} = \frac{\mu}{2a}\frac{1-e^2-2+2e}{1-2e+e^2} = \frac{\mu}{2a}\frac{-(1-2e+e^2)}{1-2e+e^2} \tag{2.57}$$

$$\boxed{\varepsilon = -\frac{\mu}{2a}} \tag{2.58}$$

Equation 2.58 is an extremely powerful relationship. The specific mechanical energy determines the size of the orbit. This relationship also serves as a demonstration of how the mechanical energy affects the two orbiting bodies.

When the mechanical energy is negative, gravitational potential energy dominates, and the resulting semi-major axis length is positive. Positive semi-major axis lengths, mathematically, correspond to closed orbit types. In order for an orbit to be closed, the gravitational potential energy must be greater than the kinetic energy for any given location in the orbit. Closed orbits can be thought of as the speed of the two bodies never moving fast enough to break free from one another's gravitational influence.

When the mechanical energy is zero, both kinetic energy and gravitational potential energy are equal to one another, and the semi-major axis is infinite. This property of the mathematics corresponds exactly with the structure of a parabolic orbit as it is expressed by the polar form of a conic section. A parabolic orbit can be thought of as an orbit where the kinetic energy will increase the relative position of the two bodies to the very point when the gravitational potential energy becomes zero, but not enough to escape with any excess.

When the mechanical energy is positive, the kinetic energy dominates, and the resulting semi-major axis length is negative. A negative semi-major length, mathematically, correspond to hyperbolic orbit types. Hyperbolic orbits can be thought of as the speed of the two bodies moving fast enough to break free from one another's gravitational influence.

A massively important conic section parameter that has not yet been related to dynamical properties is the eccentricity. Begin by substituting equation 2.53 into equation 2.58 for a.

$$\varepsilon = -\frac{\mu}{2\left(\frac{h^2}{\mu(1-e^2)}\right)} = \frac{\mu^2(e^2-1)}{2h^2} \tag{2.59}$$

Rearranging to solve for eccentricity:

$$e^2 - 1 = \frac{2\varepsilon h^2}{\mu^2} \quad \longrightarrow \quad \boxed{e = \sqrt{1 + \frac{2\varepsilon h^2}{\mu^2}}} \tag{2.60}$$

By the geometry of a conic section, the semi-minor axis length is:

$$b = a\sqrt{1-e^2} \qquad \text{For } e \leq 1 \tag{2.61}$$

$$b = a\sqrt{e^2-1} \qquad \text{For } e \geq 1 \tag{2.62}$$

Both a and e have already been given in terms of the dynamical system properties. Therefore, by extension, so too has the semi-minor axis length. All four major conic section

parameters (p, e, a, and b) are obtainable from direct relationship to the system dynamics, thus demonstrating that orbits take the geometrical shape of a conic section.

Application

Alpha Centauri A/B is a well studied and understood binary star system. From the works of [2] (and many others), the orbital parameters of Alpha Centauri A/B are:

$$m_A = 2.145647 \cdot 10^{30}\,\text{kg} \qquad m_B = 1.808326 \cdot 10^{30}\,\text{kg}$$
$$a = 3.485481 \cdot 10^{12}\,\text{m} \qquad e = 0.51947$$

Where a is the semi-major axis of the relative orbit. The objective is to determine and plot the motion of each star from the perspective of an inertial $x-y$ plane (orbital plane) centered at the system center of mass (barycenter for binary star systems). The horizontal direction is coincident with the line that passes though both the periapsis and apoapsis when $\theta = 0°$ and $\theta = 180°$, called the line of apsides. From chapter one (equations 1.25 and 1.27), the positions of the masses with respect to the center of mass are given by the following relations:

$$\underline{r}_{A/CM} = -\frac{m_B}{m_A + m_B}\underline{r} \qquad \underline{r}_{B/CM} = \frac{m_A}{m_A + m_B}\underline{r}$$

The relative position vector can be written in familiar Cartesian form:

$$\underline{r} = r\left(\cos(\theta)\,\hat{i} + \sin(\theta)\,\hat{j}\right) = \frac{a(1-e^2)}{1 + e\cos(\theta)}\left(\cos(\theta)\,\hat{i} + \sin(\theta)\,\hat{j}\right) \tag{2.63}$$

Substituting the observed quantities, the inertial positions of either mass can be rewritten in Cartesian form as:

$$\underline{r}_{A/CM} = -\frac{1.16391 \cdot 10^{12}}{1 + 0.51947\cos(\theta)}\left(\cos(\theta)\,\hat{i} + \sin(\theta)\,\hat{j}\right)$$
$$\underline{r}_{B/CM} = \frac{1.38102 \cdot 10^{12}}{1 + 0.51947\cos(\theta)}\left(\cos(\theta)\,\hat{i} + \sin(\theta)\,\hat{j}\right)$$

Figure 2.3 shows the motion of Alpha Centauri A and B about the barycenter, as viewed from the orbital plane. Both stars follow elliptical trajectories with the same eccentricity while remaining on opposite sides of the barycenter. Because Alpha Centauri A is the more massive star, it traces the smaller orbit.

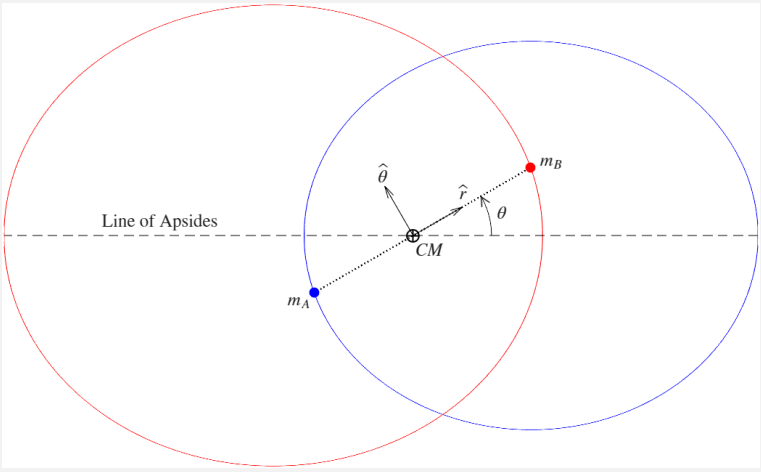


Figure 2.3: Motion of the Alpha Centauri A/B system about the center of mass.

2.3 The Eccentricity Vector

Orbits behave as conic sections, with each of the conic section parameters having an equivalence to the dynamic quantities of the system. Now that the parallels between conic section geometry and system dynamics have been established, it is time to revisit the hidden structure determined by equation 2.38. Note that the constants C_1 and C_2 have been determined by applying equations 2.28 and 2.32. Equation 2.38 has been rewritten with the constants as they were determined by equations 2.33 and 2.34.

$$\begin{pmatrix} \frac{1}{r_p} - \frac{\mu}{h^2} \\ 0 \end{pmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{pmatrix} \frac{1}{r} - \frac{\mu}{h^2} \\ -\frac{\dot{r}}{h} \end{pmatrix} \tag{2.64}$$

Factoring the left hand side:

$$\frac{\mu}{h^2} \begin{pmatrix} \frac{h^2}{\mu r_p} - 1 \\ 0 \end{pmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{pmatrix} \frac{1}{r} - \frac{\mu}{h^2} \\ -\frac{\dot{r}}{h} \end{pmatrix} \tag{2.65}$$

Dividing through by the factor and recalling equation 2.42 for e on the left hand side:

$$e \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{pmatrix} \frac{h^2}{\mu}\frac{1}{r} - 1 \\ -\frac{\dot{r}h}{\mu} \end{pmatrix} \tag{2.66}$$

Equation 2.66 is the same as that of a polar to cartesian coordinate vector rotation transformation. If this structure is to be trusted, then that would suggest a two dimensional reference frame where this vector could exist. The obvious choice is the plane defined by the $\hat{r}$ and $\hat{\theta}$ directions, as this corresponds to the orbital plane. This choice is obvious because, in general, this was the only meaningful two dimensional plane within the structure of the dynamics. The left hand side of equation 2.66 is arbitrary since it is expressed in cartesian coordinates, which leaves the vector in polar coordinates on the right hand side. These are the *same* vectors, but expressed in terms of different coordinate systems. Because the dynamics were done from the $r-\theta$ plane of the cylindrical coordinate system, it makes sense to look deeper into the polar vector on the right hand side. This vector will be denoted as $\underline{e}$ because it has magnitude e as recognized by the left hand side of equation 2.66.

$$\underline{e} = \left(\frac{h^2}{\mu}\frac{1}{r} - 1\right)\hat{r} + \left(-\frac{\dot{r}h}{\mu}\right)\hat{\theta} \tag{2.67}$$

Rearranging:

$$\underline{e} = \frac{1}{\mu}\left(\frac{h^2}{r}\hat{r} - \dot{r}h\,\hat{\theta}\right) - \hat{r} \tag{2.68}$$

The terms in parenthesis on the right hand side of equation 2.68 are recognized as the cross product between the inertial derivative of the relative position vector (recall no $\hat{z}$ dependence by definition) and the specific orbital angular momentum vector:

$$\underline{\dot{r}} \times \underline{h} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \dot{r} & r\dot{\theta} & 0 \\ 0 & 0 & h \end{vmatrix} = (r\dot{\theta}h - 0)\hat{r} - (\dot{r}h - 0)\hat{\theta} + (0 - 0)\hat{z} \tag{2.69}$$

Substituting in equation 1.85 into equation 2.69 for $\dot{\theta}$:

$$\underline{\dot{r}} \times \underline{h} = r\dot{\theta}h\,\hat{r} - \dot{r}h\,\hat{\theta} = \frac{h^2}{r}\,\hat{r} - \dot{r}h\,\hat{\theta} \tag{2.70}$$

Plugging equation 2.70 into equation 2.68 results in the constant vector expressed in terms of the $r-\theta$ plane of the center of mass fixed, non-inertial, cylindrical coordinate system.

$$\boxed{\underline{e} = \frac{\underline{\dot{r}} \times \underline{h}}{\mu} - \hat{r}} \tag{2.71}$$

The formal name of equation 2.71 is the eccentricity vector. This vector is constant throughout any two body Keplerian orbit. A property of this vector is that its magnitude is always equal to the eccentricity, as evidenced by the left hand side of equation 2.66. If the eccentricity vector is applied to the condition of periapsis ($\theta = 0$, $r = r_p$, and $\dot{r} = 0$) some more properties of the eccentricity vector become known. In this case, equation 2.67 is a convenient form of the eccentricity equation.

$$\underline{e} = \left(\frac{h^2}{\mu}\frac{1}{r_p} - 1\right)\hat{r} + \left(-\frac{(0)h}{\mu}\right)\hat{\theta} = e\,\hat{r} \tag{2.72}$$

Notice that the radial term is equal to the eccentricity by equation 2.42. So, when the (constant) eccentricity vector is applied to periapsis, it points in the radial component only. This would suggest that the eccentricity vector always points towards an orbit's periapsis with a magnitude equal to the orbital eccentricity. If this behavior is consistent over the entire orbit, then at the orbit's semi-latus rectum ($\theta = \pi/2$, $r = p$, and $\dot{r} = \dot{r}_{\to p}$), the eccentricity vector would point in the $-\hat{\theta}$ direction with magnitude equal to the orbital eccentricity. Again utilizing equation 2.42 for convenience:

$$\underline{e} = \left(\frac{h^2}{\mu}\frac{1}{p} - 1\right)\hat{r} + \left(-\frac{\dot{r}_{\to p}\,h}{\mu}\right)\hat{\theta} \tag{2.73}$$

Substituting equation 2.41 in for p:

$$\underline{e} = \left(\frac{h^2}{\mu}\frac{1}{\left(\frac{h^2}{\mu}\right)} - 1\right)\hat{r} - \frac{\dot{r}_{\to p}h}{\mu}\hat{\theta} \tag{2.74}$$

$$\underline{e} = \frac{h^2}{\mu}\big(1-1\big)\,\hat{r} - \frac{\dot{r}h}{\mu}\hat{\theta} = 0\,\hat{r} - \frac{\dot{r}_{\to p}h}{\mu}\hat{\theta} \tag{2.75}$$

$$\underline{e} = -\frac{\dot{r}_{\to p}h}{\mu}\hat{\theta} \tag{2.76}$$

To characterize the time rate of change of the relative position (at the semi-latus rectum), take a time derivative of equation 2.39.

$$\dot{r} = \frac{d}{dt}\left(\frac{p}{1+e\cos(\theta)}\right) = \frac{d\theta}{dt}\frac{d}{d\theta}\left(\frac{p}{1+e\cos(\theta)}\right) = \dot{\theta}\frac{d}{d\theta}\left(\frac{p}{1+e\cos(\theta)}\right) \tag{2.77}$$

Substituting equation 1.85 for $\dot{\theta}$.

$$\dot{r} = \frac{h}{r^2}\left(\frac{pe\sin(\theta)}{\left(1+e\cos(\theta)\right)^2}\right) = \frac{1}{r^2}\left(\frac{pe\sin(\theta)}{\left(1+e\cos(\theta)\right)^2}\right)h \tag{2.78}$$

Substitute equation 2.39 in for r and equation 2.41 in for p.

$$\dot{r} = \frac{\left(1+e\cos(\theta)\right)^2}{p^2}\left(\frac{pe\sin(\theta)}{\left(1+e\cos(\theta)\right)^2}\right)h \tag{2.79}$$

$$\dot{r} = \frac{he\sin(\theta)}{p} = \frac{he\sin(\theta)}{\left(\frac{h^2}{\mu}\right)} \tag{2.80}$$

$$\boxed{\dot{r} = \frac{\mu e\sin(\theta)}{h}} \tag{2.81}$$

Therefore:

$$\dot{r}_{\to p} = \frac{\mu e\sin\left(\frac{\pi}{2}\right)}{h} = \frac{\mu e}{h} \tag{2.82}$$

Substituting equation 2.82 into equation 2.76 for $\dot{r}_{\to p}$.

$$\underline{e} = -\frac{\left(\frac{\mu e}{h}\right)h}{\mu}\hat{\theta} = -e\,\hat{\theta} \tag{2.83}$$

The result of equation 2.83 indicates that the eccentricity vector is exactly what was predicted. It is a constant vector within the orbital plane, with magnitude equal to the orbital eccentricity and direction always pointing towards the orbit's periapsis.

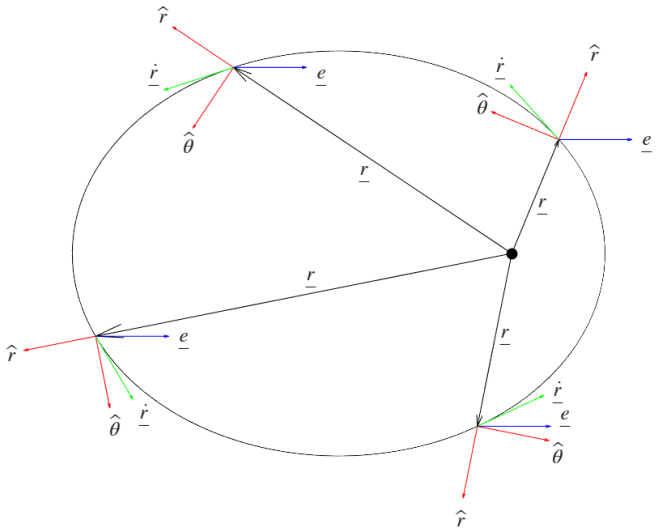


Figure 2.4: Eccentricity vector in various locations along an elliptic orbit.

Example

Determine an equation for the eccentricity vector that expresses $\underline{e}$ in terms of only the $\underline{r}$ and $\dot{\underline{r}}$ vectors without any cross product terms. Recall that the specific orbital angular momentum for the center of mass fixed, non-inertial, cylindrical reference frame was given by equation 1.84. Substitute equation 1.84 into equation 2.71.

$$\underline{e} = \frac{\dot{\underline{r}} \times \underline{h}}{\mu} - \hat{r} = \frac{\dot{\underline{r}} \times (\underline{r} \times \dot{\underline{r}})}{\mu} - \hat{r} \tag{2.84}$$

Utilizing the vector triple product rule:

$$\underline{e} = \frac{(\dot{\underline{r}} \cdot \dot{\underline{r}})\,\underline{r} - (\dot{\underline{r}} \cdot \underline{r})\,\dot{\underline{r}}}{\mu} - \hat{r} \tag{2.85}$$

The magnitude of $\dot{\underline{r}}$ is the orbital velocity v:

$$\underline{e} = \frac{v^2\,\underline{r} - (\underline{r} \cdot \dot{\underline{r}})\,\dot{\underline{r}}}{\mu} - \hat{r} = \left(\frac{v^2}{\mu} - \frac{1}{r}\right)\underline{r} - (\underline{r} \cdot \dot{\underline{r}})\,\dot{\underline{r}} \tag{2.86}$$

Rewrite equation 2.86 to gather components in terms of constants.

$$\underline{e} = \left(\frac{v^2}{\mu} - \frac{2}{r} + \frac{1}{r}\right)\underline{r} - (\underline{r} \cdot \dot{\underline{r}})\,\dot{\underline{r}} \tag{2.87}$$

Using the equation for specific mechanical energy in equation 1.82 to rewrite:

$$\underline{e} = \left(\frac{2\varepsilon}{\mu} + \frac{1}{r}\right)\underline{r} - (\underline{r} \cdot \dot{\underline{r}})\,\dot{\underline{r}} \tag{2.88}$$

An interesting fact of orbital mechanics is that, for *any* instantaneous state ($\underline{r}$ **and** $\dot{\underline{r}}$), the orbit can be completely determined. Having the eccentricity vector in the form of equation 2.88 allows for simpler calculation of the orbital eccentricity, which can be determined very easily by taking the magnitude of the eccentricity vector.

The derivation of the eccentricity vector as described in this section differs from the more standard derivation. The standard derivation of the eccentricity vector begins by taking the cross product between equation 1.86 and equation 1.84.

$$\ddot{\underline{r}} \times \underline{h} = -\frac{\mu}{r^2}\,\hat{r} \times (\underline{r} \times \dot{\underline{r}}) = -\frac{\mu}{r^3}\left(\underline{r} \times \underline{r} \times \dot{\underline{r}}\right) \tag{2.89}$$

$$\ddot{\underline{r}} \times \underline{h} = -\frac{\mu}{r^3}\left((\underline{r} \cdot \dot{\underline{r}})\,\underline{r} - (\underline{r} \cdot \underline{r})\dot{\underline{r}}\right) \tag{2.90}$$

Recognizing both sides as time derivatives:

$$\frac{d}{dt}\left(\dot{\underline{r}} \times \underline{h}\right) = \ddot{\underline{r}} \times \underline{h} + \dot{\underline{r}} \times \dot{\underline{h}} = -\frac{\mu}{r^3}\left((\underline{r} \cdot \dot{\underline{r}})\,\underline{r} - (\underline{r} \cdot \underline{r})\dot{\underline{r}}\right) = \frac{d}{dt}\left(\frac{\mu}{r}\underline{r}\right) \tag{2.91}$$

Integrating with respect to time introduces a constant vector from the integration constant.

$$\int \frac{d}{dt}\left(\dot{\underline{r}} \times \underline{h}\right) = \int \frac{d}{dt}\left(\frac{\mu}{r}\underline{r}\right) \quad \longrightarrow \quad \dot{\underline{r}} \times \underline{h} = \frac{\mu}{r}\underline{r} + \underline{C} \tag{2.92}$$

$$\underline{C} = \dot{\underline{r}} \times \underline{L} - \frac{\mu}{r}\underline{r} \tag{2.93}$$

Since the integration constant remains constant from the division of a scalar, the constant vector is once more rewritten in the familiar form of equation 2.71.

$$\underline{C} = \frac{\dot{\underline{r}} \times \underline{h}}{\mu} - \frac{\underline{r}}{r} = \frac{\dot{\underline{r}} \times \underline{h}}{\mu} - \hat{r} = \underline{e} \tag{2.94}$$

Although both approaches arrive at the same conserved vector, they differ in emphasis. The standard derivation constructs the eccentricity vector by identifying a quantity whose time derivative vanishes, while the one derived in this section comes from recognizing the rotational structure from the integration constants used to derive the orbit equation.

Example

Derive the orbit equation from the dot product between the relative position vector and the eccentricity vector (equation 2.71). The vectors $\underline{r}$ and $\underline{e}$ are separated by an angular distance equal to θ since periapsis is located at $\theta = 0$.

$$re\cos(\theta) = |\underline{r}|\,|\underline{e}|\cos(\theta) = \underline{r} \cdot \underline{e} = \underline{r} \cdot \left(\frac{\dot{\underline{r}} \times \underline{h}}{\mu} - \hat{r}\right) \tag{2.95}$$

Using the scalar triple product rule:

$$re\cos(\theta) = \frac{(\underline{r} \times \dot{\underline{r}}) \cdot \underline{h}}{\mu} - r(\hat{r} \cdot \hat{r}) = \frac{(r^2\dot{\theta}\hat{z}) \cdot h\hat{z}}{\mu} = \frac{r^2\dot{\theta}h}{\mu} - r \tag{2.96}$$

Using equation 1.85 for $\dot{\theta}$:

$$re\cos(\theta) = \frac{r^2\left(\frac{h}{r^2}\right)h}{\mu} - r = \frac{h^2}{\mu} - r \tag{2.97}$$

Rearranging to solve for r:

$$r + re\cos(\theta) = r(1 + e\cos(\theta)) = \frac{h^2}{\mu} \tag{2.98}$$

Rearranging again and recognizing the numerator as equation 2.41:

$$r = \frac{\frac{h^2}{\mu}}{1 + e\cos(\theta)} = \frac{p}{1 + e\cos(\theta)} \tag{2.99}$$

□

2.4 The Perifocal Reference Frame

Consider an inertial orthogonal coordinate system centered at the Keplerian two body system center of mass. For any Keplerian two body orbit with non-zero eccentricity, there are two directions that remain fixed in inertial space: the eccentricity vector direction and the orbital angular momentum direction. These constant directions are used to define the inertial reference frame:

$$\hat{p} = \frac{\underline{e}}{e} = \hat{e} \qquad \hat{q} = \hat{w} \times \hat{p} = \hat{h} \times \hat{e} \qquad \hat{w} = \frac{\underline{h}}{h} = \hat{h} \tag{2.100}$$

This reference frame is called the perifocal reference frame. The earlier example of the Alpha Centauri A/B binary star system actually uses this reference frame to plot the positions of the two stars relative to the barycenter, although the $\hat{w}$ direction was unnecessary for that problem. Thus the $\hat{p} - \hat{q}$ plane coincides with the two body orbital plane.

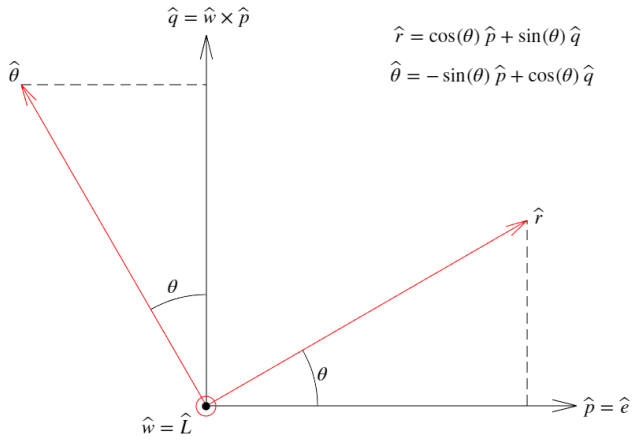


Figure 2.5: Transformation between the cylindrical and perifocal frames.

Expressing the relative position vector and its derivative in the perifocal frame:

$$\underline{r} = r\,\hat{r} = r\Big(\cos(\theta)\,\hat{p} + \sin(\theta)\,\hat{q}\Big) \tag{2.101}$$

$$\dot{\underline{r}} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta} = \dot{r}\Big(\cos(\theta)\,\hat{p} + \sin(\theta)\,\hat{q}\Big) + r\dot{\theta}\Big(-\sin(\theta)\,\hat{p} + \cos(\theta)\,\hat{q}\Big) \tag{2.102}$$

Substituting the orbit equation for r, equation 2.81 for $\dot{r}$, and equation 1.85 for $\dot{\theta}$:

$$\boxed{\underline{r} = \frac{h^2/\mu}{1 + e\cos(\theta)}\Big(\cos(\theta)\,\hat{p} + \sin(\theta)\,\hat{q}\Big)} \tag{2.103}$$

$$\dot{\underline{r}} = \frac{\mu e \sin(\theta)}{h}\left(\cos(\theta)\,\hat{p} + \sin(\theta)\,\hat{q}\right) + \frac{\mu}{h}\left(1 + e\cos(\theta)\right)\left(-\sin(\theta)\,\hat{p} + \cos(\theta)\,\hat{q}\right) \tag{2.104}$$

Combining $\hat{p}$ and $\hat{q}$ terms:

$$\dot{\underline{r}} = \frac{\mu}{h}\Big(\left(e\sin(\theta) - \sin(\theta)(1 + e\cos(\theta))\right)\hat{p} + \left(e\sin^2(\theta) + \cos(\theta)(1 + e\cos(\theta))\right)\hat{q}\Big) \tag{2.105}$$

Applying the Pythagorean trigonometric identity and simplifying:

$$\boxed{\dot{\underline{r}} = \frac{\mu}{h}\Big(-\sin(\theta)\,\hat{p} + \left(e + \cos(\theta)\right)\hat{q}\Big)} \tag{2.106}$$

2.5 The Lagrange Coefficients: $\Delta\theta$

The governing question of orbit propagation is: if the relative position and velocity is currently known, what will they look like later? The applications of this question are endless, but the question is where to begin answering it. The initial relative position and velocity are:

$$\underline{r}_0 = x_0\,\hat{p} + y_0\,\hat{q} \tag{2.107}$$

$$\underline{\dot{r}}_0 = \dot{x}_0\,\hat{p} + \dot{y}_0\,\hat{q} \tag{2.108}$$

where x_0 and y_0 are the components of the relative position expressed in the perifocal frame, and $\dot{x}_0$ and $\dot{y}_0$ are the components of the relative velocity expressed in the perifocal frame. It will be extremely useful to determine the $\hat{p}$ and $\hat{q}$ directions. From the initial state:

$$\hat{p} = \frac{1}{x_0}\underline{r}_0 - \frac{y_0}{x_0}\hat{q} \tag{2.109}$$

Plugging into the initial relative velocity for $\hat{p}$:

$$\underline{\dot{r}}_0 = \dot{x}_0\left(\frac{1}{x_0}\underline{r}_0 - \frac{y_0}{x_0}\hat{q}\right) + \dot{y}_0\,\hat{q} \tag{2.110}$$

$$\underline{\dot{r}}_0 = \frac{\dot{x}_0}{x_0}\underline{r}_0 + \left(\dot{y}_0 - \frac{\dot{x}_0 y_0}{x_0}\right)\hat{q} \tag{2.111}$$

Rearranging and seeking a common denominator:

$$\left(\frac{x_0\dot{y}_0 - \dot{x}_0 y_0}{x_0}\right)\hat{q} = \frac{x_0}{x_0}\underline{\dot{r}}_0 - \frac{\dot{x}_0}{x_0}\underline{r}_0 = \frac{-\dot{x}_0\underline{r}_0 + x_0\underline{\dot{r}}_0}{x_0} \tag{2.112}$$

Rearranging for $\hat{q}$:

$$\hat{q} = \frac{-\dot{x}_0\underline{r}_0 + x_0\underline{\dot{r}}_0}{x_0\dot{y}_0 - \dot{x}_0 y_0} \tag{2.113}$$

Plugging equation 2.113 into equation 2.109 for $\hat{q}$:

$$\hat{p} = \frac{1}{x_0}\underline{r}_0 - \frac{y_0}{x_0}\left(\frac{-\dot{x}_0\underline{r}_0 + x_0\underline{\dot{r}}_0}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right) \tag{2.114}$$

Expanding and seeking a common denominator:

$$\hat{p} = \frac{x_0\dot{y}_0\underline{r}_0 - \dot{x}_0 y_0\underline{r}_0 + \dot{x}_0 y_0\underline{r}_0 - x_0 y_0\underline{\dot{r}}_0}{x_0(x_0\dot{y}_0 - \dot{x}_0 y_0)} = \frac{(x_0\dot{y}_0 - \dot{x}_0 y_0 + \dot{x}_0 y_0)\underline{r}_0 - x_0 y_0\underline{\dot{r}}_0}{x_0(x_0\dot{y}_0 - \dot{x}_0 y_0)} \tag{2.115}$$

$$\hat{p} = \frac{\dot{y}_0\underline{r}_0 - y_0\underline{\dot{r}}_0}{x_0\dot{y}_0 - \dot{x}_0 y_0} \tag{2.116}$$

With the $\hat{p}$ and $\hat{q}$ direction determined from the initial state, the relative position and velocity can be expressed in terms of the initial state:

$$\underline{r} = x\,\hat{p} + y\,\hat{q} = x\left(\frac{\dot{y}_0\underline{r}_0 - y_0\underline{\dot{r}}_0}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right) + y\left(\frac{-\dot{x}_0\underline{r}_0 + x_0\underline{\dot{r}}_0}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right) \tag{2.117}$$

$$\boxed{\underline{r} = \left(\frac{x\dot{y}_0 - \dot{x}_0 y}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right)\underline{r}_0 + \left(\frac{x_0 y - x y_0}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right)\underline{\dot{r}}_0 \equiv f\,\underline{r}_0 + g\,\underline{\dot{r}}_0} \tag{2.118}$$

$$\underline{\dot{r}} = \dot{x}\,\hat{p} + \dot{y}\,\hat{q} = \dot{x}\left(\frac{\dot{y}_0\underline{r}_0 - y_0\underline{\dot{r}}_0}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right) + \dot{y}\left(\frac{-\dot{x}_0\underline{r}_0 + x_0\underline{\dot{r}}_0}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right) \tag{2.119}$$

$$\boxed{\underline{\dot{r}} = \left(\frac{\dot{x}\dot{y}_0 - \dot{x}_0\dot{y}}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right)\underline{r}_0 + \left(\frac{x_0\dot{y} - \dot{x}y_0}{x_0\dot{y}_0 - \dot{x}_0 y_0}\right)\underline{\dot{r}}_0 \equiv \dot{f}\,\underline{r}_0 + \dot{g}\,\underline{\dot{r}}_0} \tag{2.120}$$

The variables f, $\dot{f}$, g, and $\dot{g}$ are called the Lagrange coefficients. They are scalar functions that use the initial state as a starting point to propagate the orbit. Unfortunately, x, $\dot{x}$, y, and $\dot{y}$ are not known, which is the whole point of orbit propagation. In addition, generally the relative position and velocity vectors are not observed in the perifocal basis. Thus the Lagrange coefficients must be expressed in terms of variables that can be determined from an initial state given in any reference frame in order to extract their usefulness.

Simplifications

Express $e\cos(\theta)$ and $e\sin(\theta)$ in terms of only variables that can be recovered from an observable state. Using the orbit equation:

$$r = \frac{h^2/\mu}{1 + e\cos(\theta)} \tag{2.121}$$

$$1 + e\cos(\theta) = \frac{h^2}{\mu r} \tag{2.122}$$

$$e\cos(\theta) = \frac{h^2}{\mu r} - 1 = \frac{h^2}{\mu r} - \frac{\mu r}{\mu r} \tag{2.123}$$

$$\boxed{e\cos(\theta) = \frac{h^2 - \mu r}{\mu r}} \tag{2.124}$$

Consider the dot product between $\underline{r}$ and $\underline{\dot{r}}$.

$$\underline{r}\cdot\underline{\dot{r}} = (r\,\hat{r})\cdot(\dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta}) = r\dot{r} \tag{2.125}$$

Substituting equation 2.81 for $\dot{r}$:

$$\underline{r}\cdot\underline{\dot{r}} = r\dot{r} = r\left(\frac{\mu e\sin(\theta)}{h}\right) \quad \longrightarrow \quad \boxed{e\sin(\theta) = \frac{(\underline{r}\cdot\underline{\dot{r}})h}{\mu r} = \frac{\dot{r}h}{\mu}} \tag{2.126}$$

The orbital angular momentum can be determined by equation 1.84 and the eccentricity vector can be determined by equation 2.85. These equations require only the relative position state ($\underline{r}$ and $\underline{\dot{r}}$). From equations 2.103 and 2.106, the values of x, $\dot{x}$, y, and $\dot{y}$ for any given θ are:

$$x = \frac{h^2\cos(\theta)}{\mu\big(1 + e\cos(\theta)\big)} = r\cos(\theta) \qquad \dot{x} = \frac{-\mu\sin(\theta)}{h} \tag{2.127}$$

$$y = \frac{h^2\sin(\theta)}{\mu\big(1 + e\cos(\theta)\big)} = r\sin(\theta) \qquad \dot{y} = \frac{\mu(e + \cos(\theta))}{h} \tag{2.128}$$

All Lagrange coefficients share a common denominator. Using the components of the relative position and velocity vector expressed in the perifocal frame, the denominator is:

$$\begin{aligned} x_0\dot{y}_0 - \dot{x}_0 y_0 = &\left(\frac{h^2\cos(\theta_0)}{\mu\Big(1+e\cos(\theta_0)\Big)}\right)\left(\frac{\mu(e+\cos(\theta))}{h}\right) \\ &-\left(\frac{-\mu\sin(\theta_0)}{h}\right)\left(\frac{h^2\sin(\theta_0)}{\mu\Big(1+e\cos(\theta_0)\Big)}\right) \end{aligned} \tag{2.129}$$

$$x_0\dot{y}_0 - \dot{x}_0 y_0 = \frac{h\Big(e\cos(\theta_0)+\cos^2(\theta_0)\Big)}{\Big(1+e\cos(\theta_0)\Big)} + \frac{h\sin^2(\theta_0)}{\Big(1+e\cos(\theta_0)\Big)} \tag{2.130}$$

$$x_0\dot{y}_0 - \dot{x}_0 y_0 = h\left(\frac{\cos^2(\theta_0)+\sin^2(\theta_0)+e\cos(\theta_0)}{1+e\cos(\theta_0)}\right) = h\left(\frac{1+e\cos(\theta_0)}{1+e\cos(\theta_0)}\right) \tag{2.131}$$

$$x_0\dot{y}_0 - \dot{x}_0 y_0 = h \tag{2.132}$$

Thus the denominator of all Lagrange coefficients are known and constant. Now to determine the functional form of each of the four Lagrange coefficients using equations 2.127, 2.128, and 2.132. Starting with f:

$$\begin{aligned} f = \frac{x\dot{y}_0 - \dot{x}_0 y}{x_0\dot{y}_0 - \dot{x}_0 y_0} = &\frac{1}{h}\left(\left(\frac{h^2\cos(\theta)}{\mu\Big(1+e\cos(\theta)\Big)}\right)\left(\frac{\mu(e+\cos(\theta_0))}{h}\right)\right) \\ &-\frac{1}{h}\left(\left(\frac{-\mu\sin(\theta_0)}{h}\right)\left(\frac{h^2\sin(\theta)}{\mu\Big(1+e\cos(\theta)\Big)}\right)\right) \end{aligned} \tag{2.133}$$

$$f = \frac{h}{h}\left(\frac{\cos(\theta)(e+\cos(\theta_0))+\sin(\theta)\sin(\theta_0)}{1+e\cos(\theta)}\right) \tag{2.134}$$

$$f = \frac{e\cos(\theta)+\cos(\theta)\cos(\theta_0)+\sin(\theta)\sin(\theta_0)}{1+e\cos(\theta)} \tag{2.135}$$

Using equation 2.124 and the cosine angle difference trigonometric identity:

$$f = \frac{\left(\frac{h^2-\mu r}{\mu r}\right)+\cos(\theta-\theta_0)}{1+\left(\frac{h^2-\mu r}{\mu r}\right)} \tag{2.136}$$

$$f = \frac{\mu r}{\mu r}\frac{h^2-\mu r+\mu r\cos(\theta-\theta_0)}{\mu r+h^2-\mu r} \tag{2.137}$$

$$f = \frac{h^2-\mu r\Big(1-\cos(\theta-\theta_0)\Big)}{h^2} \tag{2.138}$$

Defining $\Delta\theta \equiv \theta - \theta_0$ and simplifying:

$$\boxed{f = 1 - \frac{\mu r}{h^2}\Big(1-\cos(\Delta\theta)\Big)} \tag{2.139}$$

The quantity $\Delta\theta$ is significant in the context of orbit propagation. One may ask the question of what the state will become after traversing some $\Delta\theta$ from the initial observation. Thus $\Delta\theta$ has meaning beyond just simplifying the mathematics.

Moving on to g:

$$g = \frac{x_0 y - x y_0}{x_0 \dot{y}_0 - \dot{x}_0 y_0} = \frac{1}{h}\left(\left(\frac{h^2\cos(\theta_0)}{\mu\Big(1+e\cos(\theta_0)\Big)}\right)\left(\frac{h^2\sin(\theta)}{\mu\Big(1+e\cos(\theta)\Big)}\right)\right) - \frac{1}{h}\left(\left(\frac{h^2\cos(\theta)}{\mu\Big(1+e\cos(\theta)\Big)}\right)\left(\frac{h^2\sin(\theta_0)}{\mu\Big(1+e\cos(\theta_0)\Big)}\right)\right) \tag{2.140}$$

$$g = \frac{h^3}{\mu^2}\left(\frac{\sin(\theta)\cos(\theta_0) - \cos(\theta)\sin(\theta_0)}{(1+e\cos(\theta_0))(1+e\cos(\theta))}\right) \tag{2.141}$$

Using the orbit equation in the denominator and the sine angle difference trigonometric identity in the numerator:

$$g = \frac{h^3}{\mu^2}\left(\frac{\sin(\theta-\theta_0)}{\left(\frac{h^2}{\mu r_0}\right)\left(\frac{h^2}{\mu r}\right)}\right) = \frac{h^3}{\mu^2}\left(\frac{\mu^2 r r_0 \sin(\Delta\theta)}{h^4}\right) \tag{2.142}$$

$$\boxed{g = \frac{r r_0}{h}\sin(\Delta\theta)} \tag{2.143}$$

Next solving for $\dot{f}$:

$$\dot{f} = \frac{\dot{x}\dot{y}_0 - \dot{x}_0\dot{y}}{x_0\dot{y}_0 - \dot{x}_0 y_0} = \frac{1}{h}\left(\left(\frac{-\mu\sin(\theta)}{h}\right)\left(\frac{\mu(e+\cos(\theta_0))}{h}\right)\right) - \frac{1}{h}\left(\left(\frac{-\mu\sin(\theta_0)}{h}\right)\left(\frac{\mu(e+\cos(\theta))}{h}\right)\right) \tag{2.144}$$

$$\dot{f} = \frac{\mu^2}{h^3}\Big(-\sin(\theta)(e+\cos(\theta_0)) + \sin(\theta_0)(e+\cos(\theta))\Big) \tag{2.145}$$

$$\dot{f} = \frac{\mu^2}{h^3}\Big(e\sin(\theta_0) - e\sin(\theta) - \sin(\theta)\cos(\theta_0) + \cos(\theta)\sin(\theta_0)\Big) \tag{2.146}$$

Using the sine angle difference trigonometric identity and substituting $\theta = \theta_0 + \Delta\theta$:

$$\dot{f} = \frac{\mu^2}{h^3}\Big(e\sin(\theta_0) - e\sin(\theta_0+\Delta\theta) - \sin(\Delta\theta)\Big) \tag{2.147}$$

Using the sine angle sum trigonometric identity:

$$\dot{f} = \frac{\mu^2}{h^3}\Big(e\sin(\theta_0) - e\Big(\sin(\theta_0)\cos(\Delta\theta) + \cos(\theta_0)\sin(\Delta\theta)\Big) - \sin(\Delta\theta)\Big) \tag{2.148}$$

$$\dot{f} = \frac{\mu^2}{h^3}\Big(e\sin(\theta_0)\Big(1-\cos(\Delta\theta)\Big) - \sin(\Delta\theta)\Big(1+e\cos(\theta_0)\Big)\Big) \tag{2.149}$$

Using equations 2.124 and 2.126:

$$\dot{f} = \frac{\mu^2}{h^3}\left(\frac{\dot{r}_0 h}{\mu}\Big(1-\cos(\Delta\theta)\Big) - \sin(\Delta\theta)\left(1+\frac{h^2-\mu r_0}{\mu r_0}\right)\right) \tag{2.150}$$

$$\dot{f} = \frac{\mu^2}{h^3}\left(\frac{\dot{r}_0 h}{\mu}\Big(1-\cos(\Delta\theta)\Big) - \sin(\Delta\theta)\left(\frac{\mu r_0 + h^2 - \mu r_0}{\mu r_0}\right)\right) \tag{2.151}$$

$$\dot{f} = \frac{\mu^2}{h^3}\left(\frac{\dot{r}_0 h}{\mu}\Big(1-\cos(\Delta\theta)\Big) - \frac{h^2}{\mu r_0}\sin(\Delta\theta)\right) \tag{2.152}$$

$$\dot{f} = \frac{\mu r_0 \dot{r}_0}{h^2 r_0}\Big(1-\cos(\Delta\theta)\Big) - \frac{\mu}{h r_0}\sin(\Delta\theta) \tag{2.153}$$

$$\boxed{\dot{f} = \frac{\mu}{h r_0}\left(\frac{r_0\dot{r}_0}{h}\Big(1-\cos(\Delta\theta)\Big) - \sin(\Delta\theta)\right)} \tag{2.154}$$

Finally solving for $\dot{g}$:

$$\dot{g} = \frac{x_0\dot{y} - \dot{x}y_0}{x_0\dot{y}_0 - \dot{x}_0 y_0} = \frac{1}{h}\left(\left(\frac{h^2\cos(\theta_0)}{\mu\big(1+e\cos(\theta_0)\big)}\right)\left(\frac{(e+\cos(\theta))}{h}\right)\right) - \frac{1}{h}\left(\left(\frac{-\mu\sin(\theta)}{h}\right)\left(\frac{h^2\sin(\theta_0)}{\mu\big(1+e\cos(\theta_0)\big)}\right)\right) \tag{2.155}$$

$$\dot{g} = \frac{e\cos(\theta_0) + \cos(\theta)\cos(\theta_0) + \sin(\theta)\sin(\theta_0)}{1+e\cos(\theta_0)} \tag{2.156}$$

Using equation 2.124 and the cosine angle difference trigonometric identity:

$$\dot{g} = \frac{\frac{h^2-\mu r_0}{\mu r_0} + \cos(\theta-\theta_0)}{1+\frac{h^2-\mu r_0}{\mu r_0}} \tag{2.157}$$

$$\dot{g} = \frac{\mu r_0}{\mu r_0}\frac{h^2 - \mu r_0 + \mu r_0\cos(\Delta\theta)}{\mu r_0 + h^2 - \mu r_0} \tag{2.158}$$

$$\dot{g} = \frac{h^2 - \mu r_0\Big(1-\cos(\Delta\theta)\Big)}{h^2} \tag{2.159}$$

$$\boxed{\dot{g} = 1 - \frac{\mu r_0}{h^2}\Big(1-\cos(\Delta\theta)\Big)} \tag{2.160}$$

The astute reader may notice that the Lagrange coefficients f and g contain an r term (not to be confused with r_0). In fact, r is the magnitude of the propagated relative position. At first glance, the Lagrange coefficients can not be determined without knowledge of the propagated orbit, orbit propagation can not proceed due to a circular dependency. However, consider that $\theta = \theta_0 + \Delta\theta$, and the relative position can be rewritten using the orbit equation.

$$r = \frac{h^2/\mu}{1+e\cos(\theta_0+\Delta\theta)} \tag{2.161}$$

Using the cosine angle sum trigonometric identity:

$$r = \frac{h^2/\mu}{1+e\cos(\theta_0)\cos(\Delta\theta) - e\sin(\theta_0)\sin(\Delta\theta)} \tag{2.162}$$

Using equations 2.124 and 2.126 for $e\cos(\theta_0)$ and $e\sin(\theta_0)$:

$$r = \frac{h^2/\mu}{1+\left(\frac{h^2-\mu r_0}{\mu r_0}\right)\cos(\Delta\theta) - \left(\frac{(\underline{r}_0\cdot\underline{\dot{r}}_0)h}{\mu r_0}\right)\sin(\Delta\theta)} \tag{2.163}$$

Vector Relationships

Determine relationships between the dot products of the eccentricity vector, the relative position vector, and the relative velocity vectors. Start by expanding the eccentricity vector with equation 2.70

$$\underline{e} = \frac{\underline{\dot{r}} \times \underline{h}}{\mu} - \hat{r} = \frac{\frac{h^2}{r}\hat{r} - \dot{r}h\,\hat{\theta}}{\mu} - \hat{r} = \left(\frac{h^2}{\mu r} - 1\right)\hat{r} - \frac{\dot{r}h}{\mu}\hat{\theta} \tag{2.164}$$

Taking the dot product of $\underline{e}$ and $\underline{r}$:

$$\underline{e} \cdot \underline{r} = \left(\left(\frac{h^2}{\mu r} - 1\right)\hat{r} - \frac{\dot{r}h}{\mu}\hat{\theta}\right) \cdot r\,\hat{r} = r\left(\frac{h^2}{\mu r} - 1\right) \tag{2.165}$$

$$\boxed{\underline{e} \cdot \underline{r} = r\left(\frac{h^2 - \mu r}{\mu r}\right)} \tag{2.166}$$

Taking the dot product of $\underline{e}$ and $\underline{\dot{r}}$:

$$\underline{e} \cdot \underline{\dot{r}} = \left(\left(\frac{h^2}{\mu r} - 1\right)\hat{r} - \frac{\dot{r}h}{\mu}\hat{\theta}\right) \cdot \left(\dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta}\right) \tag{2.167}$$

$$\underline{e} \cdot \underline{\dot{r}} = \left(\left(\frac{h^2}{\mu r} - 1\right)\hat{r} - \frac{\dot{r}h}{\mu}\hat{\theta}\right) \cdot \left(\dot{r}\,\hat{r} + r\left(\frac{h}{r^2}\right)\hat{\theta}\right) \tag{2.168}$$

$$\underline{e} \cdot \underline{\dot{r}} = \left(\frac{h^2}{\mu r} - 1\right)(\dot{r}) - \left(\frac{\dot{r}h}{\mu}\right)\left(r\left(\frac{h}{r^2}\right)\right) \tag{2.169}$$

$$\underline{e} \cdot \underline{\dot{r}} = \frac{h^2}{\mu}\frac{\dot{r}}{r} - \dot{r} - \frac{h^2}{\mu}\frac{\dot{r}}{r} \tag{2.170}$$

$$\boxed{\underline{e} \cdot \underline{\dot{r}} = -\dot{r}} \tag{2.171}$$

Recalling equation 2.125:

$$\dot{r} = \frac{\underline{r} \cdot \underline{\dot{r}}}{r} = -\underline{e} \cdot \underline{\dot{r}} \qquad \longrightarrow \qquad \boxed{\underline{r} \cdot \underline{\dot{r}} = -r\,\underline{e} \cdot \underline{\dot{r}}} \tag{2.172}$$

Substituting equations 2.166 and 2.172 into equation 2.163:

$$r = \frac{h^2/\mu}{1 + \frac{\underline{e}\cdot\underline{r}_0}{r_0}\cos(\Delta\theta) - \left(\frac{-r_0(\underline{e}\cdot\underline{\dot{r}}_0)h}{\mu r_0}\right)\sin(\Delta\theta)} \tag{2.173}$$

$$\boxed{r = \frac{\frac{h^2}{\mu}}{1 + \underline{e} \cdot \left(\frac{\cos(\Delta\theta)}{r_0}\underline{r}_0 + \frac{h\sin(\Delta\theta)}{\mu}\underline{\dot{r}}_0\right)}} \tag{2.174}$$

With the inclusion of equation 2.174, each of the Lagrange coefficients are given in terms of only the initial state, masses, and $\Delta\theta$. So, to the question as to whether or not orbit propagation can be done from initial observation, it absolutely can.

Example

Determine the orbital angular momentum from the initial state using the Lagrange coefficients, as well as any useful relationships. From equation 1.84:

$$\underline{h} = \underline{r} \times \underline{\dot{r}} = \left(f\,\underline{r}_0 + g\,\underline{\dot{r}}_0\right) \times \left(\dot{f}\,\underline{r}_0 + \dot{g}\,\underline{\dot{r}}_0\right) \tag{2.175}$$

$$\underline{h} = \left(f\dot{f}\,\underline{r}_0 \times \underline{r}_0 + f\dot{g}\,\underline{r}_0 \times \underline{\dot{r}}_0 + \dot{f}g\,\underline{\dot{r}}_0 \times \underline{r}_0 + g\dot{g}\,\underline{\dot{r}}_0 \times \underline{\dot{r}}_0\right) \tag{2.176}$$

Since the cross product of a vector with itself is the zero vector and $\underline{\dot{r}}_0 \times \underline{r}_0 = -\underline{r}_0 \times \underline{\dot{r}}_0$:

$$\underline{h} = \left(f\dot{g}\,\underline{r}_0 \times \underline{\dot{r}}_0 - \dot{f}g\,\underline{r}_0 \times \underline{\dot{r}}_0\right) \tag{2.177}$$

$$\underline{h} = \left(f\dot{g} - \dot{f}g\right)\underline{r}_0 \times \underline{\dot{r}}_0 \tag{2.178}$$

The orbital angular momentum of a Keplerian two body system is constant. Therefore, it can also be determined from the initial state.

$$\underline{h} = \underline{r}_0 \times \underline{\dot{r}}_0 \tag{2.179}$$

Equating orbital angular momenta:

$$\underline{h} = \left(f\dot{g} - \dot{f}g\right)\underline{r}_0 \times \underline{\dot{r}}_0 = \underline{r}_0 \times \underline{\dot{r}}_0 \tag{2.180}$$

Rearranging:

$$\left(f\dot{g} - \dot{f}g - 1\right)\underline{r}_0 \times \underline{\dot{r}}_0 = \underline{0} \tag{2.181}$$

For any orbit with non-zero orbital angular momentum, $\underline{r}_0 \times \underline{\dot{r}}_0 \neq \underline{0}$. In order for orbital angular momentum to remain constant $f\dot{g} - \dot{f}g - 1 = 0$ must hold, and therefore:

$$\boxed{f\dot{g} - \dot{f}g = 1} \tag{2.182}$$

The relationship of the Lagrange coefficients, as shown in equation 2.182, is consistent with the conservation of orbital angular momentum in two body Keplerian motion. This relation can also be used as an internal consistency check for numerical implementations of the Lagrange coefficients to verify orbital angular momentum conservation throughout the propagation.

The propagation of orbits can now proceed using only an initial observation and a desired change in θ. However, this method of propagation does not yet account for time, which is an essential parameter in practical orbit propagation. The next chapter will address this limitation by introducing the necessary tools to incorporate time into orbit propagation. In addition, a new formulation for the Lagrange coefficients can subsequently be determined for orbit propagation given a time interval instead of a change in true anomaly.

2.6 Summary

Orbit conic section parameters:

$$p = a(1 - e^2) = \frac{h^2}{\mu}$$

$$a = \frac{p}{1 - e^2} = -\frac{\mu}{2\varepsilon}$$

$$e = \sqrt{1 + \frac{2\varepsilon h^2}{\mu^2}}$$

The orbit equation:

$$r = \frac{p}{1 + e\cos(\theta)}$$

Constant vectors:

$$\underline{h} = \underline{r} \times \underline{\dot{r}}$$

$$\underline{e} = \frac{\underline{\dot{r}} \times \underline{h}}{\mu} - \hat{r}$$

The perifocal reference frame:

$$\hat{p} = \frac{\underline{e}}{e} \qquad \hat{q} = \hat{w} \times \hat{p} \qquad \hat{w} = \frac{\underline{h}}{h}$$

$$\underline{r} = \frac{h^2/\mu}{1 + e\cos(\theta)}\Big(\cos(\theta)\,\hat{p} + \sin(\theta)\,\hat{q}\Big)$$

$$\underline{\dot{r}} = \frac{\mu}{h}\Big(-\sin(\theta)\,\hat{p} + (e + \cos(\theta))\,\hat{q}\Big)$$

Orbit propagation via an inital state and $\Delta\theta$ with the Lagrange coefficients:

$$\underline{r} = f\,\underline{r}_0 + g\,\underline{\dot{r}}_0 \qquad \underline{\dot{r}} = \dot{f}\,\underline{r}_0 + \dot{g}\,\underline{\dot{r}}_0$$

$$r = \frac{h^2/\mu}{1 + \underline{e} \cdot \left(\dfrac{\cos(\Delta\theta)}{r_0}\underline{r}_0 + \dfrac{h\sin(\Delta\theta)}{\mu}\underline{\dot{r}}_0\right)}$$

$$f = 1 - \frac{\mu r}{h^2}\Big(1 - \cos(\Delta\theta)\Big) \qquad \dot{f} = \frac{\mu}{h r_0}\left(\frac{r_0 \dot{r}_0}{h}\Big(1 - \cos(\Delta\theta)\Big) - \sin(\Delta\theta)\right)$$

$$g = \frac{r r_0}{h}\sin(\Delta\theta) \qquad \dot{g} = 1 - \frac{\mu r_0}{h^2}\Big(1 - \cos(\Delta\theta)\Big)$$

$$f\dot{g} - \dot{f}g = 1$$

References

[1] Eric W. Weisstein. *Conic Section Directrix*. `https://mathworld.wolfram.com/ConicSectionDirectrix.html`. From MathWorld—A Wolfram Resource. (Visited on 04/10/2026).

[2] Akeson et. al. *Precision Millimeter Astrometry of the a Centauri AB System*. `iopscience.iop.org/article/10.3847/1538-3881/abfaff`. From IOPscience. (Visited on 07/20/2026).

Chapter 3

Time Evolution

Introduction

The previous chapter has laid out the geometry of orbital motion, though it does nothing to make predictions about where objects will be located after a certain amount of time. Knowing how orbits evolve in time allows for intricate mission design and the planning of celestial events. Though the mathematics is fairly straightforward, orbit time evolution can be somewhat convoluted.

The time rate of change of θ has been an invaluable tool in formulating useful relationships regarding the two body Kepler problem. With the inclusion of the orbit equation, $\dot{\theta}$ will provide the starting place for determining the equations of orbit time dependency.

$$\frac{d\theta}{dt} = \dot{\theta} = \frac{h}{r^2} = \frac{h}{\left(\frac{h^2/\mu}{1+e\cos(\theta)}\right)^2} = \frac{\mu^2}{h^3}\Big(1 + e\cos(\theta)\Big)^2 \tag{3.1}$$

Separating variables:

$$dt = \frac{h^3}{\mu^2}\frac{1}{\Big(1 + e\cos(\theta)\Big)^2}\, d\theta \tag{3.2}$$

$\theta = 0$ at periapsis, which provides a convenient lower bound for the integration that follows.

$$\int_{t_p}^{t} dt = \frac{h^3}{\mu^2}\int_0^{\theta} \frac{1}{\Big(1 + e\cos(\theta)\Big)^2}\, d\theta \tag{3.3}$$

Depending on the value of e, the integral on the right hand side of equation 3.3 can have three different solutions. This suggests that the solution to this equation is dependent on the type of orbit (closed, parabolic, or hyperbolic).

$$\boxed{t - t_p = \frac{h^3}{\mu^2}\int_0^{\theta} \frac{1}{\Big(1 + e\cos(\theta)\Big)^2}\, d\theta} \tag{3.4}$$

3.1 Closed Orbits: $e < 1$

Equation 3.4 is a rather tricky integral to solve, and thus use of an integral table is helpful. The CRC standard mathematical tables [1] provide a solution.

$$t - t_p = \frac{h^3}{\mu^2(e^2-1)}\left(\frac{e\cdot\sin(\theta)}{1+e\cdot\cos(\theta)} - \int_0^\theta \frac{1}{1+e\cdot\cos(\theta)}d\theta\right) \tag{3.5}$$

The new integral has two different solutions based on the numerical value of e. In the case of closed orbits, $e < 1$, once again using the CRC standard mathematical tables [1]:

$$t - t_p = \frac{h^3}{\mu^2(e^2-1)}\left(\frac{e\cdot\sin(\theta)}{1+e\cdot\cos(\theta)} - \frac{2}{\sqrt{1-e^2}}\tan^{-1}\left(\frac{\sqrt{1-e^2}}{1+e}\tan\left(\frac{\theta}{2}\right)\right)\right) \tag{3.6}$$

Simplifying this expression results in:

$$t - t_p = \frac{h^3}{\mu^2(1-e^2)^{3/2}}\left(2\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\left(\frac{\theta}{2}\right)\right) - \frac{e\sqrt{1-e^2}\sin(\theta)}{1+e\cdot\cos(\theta)}\right) \tag{3.7}$$

Equation 3.7 is a valid expression for the time required to travel from periapsis to a given true anomaly θ. However, the result is cumbersome and can be greatly simplified by introducing the eccentric anomaly E.

From the geometry of an ellipse, shown in figure 3.1, it is known that the distance from the center of the ellipse to the focus is $a\,e$. So, the horizontal distance from the center of the ellipse to the vertical line that passes through the location on the orbit can be expressed two different ways: one in terms of θ and another in terms of the eccentric anomaly angle E. Since both expressions represent the same horizontal distance, they may be equated to obtain an expression for $\cos(E)$.

$$a\cos(E) = a\,e + r\cos(\theta) = a\,e + \frac{a(1-e^2)\cos(\theta)}{1+e\cos(\theta)} \tag{3.8}$$

$$\cos(E) = e + \frac{1-e^2}{1+e\cos(\theta)} = \frac{e+e^2\cos(\theta)+\cos(\theta)-e^2\cos(\theta)}{1+e\cos(\theta)} \tag{3.9}$$

$$\cos(E) = \frac{e+\cos(\theta)}{1+e\cos(\theta)} \tag{3.10}$$

The Pythagorean trigonometric identity is then used to obtain an expression for $\sin(\theta)$.

$$\sin^2(E) = 1-\cos^2(E) = 1-\left(\frac{e+\cos(\theta)}{1+e\cos(\theta)}\right)^2 = \frac{(1+e\cos(\theta))^2-(e^2+2e\cos(\theta)+\cos^2(\theta))}{(1+e\cos(\theta))^2} \tag{3.11}$$

$$\sin^2(E) = \frac{1+e^2\cos^2(\theta)-e^2-\cos^2(\theta)}{(1+e\cos(\theta))^2} = \frac{(1-e^2)(1-\cos^2(\theta))}{(1+e\cos(\theta))^2} = \frac{(1-e^2)\sin^2(\theta)}{(1+e\cos(\theta))^2} \tag{3.12}$$

$$\sin(E) = \frac{\sqrt{1-e^2}\sin(\theta)}{1+e\cos(\theta)} \tag{3.13}$$

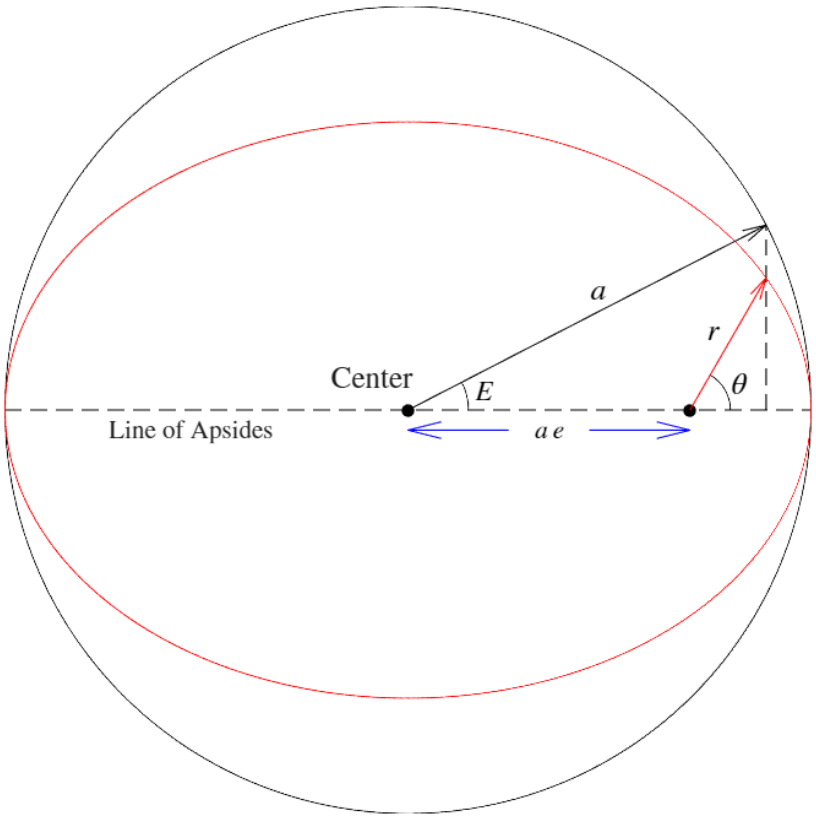


Figure 3.1: The geometric relationship between the eccentric and true anomaly angles.

Recall from equation 3.7 that there was a term that relies on a tangent half angle in θ. To get an expression for the tangent half angle in E, use equations 3.10 along with half angle trigonometric formulae.

$$\tan^2\left(\frac{E}{2}\right) = \frac{\sin^2\left(\frac{E}{2}\right)}{\cos^2\left(\frac{E}{2}\right)} = \frac{\frac{1-\cos(E)}{2}}{\frac{1+\cos(E)}{2}} = \frac{1-\cos(E)}{1+\cos(E)} \tag{3.14}$$

$$\tan^2\left(\frac{E}{2}\right) = \frac{1-\frac{e+\cos(\theta)}{1+e\cos(\theta)}}{1+\frac{e+\cos(\theta)}{1+e\cos(\theta)}} = \frac{1+e\cos(\theta)}{1+e\cos(\theta)} \cdot \frac{1+e\cos(\theta)-e-\cos(\theta)}{1+e\cos(\theta)+e+\cos(\theta)} = \frac{1-e}{1+e} \cdot \frac{1-\cos(\theta)}{1+\cos(\theta)} \tag{3.15}$$

Recognizing that equation 3.15 takes the exact form that equation 3.14 takes, but in θ:

$$\tan^2\left(\frac{E}{2}\right) = \frac{1-e}{1+e}\tan^2\left(\frac{\theta}{2}\right) \tag{3.16}$$

$$\tan\left(\frac{E}{2}\right) = \sqrt{\frac{1-e}{1+e}}\tan\left(\frac{\theta}{2}\right) \tag{3.17}$$

Isolating the eccentric anomaly:

$$E = 2\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\left(\frac{\theta}{2}\right)\right) \tag{3.18}$$

The relationship between the true anomaly and eccentric anomaly is nonlinear, as shown in Figure 3.2a. However, the two angles coincide at the apsides regardless of eccentricity.

$$E = \theta = 0^\circ \quad \text{Periapsis} \tag{3.19}$$

$$E = \theta = 180^\circ \quad \text{Apoapsis} \tag{3.20}$$

Substituting equation 3.13 and equation 3.18 into equation 3.7 expresses the time since periapsis in terms of the eccentric anomaly.

$$t - t_p = \frac{h^3}{\mu^2(1-e^2)^{3/2}}\Big(E - e\sin(E)\Big) \tag{3.21}$$

Further simplification can be made by examining the coefficient.

$$\frac{h^3}{\mu^2(1-e^2)^{3/2}} = \sqrt{\frac{h^6}{\mu^4(1-e^2)^3}} = \sqrt{\frac{1}{\mu}\left(\frac{h^2}{\mu(1-e^2)}\right)^3} \tag{3.22}$$

The term in parenthesis under the square root is the semi-major axis length per equation 2.53. Substituting equation 3.22 back into equation 3.21:

$$\boxed{t - t_p = \sqrt{\frac{a^3}{\mu}}\Big(E - e\sin(E)\Big)} \tag{3.23}$$

Example

Determine the period of a closed orbit. A complete orbit occurs when θ evolves from 0° to 360°. Both of these θ correspond to periapsis. Since the eccentric anomaly and true anomaly coincide at periapsis,:

$$E_1 = \theta_1 = 0^\circ = 0 \text{ Rad}$$
$$E_2 = \theta_2 = 360^\circ = 2\pi \text{ Rad}$$

Then use equation 3.23 to solve for the two times since periapsis.

$$t_1 - t_p = \sqrt{\frac{a^3}{\mu}}\Big(0 - e\sin(0)\Big) = 0$$
$$t_2 - t_p = \sqrt{\frac{a^3}{\mu}}\Big(2\pi - e\sin(2\pi)\Big) = 2\pi\sqrt{\frac{a^3}{\mu}}$$

The orbital period is the time elapsed between the two periapsis passages. Therefore,

$$P = t_2 - t_1 = (t_2 - t_p) - (t_1 - t_p) = 2\pi\sqrt{\frac{a^3}{\mu}} - \Big(0\Big)$$

$$\boxed{P = 2\pi\sqrt{\frac{a^3}{\mu}}} \tag{3.24}$$

A common approach in describing orbital motion is to relate the time since periapsis to the angle swept out by an object undergoing uniform circular motion. This is done by introducing two variables: the mean motion n and the mean anomaly angle M. The mean motion is defined as the constant angular velocity of a fictitious body moving in uniform circular motion with the same orbital period. The mean anomaly is the angle that this fictitious body would have swept out after the same elapsed time since periapsis.

$$t - t_p = \frac{M}{n} \tag{3.25}$$

Since the motion is assumed to be uniform, the mean anomaly increases linearly with time. The coefficient in front of equation 3.23 depends only on the orbit itself and not on the spacecraft's current location. It is therefore convenient to absorb the coefficient into the mean motion if the units allow it.

$$\sqrt{\frac{\mathrm{m}^3}{\left(\mathrm{m}^3\mathrm{kg}^{-1}\mathrm{s}^{-2}\right)\mathrm{kg}}} = \mathrm{s} \tag{3.26}$$

Since the coefficient has units of seconds, its inverse has the required units of radians per second (or equivalently s^{-1}). This motivates the definition:

$$n = \sqrt{\frac{\mu}{a^3}} \tag{3.27}$$

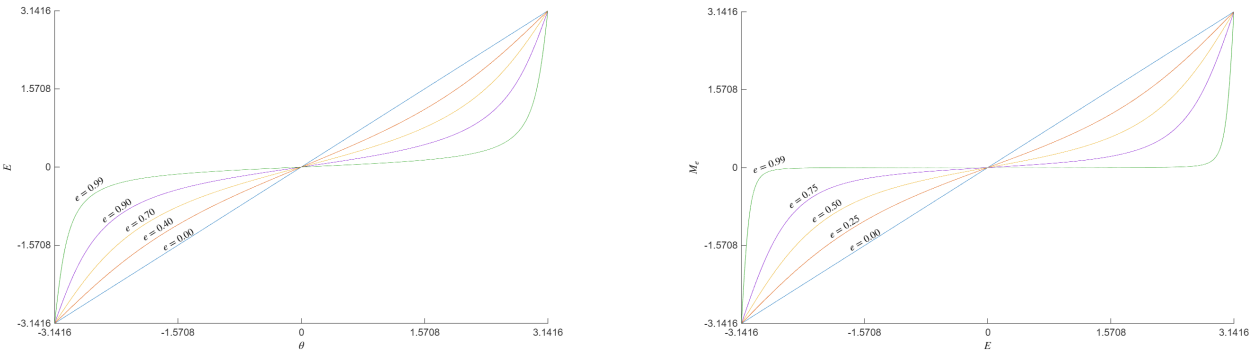
Substituting this definition of the mean motion into equation 3.23 gives:

$$t - t_p = \frac{M_e}{n} = \frac{E - e\sin(E)}{n} \tag{3.28}$$

Where M_e is the elliptical mean anomaly, which yields the relationship:

$$\boxed{M_e = E - e\sin(E)} \tag{3.29}$$

Equation 3.29 is known as the Kepler equation for elliptical orbits.



(a) Relationship between E and θ.

(b) Relationship between M and E.

Figure 3.2: Anomaly relations for elliptic orbits.

3.2 Parabolic Orbits: $e = 1$

Like in the case of closed orbits, the integral given in equation 3.4 can be looked up in the CRC standard mathematical tables [1] for the case when $e = 1$.

$$t - t_p = \frac{h^3}{\mu^2}\left(\frac{1}{2}\tan\left(\frac{\theta}{2}\right) + \frac{1}{6}\tan^3\left(\frac{\theta}{2}\right)\right) \tag{3.30}$$

The coefficient can be be simplified by closer inspection.

$$\frac{h^3}{\mu^2} = \sqrt{\frac{h^6}{\mu^4}} = \sqrt{\frac{1}{\mu}\left(\frac{h^2}{\mu}\right)^3} \tag{3.31}$$

The term in parenthesis under the square root is the semi-latus rectum per equation 2.41. Substituting equation 3.31 back into equation 3.30:

$$\boxed{t - t_p = \sqrt{\frac{p^3}{\mu}}\left(\frac{1}{2}\tan\left(\frac{\theta}{2}\right) + \frac{1}{6}\tan^3\left(\frac{\theta}{2}\right)\right)} \tag{3.32}$$

Although a parabolic orbit has no orbital period, it is still convenient to introduce a quantity analogous to the mean anomaly. Performing unit analysis on the coefficient of equation 3.32:

$$\sqrt{\frac{\mathrm{m}^3}{\left(\mathrm{m}^3\mathrm{kg}^{-1}\mathrm{s}^{-2}\right)\mathrm{kg}}} = \mathrm{s} \tag{3.33}$$

Which has units of seconds. Although a parabolic trajectory has no orbital period, it is nevertheless convenient to define a quantity n with the dimensions of angular frequency. This allows the time of flight equation to be written in a form analogous to the elliptic case. The inverse, similar to closed orbits, has the required units of s^{-1}, which motivates the definition:

$$n = \sqrt{\frac{\mu}{p^3}} \tag{3.34}$$

Substituting this definition of the mean motion into equation 3.32 gives:

$$t - t_p = \frac{M_p}{n} = \frac{1}{n}\left(\frac{1}{2}\tan\left(\frac{\theta}{2}\right) + \frac{1}{6}\tan^3\left(\frac{\theta}{2}\right)\right) \tag{3.35}$$

Where M_p is the parabolic mean anomaly, which yields the relationship:

$$\boxed{M_p = \frac{1}{2}\tan\left(\frac{\theta}{2}\right) + \frac{1}{6}\tan^3\left(\frac{\theta}{2}\right)} \tag{3.36}$$

Equation 3.36 is known as the Barker equation for parabolic orbits.

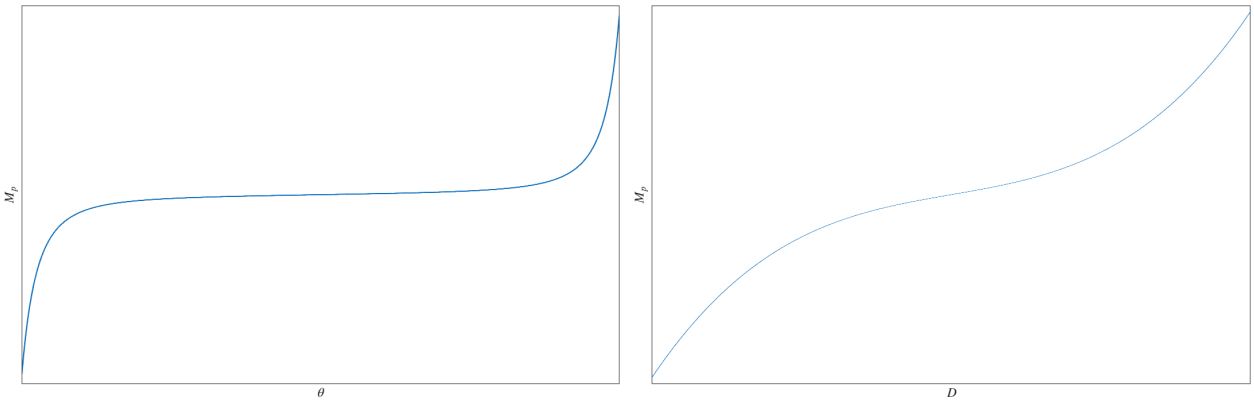
For convenience, Barker's equation is often written in terms of the auxiliary variable (Barker's Anomaly) defined as:

$$D = \tan\left(\frac{\theta}{2}\right) \tag{3.37}$$

Such that the Barker equation can be rewritten as:

$$M_p = \frac{1}{2}D + \frac{1}{6}D^3 \tag{3.38}$$

The parabolic time of flight equation represents the limiting case between closed and open trajectories. As the eccentricity of an elliptical orbit approaches unity, the elliptic time of flight equation transitions into the parabolic form, while the orbital period approaches infinity. Similarly, the parabolic trajectory represents the boundary between trajectories that remain gravitationally bound and those that are not. Although the mean motion and mean anomaly no longer represent a physical angular rate and swept angle, respectively, they provide a convenient notation for expressing the time since periapsis.



(a) Relationship between M and θ.

(b) Relationship between M and D.

Figure 3.3: Anomaly relations for parabolic orbits.

3.3 Hyperbolic Orbits: $e > 1$

Like in the previous cases, the integral given in equation 3.4 can be looked up in the CRC standard mathematical tables [1]. For the case of $e > 1$.

$$t - t_p = \frac{h^3}{\mu^2(e^2-1)}\left(\frac{e \cdot \sin(\theta)}{1 + e \cdot \cos(\theta)} - \int_0^\theta \frac{1}{1 + e \cdot \cos(\theta)} d\theta\right) \tag{3.39}$$

The new integral, as stated before, has two different solutions based on the numerical value of e. In the case of hyperbolic orbits, $e > 1$, once more using the CRC standard mathematical tables [1] taking into account the implicit absolute value:

$$t - t_p = \frac{h^3}{\mu^2(e^2-1)^{3/2}}\left(\frac{e\sqrt{e^2-1}\sin(\theta)}{1 + e \cdot \cos(\theta)} - \ln\left(\frac{\sqrt{e+1} + \sqrt{e-1}\tan\left(\frac{\theta}{2}\right)}{\sqrt{e+1} - \sqrt{e-1}\tan\left(\frac{\theta}{2}\right)}\right)\right) \tag{3.40}$$

The expression provided by equation 3.40 is a valid expression for the time since periapsis. However, like the closed orbit case, the result in cumbersome and can be simplified. This simplification in achieved by introducing another auxiliary parameter called the hyperbolic eccentric anomaly F. The hyperbolic eccentric anomaly is a fictitious angle such that:

$$\cosh(F) = \frac{x}{|a|} \quad , \quad \sinh(F) = \frac{y}{b} \tag{3.41}$$

which is consistent with the equation of a hyperbola.

$$\cosh^2(F) - \sinh^2(F) = \frac{x^2}{|a|^2} - \frac{y^2}{b^2} = 1 \tag{3.42}$$

Note that x and y are the horizontal and vertical projections of the relative position vector, at the current true anomaly, with respect to the conic section center (see figure 3.4).

In figure 3.4, the shaded area is related to F via the relation: $F = \frac{2 \cdot \text{Area}}{|a|^2}$ where the area is defined as the area between the line of apsides, the equilateral hyperbola ($e = \sqrt{2}$), and the line from periapsis to the center. This area can be tricky to determine in general, so this definition will only be mentioned in passing.

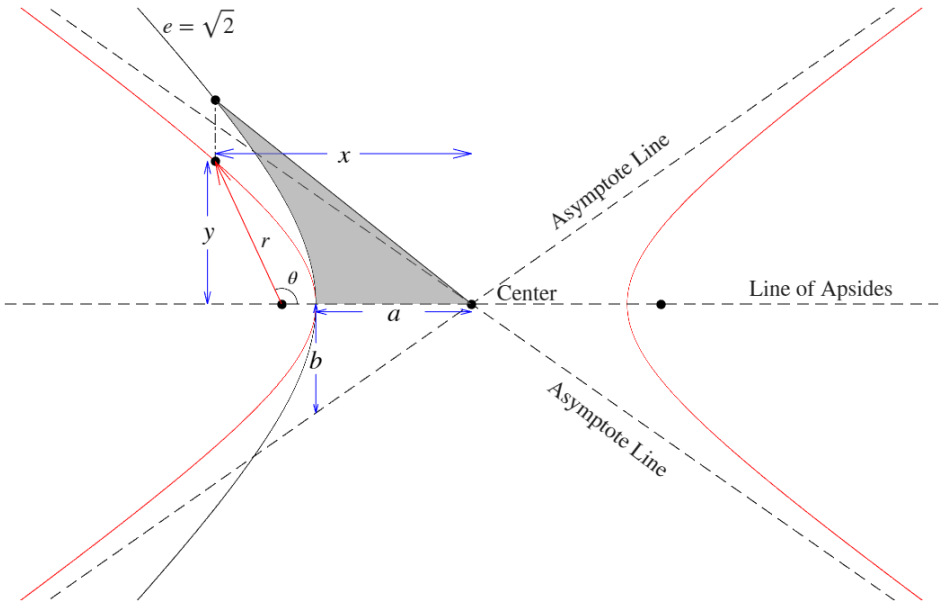


Figure 3.4: Geometric parameters relating to the hyperbolic eccentric anomaly.

From the conic section relationship, the semi-major axis associated with the polar form of a hyperbolic orbit is negative. This signed definition is commonly used because it allows the same orbital relationships to apply to any orbit. However, in the cartesian equation of a hyperbola, the semi-major axis length is defined as a positive geometric length. To avoid confusion between these two conventions, absolute values will be used when referring to geometric lengths.

Consider the hyperbolic sine of F.

$$\sinh(F) = \frac{y}{b} = \frac{r\sin(\theta)}{|a|\sqrt{e^2-1}} = \frac{|a|(e^2-1)}{1+e\cos(\theta)}\frac{\sin(\theta)}{|a|\sqrt{e^2-1}} \tag{3.43}$$

$$\sinh(F) = \frac{\sqrt{e^2-1}\sin(\theta)}{1+e\cos(\theta)} \tag{3.44}$$

Solve for F by taking the inverse hyperbolic sine of equation 3.44.

$$F = \sinh^{-1}\left(\frac{\sqrt{e^2-1}\sin(\theta)}{1+e\cos(\theta)}\right) = \ln\left(\frac{\sqrt{e^2-1}\sin(\theta)}{1+e\cos(\theta)} + \sqrt{\frac{(e^2-1)\sin^2(\theta)}{(1+e\cos(\theta))^2}+1}\right) \tag{3.45}$$

$$F = \ln\left(\frac{\sqrt{e^2-1}\sin(\theta) + \sqrt{(e^2-1)\sin^2(\theta)+(1+e\cos(\theta))^2}}{1+e\cos(\theta)}\right) \tag{3.46}$$

$$F = \ln\left(\frac{\sqrt{e^2-1}\sin(\theta) + \sqrt{e^2\sin^2(\theta)-\sin^2(\theta)+1+2e\cos(\theta)+e^2\cos^2(\theta)}}{1+e\cos(\theta)}\right) \tag{3.47}$$

$$F = \ln\left(\frac{\sqrt{e^2-1}\sin(\theta) + \sqrt{e^2(\sin^2(\theta)+\cos^2(\theta))+(1-\sin^2(\theta))+2e\cos(\theta)}}{1+e\cos(\theta)}\right) \tag{3.48}$$

$$F = \ln\left(\frac{\sqrt{e^2-1}\sin(\theta) + \sqrt{e^2+2e\cos(\theta)+\cos^2(\theta)}}{1+e\cos(\theta)}\right) \tag{3.49}$$

$$F = \ln\left(\frac{\sqrt{e^2-1}\sin(\theta) + \sqrt{(e+\cos(\theta))^2}}{1+e\cos(\theta)}\right) \tag{3.50}$$

The form that equation 3.40 takes relies on terms relating to $\tan\left(\frac{\theta}{2}\right)$. Substitute in tangent half angle formulae for $\sin(\theta)$ and $\cos(\theta)$ to relate them to the tangent half angle.

$$\sin(\theta) = \frac{2\tan\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)} \quad , \quad \cos(\theta) = \frac{1-\tan^2\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)} \tag{3.51}$$

Plug $\sin(\theta)$ and $\cos(\theta)$ into equation 3.50.

$$F = \ln\left(\frac{\sqrt{e^2-1}\frac{2\tan\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)} + e + \frac{1-\tan^2\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)}}{1+e\frac{1-\tan^2\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)}}\right) \tag{3.52}$$

$$F = \ln\left(\frac{2\sqrt{e^2-1}\tan\left(\frac{\theta}{2}\right) + e + e\tan^2\left(\frac{\theta}{2}\right) + 1 - \tan^2\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right) + e - e\tan^2\left(\frac{\theta}{2}\right)}\right) \tag{3.53}$$

$$F = \ln\left(\frac{(e+1) + 2\sqrt{e^2-1}\tan\left(\frac{\theta}{2}\right) + (e-1)\tan^2\left(\frac{\theta}{2}\right)}{(e+1) - (e-1)\tan^2\left(\frac{\theta}{2}\right)}\right) \tag{3.54}$$

$$F = \ln\left(\frac{(\sqrt{e+1})^2 + 2\sqrt{(e+1)(e-1)}\tan\left(\frac{\theta}{2}\right) + (\sqrt{e-1})^2\tan^2\left(\frac{\theta}{2}\right)}{(\sqrt{e+1})^2 - (\sqrt{e-1})^2\tan^2\left(\frac{\theta}{2}\right)}\right) \tag{3.55}$$

$$F = \ln\left(\frac{\left(\sqrt{e+1}+\sqrt{e-1}\tan\left(\frac{\theta}{2}\right)\right)^2}{\left(\sqrt{e+1}+\sqrt{e-1}\tan\left(\frac{\theta}{2}\right)\right)\left(\sqrt{e+1}-\sqrt{e-1}\tan\left(\frac{\theta}{2}\right)\right)}\right) \tag{3.56}$$

$$F = \ln\left(\frac{\sqrt{e+1}+\sqrt{e-1}\tan\left(\frac{\theta}{2}\right)}{\sqrt{e+1}-\sqrt{e-1}\tan\left(\frac{\theta}{2}\right)}\right) \tag{3.57}$$

While equation 3.57 is enough to relate θ to F, it too is somewhat inconvenient, and a simpler relation can be determined. Taking a closer look at equation 3.41 for $\cosh(F)$:

$$\cosh(F) = \frac{x}{|a|} = \frac{|a|\,e - r\cos(\theta)}{|a|} = \left(|a|\,e - \frac{|a|(e^2-1)\cos(\theta)}{1+e\cos(\theta)}\right)\frac{1}{|a|} \tag{3.58}$$

$$\cosh(F) = \frac{e(1+e\cos(\theta)) - (e^2-1)\cos(\theta)}{1+e\cos(\theta)} = \frac{e+e^2\cos(\theta) - e^2\cos(\theta) + \cos(\theta)}{1+e\cos(\theta)} \tag{3.59}$$

$$\cosh(F) = \frac{e+\cos(\theta)}{1+e\cos(\theta)} \tag{3.60}$$

Equations 3.44 and 3.60 can be used to define the hyperbolic tangent half angle in F. Using the half angle trigonometric identity for hyperbolic tangent.

$$\tanh\left(\frac{F}{2}\right) = \frac{\sinh(F)}{1+\cosh(F)} = \frac{\frac{\sqrt{e^2-1}\sin(\theta)}{1+e\cos(\theta)}}{\frac{1+e\cos(\theta)}{1+e\cos(\theta)} + \frac{e+\cos(\theta)}{1+e\cos(\theta)}} \tag{3.61}$$

$$\tanh\left(\frac{F}{2}\right) = \frac{\sqrt{e^2-1}\sin(\theta)}{(e+1)(1+\cos(\theta))} = \frac{\sqrt{(e-1)(e+1)}}{\sqrt{(e+1)(e+1)}}\frac{\sin(\theta)}{1+\cos(\theta)} \tag{3.62}$$

Substituting in the tangent half angle trigonometric identity and simplifying the coefficient:

$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin(\theta)}{1+\cos(\theta)} \tag{3.63}$$

$$\tanh\left(\frac{F}{2}\right) = \sqrt{\frac{e-1}{e+1}}\tan\left(\frac{\theta}{2}\right) \tag{3.64}$$

Which is a far simpler relation between F and θ.

Substituting equations 3.44 and 3.57 into equation 3.40 results in a new expression for hyperbolic time of flight.

$$t - t_p = \frac{h^3}{\mu(e^2-1)^{3/2}}\Big(e\sinh(F) - F\Big) \tag{3.65}$$

Taking a closer look at the constant:

$$\frac{h^3}{\mu^2(e^2-1)^{3/2}} = \sqrt{\frac{h^6}{\mu^4(e^2-1)^3}} = \sqrt{\frac{1}{\mu}\left(-\frac{h^2}{\mu(1-e^2)}\right)^3} \tag{3.66}$$

The term in parenthesis under the square root is the negative of the semi-major axis length given by equation 2.53. Substituting equation 3.66 back into equation 3.65:

$$\boxed{t - t_p = \sqrt{\frac{-a^3}{\mu}}\Big(e\sinh(F) - F\Big)} \tag{3.67}$$

Although a hyperbolic orbit (similar to the parabolic orbit) has no orbital period, it remains convenient to introduce a quantity analogous to the mean anomaly. Performing unit analysis on the coefficient of equation 3.67:

$$\sqrt{\frac{\mathrm{m}^3}{\left(\mathrm{m}^3\mathrm{kg}^{-1}\mathrm{s}^{-2}\right)\mathrm{kg}}} = \mathrm{s} \tag{3.68}$$

Which has units of seconds. This allows the time of flight equation to be written in a form analogous to the elliptic and parabolic case. The inverse of the coefficient has the required units of s^{-1}, which motivates a hyperbolic mean motion:

$$n = \sqrt{\frac{\mu}{-a^3}} \tag{3.69}$$

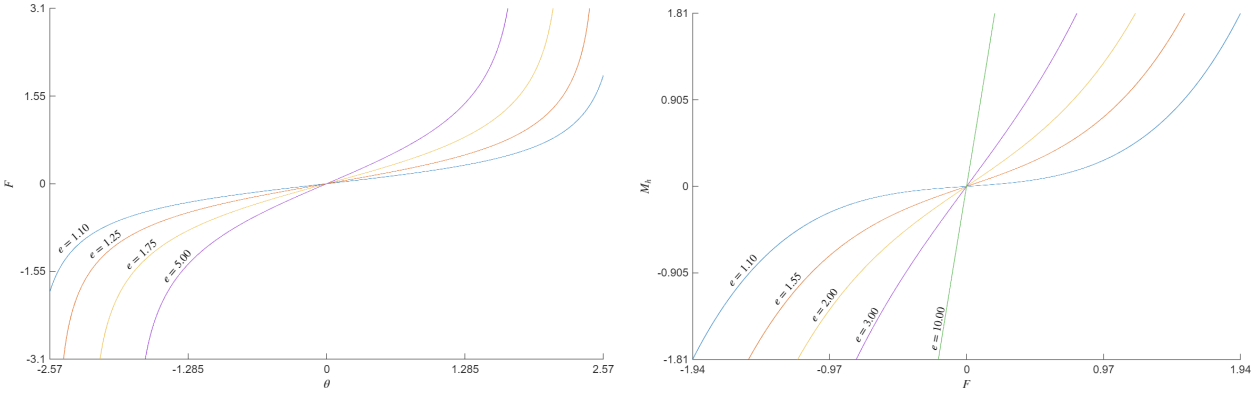
Substituting this definition of the hyperbolic mean motion into equation 3.67 gives:

$$t - t_p = \frac{M_h}{n} = \frac{1}{n}\Big(e\sinh(F) - F\Big) \tag{3.70}$$

Where M_h is the hyperbolic mean anomaly, which yields the relationship:

$$\boxed{M_h = e\sinh(F) - F} \tag{3.71}$$

Equation 3.71 is known as the Kepler equation for hyperbolic orbits.



(a) Relationship between F and θ.

(b) Relationship between M_h and F.

Figure 3.5: Anomaly relations for hyperbolic orbits.

3.4 Universal Variables Formulation

The universal variables formulation begins by introducing a change of variables to the relative position equation of motion. The changed variable is, itself, the universal variable:

$$\frac{d\chi}{dt} = \frac{\sqrt{\mu}}{r} \tag{3.72}$$

where χ is the Battin universal anomaly. Multiplying each side by $d/d\chi$:

$$\frac{d}{dt} = \frac{\sqrt{\mu}}{r}\frac{d}{d\chi} \tag{3.73}$$

which is how the general time derivative relates to the χ derivative. Consider now the conservation of mechanical energy.

$$\varepsilon = \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{\mu}{r} \tag{3.74}$$

Rearranging and substituting equation 1.85 yields the following for $\dot{r}$:

$$\dot{r}^2 = 2\varepsilon - \frac{h^2}{r^2} + \frac{2\mu}{r} \tag{3.75}$$

Multiplying through by r^2:

$$r^2\dot{r}^2 = 2\varepsilon r^2 - h^2 + 2\mu r \tag{3.76}$$

Since $\dot{r}$ is the time derivative of the relative position, it can be rewritten in terms of χ via equation 3.73:

$$\dot{r}^2 = \left(\frac{\sqrt{\mu}}{r}\frac{d}{d\chi}\left(r\right)\right)^2 = \frac{\mu}{r^2}\left(r'\right)^2 \tag{3.77}$$

Where r' is the χ derivative of the relative position. Substituting into equation 3.76:

$$\mu(r')^2 = 2\varepsilon r^2 - h^2 + 2\mu r \tag{3.78}$$

Differentiating with respect to χ:

$$2\mu r'\, r'' = 4\varepsilon r\, r' + 2\mu r' \tag{3.79}$$

Dividing by $2\mu r'$:

$$r'' = \frac{2\varepsilon}{\mu}r + 1 \tag{3.80}$$

Defining a quantity $\alpha \equiv -\frac{2\varepsilon}{\mu} = \frac{1}{a}$ (by equation 2.58):

$$r'' = 1 - \alpha\, r \tag{3.81}$$

Taking a χ derivative and rearranging:

$$\boxed{r''' + \alpha r' = 0} \tag{3.82}$$

Equation 3.82 is a third order, linear, homogeneous, differential equation. The solution to this equation involves expressing r as a power series:

$$r = \sum_{i=0}^{\infty} c_i \chi^i \tag{3.83}$$

Determining the χ derivatives of the power series:

$$r' = \sum_{i=1}^{\infty} i c_i \chi^{i-1} \tag{3.84}$$

$$r'' = \sum_{i=2}^{\infty} i(i-1) c_i \chi^{i-2} \tag{3.85}$$

$$r''' = \sum_{i=3}^{\infty} i(i-1)(i-2) c_i \chi^{i-3} \tag{3.86}$$

Substituting the χ derivatives into equation 3.82:

$$\sum_{i=3}^{\infty} i(i-1)(i-2) c_i \chi^{i-3} + \alpha \sum_{i=1}^{\infty} i c_i \chi^{i-1} = 0 \tag{3.87}$$

Switching indices to $i = j+3$ on the left summation and $i = j+1$ on the right summation in order to express each summation from $j = 0$ to ∞:

$$\sum_{j=0}^{\infty} (j+3)(j+2)(j+1) c_{j+3} \chi^j + \alpha \sum_{j=0}^{\infty} (j+1) c_{j+1} \chi^j = 0 \tag{3.88}$$

Rewriting:

$$\sum_{j=0}^{\infty} \Big((j+3)(j+2)(j+1) c_{j+3} + \alpha (j+1) c_{j+1} \Big) \chi^j = 0 \tag{3.89}$$

Since the power series is identically zero for every value of χ, each coefficient of χ^j must vanish independently. Therefore:

$$(j+3)(j+2)(j+1) c_{j+3} + \alpha (j+1) c_{j+1} = 0 \tag{3.90}$$

Simplifying:

$$c_{j+3} = -\frac{\alpha}{(j+3)(j+2)} c_{j+1} \tag{3.91}$$

Solving for the constants:

$$j = 0: \quad c_{0+3} = \frac{-\alpha}{(0+3)(0+2)} c_{0+1} \quad \rightarrow \quad c_3 = \frac{-\alpha}{(3)(2)(1)} c_1 \tag{3.92}$$

$$j = 1: \quad c_{1+3} = \frac{-\alpha}{(1+3)(1+2)} c_{1+1} \quad \rightarrow \quad c_4 = \frac{-\alpha}{(4)(3)} c_2 \tag{3.93}$$

$$j = 2: \quad c_{2+3} = \frac{-\alpha}{(2+3)(2+2)} c_{2+1} \quad \rightarrow \quad c_5 = \frac{-\alpha}{(5)(4)} c_3 = \frac{(-\alpha)^2}{(5)(4)(3)(2)(1)} c_1 \tag{3.94}$$

$$j = 3: \quad c_{3+3} = \frac{-\alpha}{(3+3)(3+2)} c_{3+1} \quad \rightarrow \quad c_6 = \frac{-\alpha}{(6)(5)} c_4 = \frac{(-\alpha)^2}{(6)(5)(4)(3)} c_2 \tag{3.95}$$

$$j=4: \quad c_{4+3}=\frac{-\alpha}{(4+3)(4+2)}c_{4+1} \quad \rightarrow \quad c_7=\frac{-\alpha}{(7)(6)}c_5=\frac{(-\alpha)^3}{(7)(6)(5)(4)(3)(2)(1)}c_1 \tag{3.96}$$

$$j=5: \quad c_{5+3}=\frac{-\alpha}{(5+3)(5+2)}c_{5+1} \quad \rightarrow \quad c_8=\frac{-\alpha}{(6)(5)}c_6=\frac{(-\alpha)^3}{(8)(7)(6)(5)(4)(3)}c_2 \tag{3.97}$$

The pattern is evident. There are two scalar coefficients being formed by the recursive relation: one for even indices and another for odd indices. They are constructed, and predicted from above, as the following:

$$c_{2k+1}=\frac{(-\alpha)^k}{(2k+1)!}c_1 \qquad \& \qquad c_{2k}=\frac{2(-\alpha)^k}{(2k+2)!}c_2 \tag{3.98}$$

The recurrence relation generates two sequences of coefficients: an odd-indexed sequence beginning with c_1, and an even-indexed sequence beginning with c_2. It is important to note, however, that the even-indexed sequence begins at c_2, not c_0. The coefficient c_0 is never generated by the recursive relation because no value of $j \geq 0$ causes the recurrence to produce c_0. Therefore, c_0 remains an independent coefficient that must be included separately in the power series.

Therefore, the series solution to equation 3.82 is written as:

$$r=\sum_{i=0}^{\infty} c_i\,\chi^i = c_0\chi^0+\sum_{k=0}^{\infty}\frac{(-\alpha)^k}{(2k+1)!}c_1\chi^{2k+1}+\sum_{k=0}^{\infty}\frac{2(-\alpha)^k}{(2k+2)!}c_2\chi^{2k+2} \tag{3.99}$$

Or, more simply:

$$\boxed{r=c_0+c_1\chi\left(\sum_{k=0}^{\infty}\frac{(-\alpha\chi^2)^k}{(2k+1)!}\right)+2c_2\chi^2\left(\sum_{k=0}^{\infty}\frac{(-\alpha\chi^2)^k}{(2k+2)!}\right)} \tag{3.100}$$

Since this comes from a third order differential equation, three initial conditions are required to determine the three coefficients. The simplest initial conditions to utilize are the three χ derivatives of the relative scalar position. From equation 3.73:

$$\dot{r}=\dot{r}_0=\frac{d}{dt}(r_0)=\frac{\sqrt{\mu}}{r_0}\frac{d}{d\chi}(r_0)=\frac{\sqrt{\mu}}{r_0}r_0' \tag{3.101}$$

$$r'=\frac{r_0\dot{r}_0}{\sqrt{\mu}} \tag{3.102}$$

and from equation 3.81: $r''=1-\alpha r$

Thus the three initial conditions that must be applied are:

$$r=r_0 \qquad \& \qquad r'=\frac{r_0\dot{r}_0}{\sqrt{\mu}} \qquad \& \qquad r''=1-\alpha r_0 \tag{3.103}$$

By the definition of universal anomaly (equation 3.73), at the initial time t_0, χ_0 can be given any value. This is because χ is defined by how much time has passed since t_0.

$$\Delta\chi=\chi-\chi_0=\int_{t_0}^{t}\frac{\sqrt{\mu}}{r}dt \tag{3.104}$$

The physically relevant quantity is the change in universal anomaly over the time interval, rather than the initial value assigned to χ_0. For this reason, the origin of the universal anomaly is chosen such that $\chi_0 = 0$.

The first initial condition considers only the relative position. For convenience, absorb the extra factor of χ into the summation:

$$r = c_0 + c_1\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k+1}}{(2k+1)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k+2}}{(2k+2)!}\right) \tag{3.105}$$

Apply the initial condition $\chi = \chi_0$:

$$r_0 = c_0 + c_1\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi_0)^{2k+1}}{(2k+1)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi_0)^{2k+2}}{(2k+2)!}\right) \tag{3.106}$$

Expanding:

$$r_0 = c_0 + c_1\left(\frac{\chi_0^1}{1!} + \frac{\chi_0^3}{3!} + \frac{\chi_0^5}{5!} + \dots\right) + 2c_2\left(\frac{\chi_0^2}{2!} + \frac{\chi_0^4}{4!} + \frac{\chi_0^6}{6!} + \dots\right) \tag{3.107}$$

$$\boxed{r_0 = c_0} \tag{3.108}$$

The next initial condition requires the first χ derivative of the relative position:

$$r' = \frac{d}{d\chi}(r) = \frac{d}{d\chi}\left(c_0 + c_1\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k+1}}{(2k+1)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k+2}}{(2k+2)!}\right)\right) \tag{3.109}$$

$$r' = c_1\left(\sum_{k=0}^{\infty}\frac{(2k+1)(-\alpha)^k\chi^{2k}}{(2k+1)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(2k+2)(-\alpha)^k\chi^{2k+1}}{(2k+2)!}\right) \tag{3.110}$$

$$r' = c_1\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k}}{(2k)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k+1}}{(2k+1)!}\right) \tag{3.111}$$

Apply the initial condition $\chi = \chi_0$:

$$r' = c_1\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi_0)^{2k}}{(2k)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi_0)^{2k+1}}{(2k+1)!}\right) \tag{3.112}$$

Expanding:

$$r' = c_1\left(\frac{\chi_0^0}{0!} + \frac{\chi_0^2}{2!} + \frac{\chi_0^4}{4!} + \dots\right) + 2c_2\left(\frac{\chi_0}{1!} + \frac{\chi_0^3}{3!} + \frac{\chi_0^5}{5!} + \dots\right) \tag{3.113}$$

$\chi^0 = 1$ in general and $0! = 1$ due to the empty product convention. Therefore:

$$r' = c_1 \tag{3.114}$$

$$\boxed{c_1 = \frac{r_0\dot{r}_0}{\sqrt{\mu}}} \tag{3.115}$$

Finally, the last initial condition where the second χ derivative of the relative position is required:

$$r'' = \frac{d}{d\chi}(r') = \frac{d}{d\chi}\left(c_1\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k}}{(2k)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k+1}}{(2k+1)!}\right)\right) \tag{3.116}$$

$$r'' = c_1\left(\sum_{k=0}^{\infty}\frac{(2k)(-\alpha)^k\chi^{2k-1}}{(2k)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(2k+1)(-\alpha)^k\chi^{2k}}{(2k+1)!}\right) \tag{3.117}$$

$$r'' = c_1\left(\sum_{k=1}^{\infty}\frac{(-\alpha)^k\chi^{2k-1}}{(2k-1)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k\chi^{2k}}{(2k)!}\right) \tag{3.118}$$

Apply the initial condition $\chi = \chi_0$:

$$r_0'' = c_1\left(\sum_{k=1}^{\infty}\frac{(-\alpha)^k(\chi_0)^{2k-1}}{(2k-1)!}\right) + 2c_2\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi_0)^{2k}}{(2k)!}\right) \tag{3.119}$$

Expanding:

$$r_0'' = c_1\left(\frac{\chi_0}{1!} + \frac{\chi_0^3}{3!} + \frac{\chi_0^5}{5!} + \dots\right) + 2c_2\left(\frac{\chi_0^0}{0!} + \frac{\chi_0^2}{2!} + \frac{\chi_0^3}{6!} + \dots\right) \tag{3.120}$$

Once again, in general $\chi^0 = 1$ and $0! = 1$ due to the empty product convention. Therefore:

$$r_0'' = 2c_2 \tag{3.121}$$

$$\boxed{2c_2 = 1 - \alpha r_0} \tag{3.122}$$

All three of the coefficients to equation 3.100 have been determined from the initial conditions. With these coefficients collected, the relative position can be expressed as:

$$r = r_0 + \chi\frac{r_0\dot{r}_0}{\sqrt{\mu}}\left(\sum_{k=0}^{\infty}\frac{\left(-\alpha\chi^2\right)^k}{(2k+1)!}\right) + \chi^2\left(1 - \alpha r_0\right)\left(\sum_{k=0}^{\infty}\frac{\left(-\alpha\chi^2\right)^k}{(2k+2)!}\right) \tag{3.123}$$

Now that the relative position has been determined as a function of χ, the question can now be posed as to how time should be expressed. Rearranging the definition of the universal variable provides the starting point of the determination of time.

$$\sqrt{\mu}\,dt = r\,d\chi \tag{3.124}$$

Integrating from the initial time to the final time:

$$\int_{t_0}^{t}\sqrt{\mu}\,dt = \int_{\chi_0}^{\chi} r\,d\chi \tag{3.125}$$

The left hand side integral is trivial since its argument is constant.

$$\int_{t_0}^{t}\sqrt{\mu}\,dt = \sqrt{\mu}(t - t_0) \tag{3.126}$$

The right hand side integral can be carried out now that an expression for the relative position has been determined that is explicit in χ.

$$\int_{\chi_0}^{\chi} r\, d\chi = \int_{\chi_0}^{\chi} \left(r_0 + \chi \frac{r_0 \dot{r}_0}{\sqrt{\mu}} \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+1)!} \right) + \chi^2 \left(1 - \alpha r_0\right) \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+2)!} \right) \right) d\chi \quad (3.127)$$

Once again for convenience, absorb the factors of χ into the summations:

$$\int_{\chi_0}^{\chi} r\, d\chi = \int_{\chi_0}^{\chi} \left(r_0 + \frac{r_0 \dot{r}_0}{\sqrt{\mu}} \left(\sum_{k=0}^{\infty} \frac{(-\alpha)^k (\chi)^{2k+1}}{(2k+1)!} \right) + \left(1 - \alpha r_0\right) \left(\sum_{k=0}^{\infty} \frac{(-\alpha)^k (\chi)^{2k+2}}{(2k+2)!} \right) \right) d\chi \quad (3.128)$$

Carrying out the integration and simplifying:

$$\int_{\chi_0}^{\chi} r\, d\chi = r_0\chi + \frac{r_0 \dot{r}_0}{\sqrt{\mu}} \left(\sum_{k=0}^{\infty} \frac{(-\alpha)^k (\chi)^{2k+2}}{(2k+2)(2k+1)!} \right) + \left(1 - \alpha r_0\right) \left(\sum_{k=0}^{\infty} \frac{(-\alpha)^k (\chi)^{2k+3}}{(2k+3)(2k+2)!} \right) \Bigg|_{\chi_0}^{\chi} \quad (3.129)$$

$$\int_{\chi_0}^{\chi} r\, d\chi = r_0\chi + \chi^2 \frac{r_0 \dot{r}_0}{\sqrt{\mu}} \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+2)!} \right) + \chi^3 \left(1 - \alpha r_0\right) \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+3)!} \right) \Bigg|_{\chi_0}^{\chi} \quad (3.130)$$

Since every term has a factor of χ, every term vanishes if $\chi = \chi_0$. Therefore:

$$\int_{\chi_0}^{\chi} r\, d\chi = r_0\chi + \chi^2 \frac{r_0 \dot{r}_0}{\sqrt{\mu}} \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+2)!} \right) + \chi^3 \left(1 - \alpha r_0\right) \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+3)!} \right) \quad (3.131)$$

Thus the time relationship given by the universal variables formulation is:

$$\sqrt{\mu}(t - t_0) = r_0\chi + \chi^2 \frac{r_0 \dot{r}_0}{\sqrt{\mu}} \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+2)!} \right) + \chi^3 \left(1 - \alpha r_0\right) \left(\sum_{k=0}^{\infty} \frac{\left(-\alpha\chi^2\right)^k}{(2k+3)!} \right) \quad (3.132)$$

Stumpff Functions

The Stumpff functions were discovered in 1947 by the German astronomer Karl Stumpff while he was researching celestial mechanics. The Stumpff functions are defined as:

$$C_n(z) = \sum_{k=0}^{\infty} (-1)^k \frac{z^k}{(2k+n)!} \quad (3.133)$$

These are defined for any $n = 0, 1, 2, ..., \infty$. Several of the Stumpff functions are quite useful to the discussion of universal variables. No doubt equation 3.133 already appears familiar. Stumpff functions also have a very important recurrence relation [2], which is helpful if one Stumpff function is preferred over another:

$$C_n(z) = \frac{1}{n!} - zC_{n+2}(z) \quad (3.134)$$

Even more incredible is that the Stumpff functions $0 \leq n \leq 3$ are related to elementary

functions in the different regimes of z [3].

$$C_0(z) = \begin{cases} \cos(\sqrt{z}), & z > 0, \\ 1, & z = 0, \\ \cosh(\sqrt{-z}), & z < 0, \end{cases} \qquad C_1(z) = \begin{cases} \dfrac{\sin(\sqrt{z})}{\sqrt{z}}, & z > 0, \\ 1, & z = 0, \\ \dfrac{\sinh(\sqrt{-z})}{\sqrt{-z}}, & z < 0, \end{cases}$$

$$C_2(z) = \begin{cases} \dfrac{1 - \cos(\sqrt{z})}{z}, & z > 0, \\ \dfrac{1}{2}, & z = 0, \\ \dfrac{\cosh(\sqrt{-z}) - 1}{-z}, & z < 0, \end{cases} \qquad C_3(z) = \begin{cases} \dfrac{\sqrt{z} - \sin(\sqrt{z})}{z^{3/2}}, & z > 0, \\ \dfrac{1}{6}, & z = 0, \\ \dfrac{\sinh(\sqrt{-z}) - \sqrt{-z}}{(-z)^{3/2}}, & z < 0. \end{cases} \tag{3.135}$$

The Stumpff functions behave as depicted in the following figures.

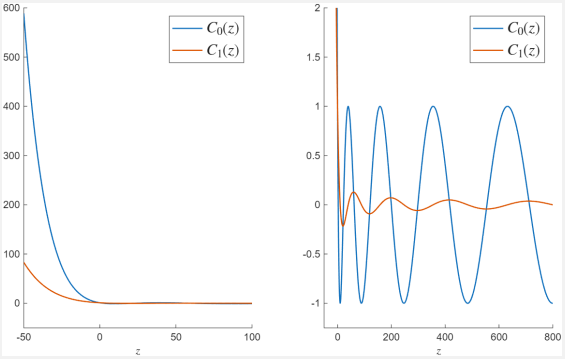


Figure 3.6: $n = 0$ and $n = 1$.

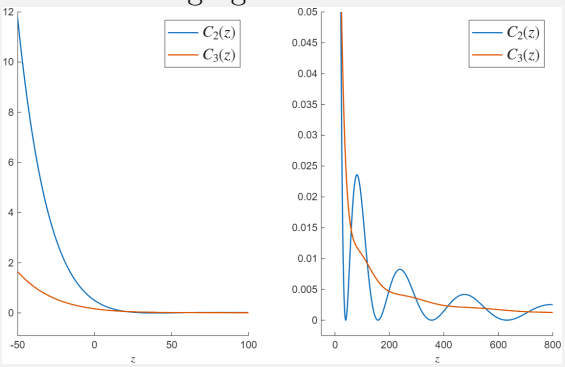


Figure 3.7: $n = 2$ and $n = 3$.

To connect the results derived in this section to other reference texts, the Stumpff functions (with $z = \alpha\chi^2$) are written in place of the infinite sums. The relative position is therefore rewritten using the Stumpff function notation:

$$\boxed{r = r_0 + \chi \frac{r_0 \dot{r}_0}{\sqrt{\mu}} C_1(\alpha\chi^2) + \chi^2 (1 - \alpha r_0) C_2(\alpha\chi^2)} \tag{3.136}$$

The time relation is similarly written as:

$$\boxed{\sqrt{\mu}(t - t_0) = r_0\chi + \chi^2 \frac{r_0 \dot{r}_0}{\sqrt{\mu}} C_2(\alpha\chi^2) + \chi^3 (1 - \alpha r_0) C_3(\alpha\chi^2)} \tag{3.137}$$

Thus, the universal variable formulation may be expressed entirely in terms of the Stumpff functions, which provide a compact representation of the power series solution.

3.5 The Lagrange Coefficients: Δt

Consider the vector equation of motion given by equation 1.86.

$$\underline{\ddot{r}} + \frac{\mu}{r^3}\underline{r} = \underline{0} \tag{3.138}$$

Changing the second order derivative to be in respect to χ:

$$\underline{\dot{r}} = \frac{d}{dt}(\underline{r}) = \frac{\sqrt{\mu}}{r}\frac{d}{d\chi}(\underline{r}) = \frac{\sqrt{\mu}}{r}\underline{r}' \tag{3.139}$$

Where $\underline{r}'$ is the χ derivative of $\underline{r}$. Differentiating once more:

$$\frac{d}{dt}(\underline{\dot{r}}) = \frac{\sqrt{\mu}}{r}\frac{d}{d\chi}\left(\frac{\sqrt{\mu}}{r}\underline{r}'\right) = \frac{\mu}{r}\frac{d}{d\chi}\left(\frac{1}{r}\underline{r}'\right) \tag{3.140}$$

$$\underline{\ddot{r}} = \frac{\mu}{r}\left(-\frac{r'}{r^2}\underline{r}' + \frac{1}{r}\underline{r}''\right) \tag{3.141}$$

Substituting back into equation 3.138:

$$\frac{\mu}{r^2}\underline{r}'' - \frac{\mu r'}{r^3}\underline{r}' + \frac{\mu}{r^3}\underline{r} = \underline{0} \tag{3.142}$$

Which simplifies to:

$$r\,\underline{r}'' - r'\,\underline{r}' + \underline{r} = \underline{0} \tag{3.143}$$

Differentiating once with respect to χ:

$$r'\,\underline{r}'' + r\,\underline{r}''' - r''\,\underline{r}' - r'\,\underline{r}'' + \underline{r}' = \underline{0} \tag{3.144}$$

Simplifying:

$$r\,\underline{r}''' + (1 - r'')\,\underline{r}' = \underline{0} \tag{3.145}$$

Substituting equation 3.81 into equation 3.145 for the scalar r'':

$$r\,\underline{r}''' + (1 - 1 + \alpha\, r)\,\underline{r}' = \underline{0} \tag{3.146}$$

$$r\,\underline{r}''' + \alpha\, r\,\underline{r}' = \underline{0} \tag{3.147}$$

Dividing by r:

$$\boxed{\underline{r}''' + \alpha\,\underline{r}' = \underline{0}} \tag{3.148}$$

Notice that the resulting homogeneous vector differential equation has the **exact** same form as the homogeneous scalar differential equation used to solve for the relative position magnitude (equation 3.82). As such, the general solution has the exact same form, but with one notable difference. The constants are no longer scalar coefficients, but instead coefficient vectors:

$$\underline{r} = \sum_{i=0}^{\infty}\underline{c}_i\chi^i = \underline{c}_0 + \underline{c}_1\chi\left(\sum_{k=0}^{\infty}\frac{(-\alpha\chi^2)^k}{(2k+1)!}\right) + 2\underline{c}_2\chi^2\left(\sum_{k=0}^{\infty}\frac{(-\alpha\chi^2)^k}{(2k+2)!}\right) \tag{3.149}$$

The relative position vector will not have the same initial conditions as the relative position magnitude solution. For this reason, additional care will be required to determine the constant vectors for this solution.

The coefficient vectors are determined by evaluating the initial relative position vector and the first two χ derivatives of the relative position vector. Obviously $\underline{r} = \underline{r}_0$ initially. The first χ derivative of the relative position vector as determined by equation 3.139 is:

$$\underline{r}' = \frac{r}{\sqrt{\mu}}\,\underline{\dot{r}} \tag{3.150}$$

By equation 3.143, the second χ derivative of the relative position vector is:

$$\underline{r}'' = \frac{r'}{r}\,\underline{r}' - \frac{1}{r}\,\underline{r} \tag{3.151}$$

Substituting equation 3.102 for the scalar r' and equation 3.150 for the vector $\underline{r}'$:

$$\underline{r}'' = \frac{\dot{r}}{\sqrt{\mu}}\left(\frac{r}{\sqrt{\mu}}\,\underline{\dot{r}}\right) - \frac{1}{r}\underline{r} \tag{3.152}$$

$$\underline{r}'' = \frac{r\,\dot{r}}{\mu}\underline{\dot{r}} - \frac{1}{r}\underline{r} \tag{3.153}$$

Thus the initial conditions that must be satisfied are:

$$\underline{r} = \underline{r}_0 \qquad \& \qquad \underline{r}' = \frac{r_0}{\sqrt{\mu}}\,\underline{\dot{r}}_0 \qquad \& \qquad \underline{r}'' = \frac{r_0\,\dot{r}_0}{\mu}\underline{\dot{r}}_0 - \frac{1}{r_0}\underline{r}_0 \tag{3.154}$$

Part of determining the coefficient vectors that does remain the same from the previous relative position magnitude solution is that:

$$\underline{r}_0 = \underline{c}_0 \qquad \& \qquad \underline{r}_0' = \underline{c}_1 \qquad \& \qquad \underline{r}_0'' = 2\underline{c}_2 \tag{3.155}$$

The process is **exactly** the same as before, with no differences in result due to the coefficient vectors being constant. Using these results:

$$\boxed{\begin{aligned} \underline{c}_0 &= \underline{r}_0 \\ \underline{c}_1 &= \frac{r_0}{\sqrt{\mu}}\,\underline{\dot{r}}_0 \\ 2\underline{c}_2 &= \frac{r_0\,\dot{r}_0}{\mu}\underline{\dot{r}}_0 - \frac{1}{r_0}\underline{r}_0 \end{aligned}} \tag{3.156}$$

Thus the series solution to the relative position vector is:

$$\underline{r} = \underline{r}_0 + \chi\frac{r_0}{\sqrt{\mu}}\,\underline{\dot{r}}_0\left(\sum_{k=0}^{\infty}\frac{(-\alpha\chi^2)^k}{(2k+1)!}\right) + \chi^2\left(\frac{r_0\,\dot{r}_0}{\mu}\underline{\dot{r}}_0 - \frac{1}{r_0}\underline{r}_0\right)\left(\sum_{k=0}^{\infty}\frac{(-\alpha\chi^2)^k}{(2k+2)!}\right) \tag{3.157}$$

Substituting the appropriate Stumpff functions:

$$\underline{r} = \underline{r}_0 + \chi\frac{r_0}{\sqrt{\mu}}\,\underline{\dot{r}}_0\,C_1(\alpha\chi^2) + \chi^2\left(\frac{r_0\,\dot{r}_0}{\mu}\underline{\dot{r}}_0 - \frac{1}{r_0}\underline{r}_0\right)C_2(\alpha\chi^2) \tag{3.158}$$

Grouping terms by common vector:

$$\underline{r} = \left(1 - \frac{\chi^2}{r_0} C_2(\alpha\chi^2)\right)\underline{r}_0 + \left(\frac{r_0\chi}{\sqrt{\mu}}\, C_1(\alpha\chi^2) + \frac{r_0\dot{r}_0\chi^2}{\mu}\, C_2(\alpha\chi^2)\right)\dot{\underline{r}}_0 \tag{3.159}$$

The coefficient for $\dot{\underline{r}}_0$ can be simplified by using the Stumpff function recurrence relation (equation 3.134) for $C_1(\alpha\chi^2)$.

$$\frac{r_0\chi}{\sqrt{\mu}}\left(\frac{1}{1!} - \alpha\chi^2 C_{1+2}(\alpha\chi^2)\right) + \frac{r_0\dot{r}_0\chi^2}{\mu}\, C_2(\alpha\chi^2) \tag{3.160}$$

$$\frac{r_0\chi}{\sqrt{\mu}}\left(1 - \alpha\chi^2 C_3(\alpha\chi^2)\right) + \frac{r_0\dot{r}_0\chi^2}{\mu}\, C_2(\alpha\chi^2) \tag{3.161}$$

$$\frac{1}{\sqrt{\mu}}\left(r_0\chi + \chi^2\frac{r_0\dot{r}_0}{\sqrt{\mu}}\, C_2(\alpha\chi^2) - \chi^3\alpha r_0\, C_3(\alpha\chi^2)\right) \tag{3.162}$$

This form is, save for the factor of $1/\sqrt{\mu}$, the universal variable time relation equation minus a single $C_3(\alpha\chi^2)$ term (equation 3.137). Setting $\Delta t \equiv t - t_0$:

$$\frac{1}{\sqrt{\mu}}\left(r_0\chi + \chi^2\frac{r_0\dot{r}_0}{\sqrt{\mu}}\, C_2(\alpha\chi^2) - \chi^3\alpha r_0\, C_3(\alpha\chi^2)\right) = \Delta t - \frac{\chi^3}{\sqrt{\mu}}\, C_3(\alpha\chi^2) \tag{3.163}$$

Therefore, the complete series solution to the relative position vector as a function of the universal variable is:

$$\boxed{\underline{r} = \left(1 - \frac{\chi^2}{r_0} C_2(\alpha\chi^2)\right)\underline{r}_0 + \left(\Delta t - \frac{\chi^3}{\sqrt{\mu}}\, C_3(\alpha\chi^2)\right)\dot{\underline{r}}_0} \tag{3.164}$$

Recall back to chapter two. Scalar functions called Lagrange coefficients were derived for the purposes of orbit propagation. In chapter two, these coefficients were given in terms of $\Delta\theta$. The Lagrange coefficients can now be expressed as functions of time via the universal variables formulation. Given that the Lagrange coefficients satisfy $\underline{r} = f\underline{r}_0 + g\dot{\underline{r}}_0$:

$$\boxed{\begin{aligned} f &= 1 - \frac{\chi^2}{r_0} C_2(\alpha\chi^2) \\ g &= \Delta t - \frac{\chi^3}{\sqrt{\mu}}\, C_3(\alpha\chi^2) \end{aligned}} \tag{3.165}$$

Also from the discussion of orbit propagation is the determination of the time derivative of the relative position vector from the relation $\dot{\underline{r}} = \dot{f}\underline{r}_0 + \dot{g}\,\dot{\underline{r}}_0$. The time derivatives of the Lagrange coefficients can be obtained via equation 3.73. The time derivative of f is:

$$\dot{f} = \frac{d}{dt}(f) = \frac{\sqrt{\mu}}{r}\frac{d}{d\chi}(f) = \frac{\sqrt{\mu}}{r}\frac{d}{d\chi}\left(1 - \frac{\chi^2}{r_0} C_2(\alpha\chi^2)\right) \tag{3.166}$$

Absorbing the factor of χ^2 into $C_2(\alpha\chi^2)$:

$$\dot{f} = \frac{\sqrt{\mu}}{r}\frac{d}{d\chi}\left(1 - \frac{1}{r_0}\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi)^{2k+2}}{(2k+2)!}\right)\right) \tag{3.167}$$

$$\dot{f} = \frac{\sqrt{\mu}}{r}\left(-\frac{1}{r_0}\left(\sum_{k=0}^{\infty}\frac{(2k+2)(-\alpha)^k(\chi)^{2k+1}}{(2k+2)!}\right)\right) \tag{3.168}$$

$$\dot{f} = -\frac{\sqrt{\mu}}{rr_0}\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi)^{2k+1}}{(2k+1)!} \tag{3.169}$$

$$\dot{f} = -\chi\frac{\sqrt{\mu}}{rr_0}C_1(\alpha\chi^2) \tag{3.170}$$

This result can be left as is, but the minus sign out front is unsightly. This can be remedied by utilizing the the Stumpff function recurrence relation (equation 3.134) for $C_1(\alpha\chi^2)$.

$$\dot{f} = -\chi\frac{\sqrt{\mu}}{rr_0}\left(\frac{1}{1!} - \alpha\chi^2 C_{1+2}(\alpha\chi^2)\right) \tag{3.171}$$

Simplifying:

$$\boxed{\dot{f} = \frac{\sqrt{\mu}}{rr_0}\left(\alpha\chi^3 C_3(\alpha\chi^2) - \chi\right)} \tag{3.172}$$

Where r in the denominator of the leading coefficient is given by equation 3.136. The time derivative of g is:

$$\dot{g} = \frac{d}{dt}(g) = \frac{d}{dt}(\Delta t) - \frac{\sqrt{\mu}}{r}\frac{d}{d\chi}\left(\frac{\chi^3}{\sqrt{\mu}}C_3(\alpha\chi^2)\right) \tag{3.173}$$

Absorbing the factor of χ^3 into $C_3(\alpha\chi^2)$:

$$\dot{g} = 1 - \frac{\sqrt{\mu}}{r}\frac{d}{d\chi}\left(\frac{1}{\sqrt{\mu}}\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi)^{2k+3}}{(2k+3)!}\right)\right) \tag{3.174}$$

$$\dot{g} = 1 - \frac{\sqrt{\mu}}{r}\left(\frac{1}{\sqrt{\mu}}\left(\sum_{k=0}^{\infty}\frac{(2k+3)(-\alpha)^k(\chi)^{2k+2}}{(2k+3)!}\right)\right) \tag{3.175}$$

$$\dot{g} = 1 - \frac{1}{r}\left(\sum_{k=0}^{\infty}\frac{(-\alpha)^k(\chi)^{2k+2}}{(2k+2)!}\right) \tag{3.176}$$

$$\boxed{\dot{g} = 1 - \frac{\chi^2}{r}C_2(\alpha\chi^2)} \tag{3.177}$$

Where r in the denominator of the leading coefficient, again, is given by equation 3.136. With all Lagrange coefficients determined, the state vector ($\underline{r}\,\&\,\underline{\dot{r}}$) can be constructed from an increment of time beyond an initial observation.

3.6 Auxiliary Parameters as they Relate to χ

The time since periapsis has different solutions depending on the type of orbit. This was shown for a closed orbit with the eccentric anomaly (E), the parabolic orbit with Barker's anomaly (D), and the hyperbolic orbit with the hyperbolic eccentric anomaly (F). The universal variables approach is not restricted to only be in reference to the periapsis. Additionally, the universal variables formulation has the same solution for any orbit type.

The time since periapsis formulations exploit the conservation of angular momentum to obtain an integral over the true anomaly with reference to the periapsis. The universal variable formulation instead uses the conservation of mechanical energy to obtain a differential equation for the radial distance in terms of the universal anomaly. Although the two approaches employ different quantities, they describe the same two body dynamics. The universal variables approach should yield a result that is analogous to, or at least proportional, to the solutions derived for time since periapsis. Enforcing that the initial state corresponds to periapsis, by equation 3.137:

$$\sqrt{\mu}(t-t_p) = r_p\chi + \chi^2\frac{r_p\dot{r}_p}{\sqrt{\mu}}C_2(\alpha\chi^2) + \chi^3\left(1-\alpha r_p\right)C_3(\alpha\chi^2) \tag{3.178}$$

As has been discussed in extensive detail in chapter 2, $\dot{r}_p = 0$. Therefore:

$$\sqrt{\mu}(t-t_p) = r_p\chi + \chi^3\left(1-\alpha r_p\right)C_3(\alpha\chi^2) \tag{3.179}$$

Now to analyze the coefficient $(1-\alpha r_p)$. From equation 3.81:

$$1-\alpha r_p = 1+\frac{2\varepsilon}{\mu}r_p \tag{3.180}$$

Substituting equation 1.82 for ε:

$$1-\alpha r_p = 1+\frac{2}{\mu}\left(\frac{1}{2}\left(\dot{r_p}^2+r_p^2\dot{\theta}^2\right)-\frac{\mu}{r_p}\right)r_p \tag{3.181}$$

Substituting equation 1.85 for $\dot{\theta}$ and $\dot{r}_p = 0$:

$$1-\alpha r_p = 1+\frac{2}{\mu}\left(\frac{1}{2}r_p^2\left(\frac{h}{r_p^2}\right)^2-\frac{\mu}{r_p}\right)r_p \tag{3.182}$$

$$1-\alpha r_p = 1+\frac{r_p^3h^2}{\mu r_p^4}-2 = \frac{h^2}{\mu r_p}-1 \tag{3.183}$$

By equation 2.42:

$$1-\alpha r_p = e \tag{3.184}$$

Plugging $1-\alpha r_p$ back into equation 3.179:

$$\sqrt{\mu}(t-t_p) = r_p\chi + e\chi^3C_3(\alpha\chi^2) \tag{3.185}$$

It is known, also from chapter 2, that closed orbits have positive semi major axis lengths. Thus $\alpha = 1/a > 0$. Therefore, $z = \alpha\chi^2 > 0$. Using the elementary functional form of the Stumpff function $C_3(z)$ corresponding to $z > 0$ given by equation 3.135:

$$\sqrt{\mu}(t - t_p) = r_p\chi + e\chi^3 \left(\frac{\sqrt{\alpha\chi^2} - \sin(\sqrt{\alpha\chi^2})}{(\alpha\chi^2)^{3/2}} \right) \tag{3.186}$$

$$\sqrt{\mu}(t - t_p) = r_p\chi + e\chi^3 \sqrt{\frac{\alpha\chi^2}{\alpha^3\chi^6}} - e\sqrt{\frac{\chi^6}{\alpha^3\chi^6}} \sin\left(\sqrt{\alpha\chi^2}\right) \tag{3.187}$$

$$\sqrt{\mu}(t - t_p) = r_p\chi + \frac{e\chi^3}{\alpha\chi^2} - e\sqrt{\frac{1}{\alpha^3}} \sin\left(\sqrt{\alpha\chi^2}\right) \tag{3.188}$$

Substituting $\alpha = 1/a$ and simplifying (as defined, $\chi_0 = 0$ and increases with time, and therefore χ is always positive):

$$\sqrt{\mu}(t - t_p) = (r_p + ae)\chi - e\sqrt{a^3} \sin\left(\frac{\chi}{\sqrt{a}}\right) \tag{3.189}$$

Using the orbit equation at periapsis to solve for $r_p + ae$:

$$r_p = \frac{p}{1 + e\cos(0)} = \frac{a(1 - e^2)}{1 + e} = a(1 - e) = a - ae \quad \rightarrow \quad r_p + ae = a \tag{3.190}$$

Therefore:

$$\sqrt{\mu}(t - t_p) = a\chi - e\sqrt{a^3} \sin\left(\frac{\chi}{\sqrt{a}}\right) \tag{3.191}$$

Dividing through by $\sqrt{\mu}$:

$$t - t_p = \frac{a}{\sqrt{\mu}}\chi - e\sqrt{\frac{a^3}{\mu}} \sin\left(\frac{\chi}{\sqrt{a}}\right) \tag{3.192}$$

Seeking a common factor:

$$t - t_p = \sqrt{\frac{a^3}{\mu}} \left(\frac{\chi}{\sqrt{a}} - e\sin\left(\frac{\chi}{\sqrt{a}}\right) \right) \tag{3.193}$$

Equating equation 3.23 and equation 3.193 results in exactly the kind of proportional relationship between the eccentric and universal anomalies one would expect from two equivilent ways of solving the same problem.

$$t - t_p = \sqrt{\frac{a^3}{\mu}} \Big(E - e\sin(E) \Big) = \sqrt{\frac{a^3}{\mu}} \left(\frac{\chi}{\sqrt{a}} - e\sin\left(\frac{\chi}{\sqrt{a}}\right) \right) \tag{3.194}$$

Or a more simple relation:

$$\boxed{\chi = \sqrt{a}\, E} \tag{3.195}$$

Therefore, the universal variable formulation reduces exactly to the Kepler equation for the special case of a closed orbit where the initial state is taken at periapsis. The universal variable χ is directly proportional to the eccentric anomaly E for this special case.

Consider the case when the orbit is parabolic and the initial state is taken at periapsis. Parabolic orbits, as discussed in chapter 2, have an infinite semi major axis length, which forces α to zero. The universal variables time relation in this case yields:

$$\sqrt{\mu}(t-t_p) = r_p\chi + \chi^2\frac{r_p\dot{r}_p}{\sqrt{\mu}}C_2((0)\chi^2) + \chi^3\left(1-(0)r_p\right)C_3((0)\chi^2) \tag{3.196}$$

Recalling $\dot{r}_p = 0$ and simplifying:

$$\sqrt{\mu}(t-t_p) = r_p\chi + \chi^3 C_3(0) \tag{3.197}$$

Substituting equation 3.135 for $C_3(0)$:

$$\sqrt{\mu}(t-t_p) = r_p\chi + \frac{1}{6}\chi^3 \tag{3.198}$$

Using the orbit equation to solve for r_p:

$$r_p = \frac{p}{1+(1)\cos(0)} = \frac{p}{2} \tag{3.199}$$

Substituting r_p into equation 3.198:

$$\sqrt{\mu}(t-t_p) = \frac{p}{2}\chi + \frac{1}{6}\chi^3 \tag{3.200}$$

Pulling a factor of $\sqrt{p^3}$ from the right hand side:

$$\sqrt{\mu}(t-t_p) = \sqrt{p^3}\left(\frac{1}{2}\frac{p}{\sqrt{p^3}}\chi + \frac{1}{6}\frac{1}{\sqrt{p^3}}\chi^3\right) \tag{3.201}$$

Simplifying:

$$\sqrt{\mu}(t-t_p) = \sqrt{p^3}\left(\frac{1}{2}\sqrt{\frac{1}{p}}\chi + \frac{1}{6}\sqrt{\frac{1}{p^3}}\chi^3\right) \tag{3.202}$$

Dividing both sides by $\sqrt{\mu}$ and simplifying:

$$t-t_p = \sqrt{\frac{p^3}{\mu}}\left(\frac{1}{2}\left(\sqrt{\frac{1}{p}}\chi\right) + \frac{1}{6}\left(\sqrt{\frac{1}{p}}\chi\right)^3\right) \tag{3.203}$$

Equating equation 3.32 and equation 3.203:

$$t-t_p = \sqrt{\frac{p^3}{\mu}}\left(\frac{1}{2}D + \frac{1}{6}D^3\right) = \sqrt{\frac{p^3}{\mu}}\left(\frac{1}{2}\left(\sqrt{\frac{1}{p}}\chi\right) + \frac{1}{6}\left(\sqrt{\frac{1}{p}}\chi\right)^3\right) \tag{3.204}$$

Or a more simple relation:

$$\boxed{\chi = \sqrt{p}\,D} \tag{3.205}$$

Therefore, the universal variable formulation reduces exactly to the Barker equation for the special case of a parabolic orbit where the initial state is taken at periapsis. The universal variable χ is directly proportional to the Barker anomaly D for this special case.

For completeness, consider also the case where the orbit is hyperbolic with the initial state corresponding to periapsis. The universal variables formulation with the initial state at periapsis has $\dot{r}_p = 0$ and by equation 3.184, $1 - \alpha r_p = e$.

$$\sqrt{\mu}(t - t_p) = r_p\chi + e\,\chi^3 C_3(\alpha\chi^2) \tag{3.206}$$

Hyperbolic orbits, again from chapter 2, have a negative semi major axis legnth. Since the semi major axis is negative, it means that $z = \alpha\chi^2 < 0$. Using the elementary functional form of the Stumpff function $C_3(z)$ corresponding to $z < 0$ given by equation 3.135:

$$\sqrt{\mu}(t - t_p) = r_p\chi + e\chi^3\left(\frac{\sinh(\sqrt{-\alpha\chi^2}) - \sqrt{-\alpha\chi^2}}{(-\alpha\chi^2)^{3/2}}\right) \tag{3.207}$$

$$\sqrt{\mu}(t - t_p) = r_p\chi - e\sqrt{\frac{-\alpha\chi^8}{-\alpha^3\chi^6}} + e\sqrt{\frac{\chi^6}{-\alpha^3\chi^6}}\sinh\left(\sqrt{-\alpha\chi^2}\right) \tag{3.208}$$

$$\sqrt{\mu}(t - t_p) = r_p\chi - e\sqrt{\frac{\chi^8}{(-1)^2\alpha^2\chi^6}} + e\sqrt{\frac{\chi^6}{-\alpha^3\chi^6}}\sinh\left(\sqrt{-\alpha\chi^2}\right) \tag{3.209}$$

$$\sqrt{\mu}(t - t_p) = r_p\chi + e\frac{\chi}{\alpha} + e\sqrt{\frac{1}{-\alpha^3}}\sinh\left(\sqrt{-\alpha\chi^2}\right) \tag{3.210}$$

Substituting $\alpha = 1/a$ and simplifying (as defined, $\chi_0 = 0$ and increases with time, and therefore χ is always positive):

$$\sqrt{\mu}(t - t_p) = (r_p + ae)\chi + e\sqrt{-a^3}\sinh\left(\frac{\chi}{\sqrt{-a}}\right) \tag{3.211}$$

Substituting equation 3.190:

$$\sqrt{\mu}(t - t_p) = a\chi + e\sqrt{-a^3}\sinh\left(\frac{\chi}{\sqrt{-a}}\right) \tag{3.212}$$

Factor $\sqrt{-a^3}$ from the right hand side:

$$\sqrt{\mu}(t - t_p) = \sqrt{-a^3}\left(\sqrt{\frac{a^2}{-a^3}}\chi + e\sinh\left(\frac{\chi}{\sqrt{-a}}\right)\right) \tag{3.213}$$

$$\sqrt{\mu}(t - t_p) = \sqrt{-a^3}\left(\sqrt{\frac{1}{(-1)^2(-a)}}\chi + e\sinh\left(\frac{\chi}{\sqrt{-a}}\right)\right) \tag{3.214}$$

Dividing each side by $\sqrt{\mu}$ and simplifying:

$$t - t_p = \sqrt{\frac{-a^3}{\mu}}\left(e\sinh\left(\frac{\chi}{\sqrt{-a}}\right) - \frac{\chi}{\sqrt{-a}}\right) \tag{3.215}$$

Equating equation 3.67 and equation 3.215:

$$t - t_p = \sqrt{\frac{-a^3}{\mu}}\Big(e\sinh(F) - F\Big) = \sqrt{\frac{-a^3}{\mu}}\left(e\sinh\left(\frac{\chi}{\sqrt{-a}}\right) - \frac{\chi}{\sqrt{-a}}\right) \tag{3.216}$$

Or a more simple relation:

$$\boxed{\chi = \sqrt{-a}\, F} \tag{3.217}$$

Therefore, the universal variable formulation reduces exactly to the hyperbolic Kepler equation for the special case of a hyperbolic orbit where the initial state is taken at periapsis. The universal variable χ is directly proportional to the hyperbolic eccentric anomaly anomaly F for this special case.

The general form of each orbit type's time equations can in fact be written in terms of the Battin universal anomaly. For example, if the initial state is not periapsis, then the relations still stand.

For closed orbit types, in general:

$$\chi - \chi_0 = \sqrt{a}E - \sqrt{a}E_0 \tag{3.218}$$

Using the fact that $\chi_0 = 0$ and simplifying:

$$\chi = \sqrt{a}\,(E - E_0) \tag{3.219}$$

For parabolic orbit types, in general:

$$\chi - \chi_0 = \sqrt{p}D - \sqrt{p}D_0 \tag{3.220}$$

Using the fact that $\chi_0 = 0$ and simplifying:

$$\chi = \sqrt{p}\,(D - D_0) \tag{3.221}$$

For hyperbolic orbit types, in general:

$$\chi - \chi_0 = \sqrt{-a}F - \sqrt{-a}F_0 \tag{3.222}$$

Using the fact that $\chi_0 = 0$ and simplifying:

$$\chi = \sqrt{-a}\,(F - F_0) \tag{3.223}$$

For any universal anomaly, in general:

$$\boxed{\chi = \begin{cases} \sqrt{a}\,(E - E_0)\,, & \alpha > 0, \\ \sqrt{p}\,(D - D_0)\,, & \alpha = 0, \\ \sqrt{-a}\,(F - F_0)\,, & \alpha < 0. \end{cases}} \tag{3.224}$$

This effectively bridges the gap between each orbit type's time since periapsis equation and the universal variables formulation. This illustrates how the Battin universal anomaly is related to the time since periapsis anomalies given how α affects the **Stumpff functions**. The universal variables approach allows for time determination for any type of orbit without switching equations, which is why the universal variables approach is such a fitting name.

3.7 Summary

Closed orbit time since periapsis:

$$\tan\left(\frac{E}{2}\right) = \sqrt{\frac{1-e}{1+e}}\tan\left(\frac{\theta}{2}\right)$$
$$n = \sqrt{\frac{\mu}{a^3}}$$
$$t - t_p = \frac{M_e}{n} = \frac{1}{n}\Big(E - e\sin(E)\Big)$$

Parabolic orbit time since periapsis:

$$D = \tan\left(\frac{\theta}{2}\right)$$
$$n = \sqrt{\frac{\mu}{p^3}}$$
$$t - t_p = \frac{M_p}{n} = \frac{1}{n}\left(\frac{1}{2}D - \frac{1}{6}D^3\right)$$

Hyperbolic orbit time since periapsis:

$$\tanh\left(\frac{F}{2}\right) = \sqrt{\frac{e-1}{e+1}}\,\tan\left(\frac{\theta}{2}\right)$$
$$n = \sqrt{\frac{\mu}{-a^3}}$$
$$t - t_p = \frac{M_h}{n} = \frac{1}{n}\Big(e\sinh(F) - F\Big)$$

Universal variables formulation:

$$r = r_0 + \chi\frac{r_0\dot{r}_0}{\sqrt{\mu}}C_1(\alpha\chi^2) + \chi^2\left(1 - \alpha r_0\right)C_2(\alpha\chi^2)$$
$$\sqrt{\mu}(t - t_0) = r_0\chi + \chi^2\frac{r_0\dot{r}_0}{\sqrt{\mu}}C_2(\alpha\chi^2) + \chi^3\left(1 - \alpha r_0\right)C_3(\alpha\chi^2)$$

The Lagrange coefficients:

$$f = 1 - \frac{\chi^2}{r_0}C_2(\alpha\chi^2) \qquad g = \Delta t - \frac{\chi^3}{\sqrt{\mu}}C_3(\alpha\chi^2)$$
$$\dot{f} = \frac{\sqrt{\mu}}{rr_0}\left(\alpha\chi^3 C_3(\alpha\chi^2) - \chi\right) \qquad \dot{g} = 1 - \frac{\chi^2}{r}C_2(\alpha\chi^2)$$

χ as it relates to E, D, or F in time since periapsis:

$$\chi = \begin{cases} \sqrt{a}\left(E - E_0\right), & \alpha > 0, \\ \sqrt{p}\left(D - D_0\right), & \alpha = 0, \\ \sqrt{-a}\left(F - F_0\right), & \alpha < 0. \end{cases}$$

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Chapter 4

Case Study: Kepler's Laws of Planetary Motion

Introduction

Chapters one through three developed the complete solution to the two body problem. This case study will demonstrated that Kepler's three empirical laws of planetary motion emerge naturally from these results. Johannes Kepler discovered that the planets of our solar system follow three laws of motion using decades of precise planetary observational data that he took from Tycho Brahe following his death. At the time, the exact reason that planets followed Kepler's laws was somewhat of a mystery that required Newton to uncover. These laws have not been considered or referenced within the formalism of the solutions that were developed in chapters one through three.

4.1 The First Law

Kepler's first law of planetary motion is "each planet's orbit about the Sun is an ellipse. The Sun's center is always located at one focus of the orbital ellipse. The Sun is at one focus. The planet follows the ellipse in its orbit, meaning that the planet to Sun distance is constantly changing as the planet goes around its orbit." [1]

From the work done in chapter two, this is obvious considering that none of the planets have been observed to exit the solar system. In other words, the planets orbit the sun in closed orbits. Perfectly circular orbits are mathematical concepts, and are not commonly observed in nature (near circular orbits are fairly common). With the shape of an orbit being explained by the orbit equation, and the planets themselves having bound orbits about the sun, it is very reasonable to state the planets take elliptical paths around the sun.

As for the sun being located at the focus, the **orbit equation** corresponds to a conic section where the relative position of m_2 is measured with respect to m_1. The sun makes up 99.86% of the mass in the solar system, so it is no far off assumption to state that the center of mass is essentially located at the center of the sun: the focus of the orbit. Cases such as this are sometimes reffered to as central force problems, where $m_1 + m_2 \approx m_1$, and the center of mass corresponds to the center of the more massive body.

4.2 The Second Law

Kepler's second law of planetary motion is "the imaginary line joining a planet and the Sun sweeps equal areas of space during equal time intervals as the planet orbits. Basically, that planets do not move with constant speed along their orbits. Rather, their speed varies so that the line joining the centers of the Sun and the planet sweeps out equal parts of an area in equal times." [1]

Sonsider the relative position vector at time t and the relative position vector at time $t + dt$. Start by determining the relative position vectors before and after a differential time element.

$$\underline{r}(t) = r\,\hat{r} \tag{4.1}$$

$$\underline{r}(t + dt) = r\,\hat{r} + (dr\,\hat{r} + r d\theta\,\hat{\theta}) = \underline{r}(t) + d\underline{r} \tag{4.2}$$

The area enclosed by the parallelogram made up by $\underline{r}(t)$ and $\underline{r}(t + dt)$ is given by their cross product. However, the area that Kepler's second law refers to is only half of the area of this parallelogram, or the area swept by the relative position vector during the differential time element.

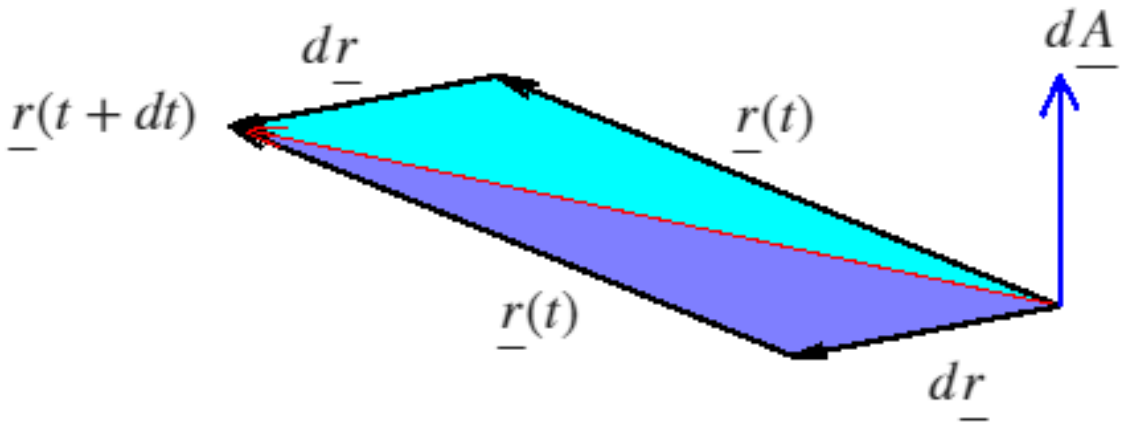


Figure 4.1: The differential area swept by the relative position vectors during dt and the area of the parallelogram formed by the two relative position vectors.

The differential area swept by the two position vectors can be calculated using half the cross product between $\underline{r}(t)$ and $\underline{r}(t + dt)$.

$$d\underline{A} = \frac{1}{2}\underline{r}(t) \times \underline{r}(t + dt) \tag{4.3}$$

$$d\underline{A} = \frac{1}{2}\underline{r}(t) \times \left(\underline{r}(t) + d\underline{r}\right) = \frac{1}{2}\left(\underline{0} + \underline{r}(t) \times d\underline{r}\right) \tag{4.4}$$

$$d\underline{A} = \frac{1}{2}\underline{r}(t) \times d\underline{r} \tag{4.5}$$

Kepler's second law of planetary motion says that the relative position vector should sweep out equal areas in equal times. So, the time derivative of equation 4.5 must be constant for Kepler's second law to hold true.

$$\frac{d\underline{A}}{dt} = \frac{1}{2}\frac{\underline{r}(t) \times d\underline{r}}{dt} = \frac{1}{2}\,\underline{r}(t) \times \frac{d\underline{r}}{dt} \tag{4.6}$$

$$\frac{d\underline{A}}{dt} = \frac{1}{2}\,\underline{r}(t) \times \dot{\underline{r}}(t) \tag{4.7}$$

This should look very familiar at this point. By equation 1.84, equation 4.7 is:

$$\boxed{\frac{d\underline{A}}{dt} = \frac{1}{2}\,\underline{h}} \tag{4.8}$$

Which of course is constant. This proves Kepler's second law of planetary motion. The time rate of change of the area swept by the relative position vector is constant. What this really means, is that Kepler's second law of planetary motion is really a statement of the conservation of angular momentum within the two body problem.

4.3 The Third Law

Kepler's third law of planetary motion is "the squares of the orbital periods of the planets are directly proportional to the cubes of the semi-major axes of their orbits." [1] In other words: $P^2 \propto a^3$.

Kepler's third law of planetary moton can be shown to be true using equation 3.24, that was derived from the time since periapsis equation for a closed orbit in chapter three.

$$P = 2\pi\sqrt{\frac{a^3}{\mu}} \tag{4.9}$$

$$\boxed{P^2 = \frac{4\pi^2}{\mu}a^3} \tag{4.10}$$

From this solution, Kepler's third law pops out straight away. Not only is it clear that the square of the period *must* be proportional to the cube of the semi-major axis length, but it also reveals the constant of proportionality. The solution to the two body problem gives an exact statement of Kepler's third law, which Kepler would not have been able to determine without Newton's laws of motion.

4.4 Summary

The solution to the two body problem not only determines the motion of celestial bodies, but also proves the laws postulated by Johannes Kepler nearly 70 years prior to Newton's laws of motion. Kepler's analysis of Brahe's data resulted in an amazingly accurate account of the motion of the planets without modern formalisms of mechanics. It does help that Kepler's third law of planetary motion was the leap that Newton needed to arrive at his universal law of gravitation, though it does not result in the full picture. Only after the equations of motion are established and the two-body problem is solved do Kepler's three empirical laws emerge as inevitable consequences of Newtonian gravitation.

References

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Part II

Orbital Mechanics

Chapter 5

Reference Frames & Transformations

Introduction

Part one of this textbook dealt with obtaining the solutions to the Kepler two body problem. Part two of this book goes into practical applications of those solutions. While these solutions do not change in form, their meaning can be thought of slightly differently depending on the specific situation. There are numerous different ways to go about applying the two body problem to the real world. This chapter outlines an important aspect of orbital mechanics that lacked specificaiton in part one: coordinate systems. Whenever a vector was denoted in part one, it reffered only to the cetner of mass fixed reference frame. This is not helpful unless a proper frame of reference has been established. But where does the origin of the coordinate system go? What are the observable quantities required to define a position or velocity? All questions that will be answered in this chapter.

5.1 The Central Force Approximation

The central force approximation has a great deal of merit in practical application. For many situations, this puts the center of mass of the system at the center of the larger of the two orbiting bodies. The origin of the center of mass fixed reference frame, in the case of the central force approximation, is therefore located at the center of the more massive body. The central force approximation is defined from the general solution to the two body problem, with the more massive of the two bodies being m_1, and $m_1 \equiv M$. Likewise, the less massive body is defined to be m_2, and $m_2 \equiv m$. The gravitational parameter of the system is approximated as:

$$\mu = G(m_1 + m_2) \equiv G(M + m) \approx GM \tag{5.1}$$

So the gravitational parameter for a system utilizing the central force approximation is essentially a statement of the gravitational strength of the central body M. In a similar fashion, the reduced mass becomes:

$$\mathscr{M} = \frac{m_1 m_2}{m_1 + m_2} \equiv \frac{Mm}{M + m} \approx m \tag{5.2}$$

The reduced mass of a central force approximated system approaches the mass of the less massive object, which makes sense because the reduced mass is the effective inertial mass of the system. This doesn't include M because the more massive body's motion is negligible when compared with that of m. This basically states that m moves around M without M having any motion as a result of the interaction.

Part one of this textbook made no approximations or assumptions for the steps leading to the solution. So, as a result, the central force approximation may seem as a stretch of the imagination. Nonetheless, the central force approximation is actually very practical in orbital mechanics.

According to the Guinness Book of World Records, the largest artificial satellite in orbit around Earth (the International Space Station) weighs up to 459025 kilograms at its heaviest [1]. To put this into perspective, the mass of the Earth is around $5.974 \cdot 10^{24}$ kilograms. Applying the central force assumption:

$$M + m = M + m = \left(M + \frac{459025 \text{ kg}}{5.974 \cdot 10^{24} \text{ kg}} M \right) = (1 + 7.6837 \cdot 10^{-20}) M \approx M \tag{5.3}$$

Clearly the assumption that $M + m \approx M$ can be a very good approximation, especially in the case of artifical satellites.

Most of the orbits of the planets around the Sun can be faithfully analyzed with the central force approximation. However, the merit of doing this must be carefully considered. Take the Sun and Jupiter system as an example. The mass of the Sun is around $1.989 \cdot 10^{30}$ kilograms and the mass of Jupiter is around $1.899 \cdot 10^{27}$ kilograms. Considering the validity of the central force approximation in this case:

$$M + m = \left(M + \frac{1.899 \cdot 10^{27}}{1.989 \cdot 10^{30}} M \right) = 1.00095475 \, M \tag{5.4}$$

The difference is approximately 0.095%, which is still sufficiently small for many preliminary analyses. However, for precise orbit determination and mission design, the exact two body formulation must be retained. The answer lies in how the analysis is being used. The reason this approximation is good for preliminary analysis is because Jupiter is so far away from the sun, that the approximation give a fairly good result. Although the Sun–Jupiter center of mass lies slightly outside the surface of the Sun (due to Jupiter's large orbital distance), Jupiter's mass is still less than a tenth of a percent of the Sun's mass. Consequently, the central force approximation remains very accurate for many applications. However, this will be insufficient for final stages of mission preparation. As a matter of fact, the center of mass of the Sun and Jupiter system is actually located just outside the surface of the Sun: clearly not at its center. It is always a good idea to be certain of the mission parameters before employing an approximation that may or may not be appropriate.

So, an important result of the central force approximation, is that the origin of the center of mass fixed reference frame is at the center of the more massive body. Perhaps the most important aspect of defining a coordinate system is locating its origin. For the International Space Station, as mentioned above, the origin of the center of mass fixed reference frame is the center of Earth.

5.2 Direction Cosine Matrices

Now that it has been established, within reasonable assumption, that the origin of the center of mass fixed reference frame is located at the center of the more massive body, the question of coordinate systems must be posed. In many applications, an observation may be made within the reference frame of a non-inertial coordinate system. Why would this be done? Because it is practical. I am sure every reader has seen the night sky and how the stars seem to move as time passes. This is because you are making that observation from the surface of Earth, which is in fact a non-inertial reference frame due to the Earth's rotation. Any ground station built on the surface of the Earth, or any other planet, will similaraly be taking measurements from a non-inertial reference frame.

In addition, these measurements are used to form vectors like position and velocity. Therefore, in order for the dynamics of part one of this textbook to have any useful interpretation, the measurements made from a ground station must be converted from the non-inertial frame into an inertial frame. Note also that vectors can be transformed via matrix multiplication. The subject of this section will be to uncover what kind of matrix is most appropriate for this transformation.

Consider two orthogonal sets of basis vectors for arbitrary coordinate reference frames, which are defined by $\mathcal{F}_A = (\hat{a}, \hat{b}, \hat{c})$ and $\mathcal{F}_U = (\hat{u}, \hat{v}, \hat{w})$. The basis vectors of frame U can be expressed as linear combinations of the basis vectors of frame A.

$$\begin{aligned} \hat{u} &= u_a\,\hat{a} + u_b\,\hat{b} + u_c\,\hat{c} \\ \hat{v} &= v_a\,\hat{a} + v_b\,\hat{b} + v_c\,\hat{c} \\ \hat{w} &= w_a\,\hat{a} + w_b\,\hat{b} + w_c\,\hat{c} \end{aligned} \tag{5.5}$$

Since basis vectors of A are orthonormal, the components of the basis vectors of U can be expressed as dot products. From linear algebra, the dot product between two mutually orthonormal vectors is $\hat{e}_i \cdot \hat{e}_j = \delta_{ij}$, where δ_{ij} is the Kronecker delta. Examining the basis vector $\hat{u}$ and its dot products with the basis vectors of A:

$$\begin{aligned} \hat{a}\cdot\hat{u} &= u_a\,(\hat{a}\cdot\hat{a}) + u_b\,(\hat{a}\cdot\hat{b}) + u_w\,(\hat{a}\cdot\hat{c}) = u_a \\ \hat{b}\cdot\hat{u} &= u_a\,(\hat{b}\cdot\hat{a}) + u_b\,(\hat{b}\cdot\hat{b}) + u_w\,(\hat{b}\cdot\hat{c}) = u_b \\ \hat{c}\cdot\hat{u} &= u_a\,(\hat{c}\cdot\hat{a}) + u_b\,(\hat{c}\cdot\hat{b}) + u_w\,(\hat{c}\cdot\hat{c}) = u_c \end{aligned} \tag{5.6}$$

The exact same process may be done for basis vectors $\hat{v}$ and $\hat{w}$. Therefore, the basis vectors of U can be rewritten as:

$$\begin{aligned} \hat{u} &= (\hat{a}\cdot\hat{u})\,\hat{a} + (\hat{b}\cdot\hat{u})\,\hat{b} + (\hat{c}\cdot\hat{u})\,\hat{c} \\ \hat{v} &= (\hat{a}\cdot\hat{v})\,\hat{a} + (\hat{b}\cdot\hat{v})\,\hat{b} + (\hat{c}\cdot\hat{v})\,\hat{c} \\ \hat{w} &= (\hat{a}\cdot\hat{w})\,\hat{a} + (\hat{b}\cdot\hat{w})\,\hat{b} + (\hat{c}\cdot\hat{w})\,\hat{c} \end{aligned} \tag{5.7}$$

Consider now that a vector $\underline{x}$ is measured in the U reference frame.

$$\underline{x}_U = x_1\,\hat{u} + x_2\,\hat{v} + x_3\,\hat{w} \tag{5.8}$$

To express $\underline{x}$ in terms of the A basis, it may be rewritten by taking advantage of the linear combinations expressed in equation 5.7.

$$\underline{x}_A = x_1 \begin{pmatrix} \hat{a}\cdot\hat{u} \\ \hat{b}\cdot\hat{u} \\ \hat{c}\cdot\hat{u} \end{pmatrix} + x_2 \begin{pmatrix} \hat{a}\cdot\hat{v} \\ \hat{b}\cdot\hat{v} \\ \hat{c}\cdot\hat{v} \end{pmatrix} + x_3 \begin{pmatrix} \hat{a}\cdot\hat{w} \\ \hat{b}\cdot\hat{w} \\ \hat{c}\cdot\hat{w} \end{pmatrix} = \begin{pmatrix} x_1(\hat{a}\cdot\hat{u}) + x_2(\hat{a}\cdot\hat{v}) + x_3(\hat{a}\cdot\hat{w}) \\ x_1(\hat{b}\cdot\hat{u}) + x_2(\hat{b}\cdot\hat{v}) + x_3(\hat{b}\cdot\hat{w}) \\ x_1(\hat{c}\cdot\hat{u}) + x_2(\hat{c}\cdot\hat{v}) + x_3(\hat{c}\cdot\hat{w}) \end{pmatrix} \tag{5.9}$$

Or, to put this into matrix form:

$$\boxed{\underline{x}_A = \begin{bmatrix} \hat{a}\cdot\hat{u} & \hat{a}\cdot\hat{v} & \hat{a}\cdot\hat{w} \\ \hat{b}\cdot\hat{u} & \hat{b}\cdot\hat{v} & \hat{b}\cdot\hat{w} \\ \hat{c}\cdot\hat{u} & \hat{c}\cdot\hat{v} & \hat{c}\cdot\hat{w} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_U \equiv \mathbf{R}^{AU}\underline{x}_U} \tag{5.10}$$

Where $\mathbf{R}^{AU}$ is the transformation matrix between frame U and frame A. Note that each frame is internally orthonormal, and a basis vector of frame A is not generally orthogonal to a basis vector of frame U. Since the magnitudes of the basis vectors are unity, by the definition of the dot product: $\hat{e}_i \cdot \hat{e}_j = |\hat{e}_i||\hat{e}_j| \cos\Big(\angle\{\hat{e}_i, \hat{e}_j\}\Big) = \cos\Big(\angle\{\hat{e}_i, \hat{e}_j\}\Big)$. Therefore:

$$\mathbf{R}^{AU} = \begin{bmatrix} \hat{a}\cdot\hat{u} & \hat{a}\cdot\hat{v} & \hat{a}\cdot\hat{w} \\ \hat{b}\cdot\hat{u} & \hat{b}\cdot\hat{v} & \hat{b}\cdot\hat{w} \\ \hat{c}\cdot\hat{u} & \hat{c}\cdot\hat{v} & \hat{c}\cdot\hat{w} \end{bmatrix} = \begin{bmatrix} \cos\Big(\angle\{\hat{a},\hat{u}\}\Big) & \cos\Big(\angle\{\hat{a},\hat{v}\}\Big) & \cos\Big(\angle\{\hat{a},\hat{w}\}\Big) \\ \cos\Big(\angle\{\hat{b},\hat{u}\}\Big) & \cos\Big(\angle\{\hat{b},\hat{v}\}\Big) & \cos\Big(\angle\{\hat{b},\hat{w}\}\Big) \\ \cos\Big(\angle\{\hat{c},\hat{u}\}\Big) & \cos\Big(\angle\{\hat{c},\hat{v}\}\Big) & \cos\Big(\angle\{\hat{c},\hat{w}\}\Big) \end{bmatrix} \tag{5.11}$$

Since each element of $\mathbf{R}^{AU}$ is the cosine of the angle between one basis vector of A and one basis vector of U, the transformation matrix is called a direction cosine matrix (DCM). This transformation was built by taking advantage of the orthonormality of the basis vectors of frame A. A similar process can be completed to build a matrix that transforms a vector expressed in the A frame into the U frame, but the orthogonality of the two sets of basis vectors can be further taken advantage of. Consider the multiplication of the transpose of $\mathbf{R}^{AU}$ with equation 5.10.

$$\Big(\mathbf{R}^{AU}\Big)^T \underline{x}_A = \Big(\mathbf{R}^{AU}\Big)^T \mathbf{R}^{AU}\underline{x}_U \tag{5.12}$$

$$\Big(\mathbf{R}^{AU}\Big)^T \underline{x}_A = \begin{bmatrix} \hat{a}\cdot\hat{u} & \hat{b}\cdot\hat{u} & \hat{c}\cdot\hat{u} \\ \hat{a}\cdot\hat{v} & \hat{b}\cdot\hat{v} & \hat{c}\cdot\hat{v} \\ \hat{a}\cdot\hat{w} & \hat{b}\cdot\hat{w} & \hat{c}\cdot\hat{w} \end{bmatrix} \begin{bmatrix} \hat{a}\cdot\hat{u} & \hat{a}\cdot\hat{v} & \hat{a}\cdot\hat{w} \\ \hat{b}\cdot\hat{u} & \hat{b}\cdot\hat{v} & \hat{b}\cdot\hat{w} \\ \hat{c}\cdot\hat{u} & \hat{c}\cdot\hat{v} & \hat{c}\cdot\hat{w} \end{bmatrix} \underline{x}_U \tag{5.13}$$

Recalling the components of the basis vectors as dot products, as expressed by equation 5.6 (for shorthand notation):

$$\Big(\mathbf{R}^{AU}\Big)^T \underline{x}_A = \begin{bmatrix} u_a & u_b & u_c \\ v_a & v_b & v_c \\ w_a & w_b & w_c \end{bmatrix} \begin{bmatrix} u_a & v_a & w_a \\ u_b & v_b & w_b \\ u_c & v_c & w_c \end{bmatrix} \underline{x}_U = \begin{bmatrix} \hat{u}^T \\ \hat{v}^T \\ \hat{w}^T \end{bmatrix} \begin{bmatrix} \hat{u} & \hat{v} & \hat{w} \end{bmatrix} \underline{x}_U \tag{5.14}$$

$$\Big(\mathbf{R}^{AU}\Big)^T \underline{x}_A = \begin{bmatrix} \hat{u}^T\hat{u} & \hat{u}^T\hat{v} & \hat{u}^T\hat{w} \\ \hat{v}^T\hat{u} & \hat{v}^T\hat{v} & \hat{v}^T\hat{w} \\ \hat{w}^T\hat{u} & \hat{w}^T\hat{v} & \hat{w}^T\hat{w} \end{bmatrix} \underline{x}_U \equiv \begin{bmatrix} \hat{u}\cdot\hat{u} & \hat{u}\cdot\hat{v} & \hat{u}\cdot\hat{w} \\ \hat{v}\cdot\hat{u} & \hat{v}\cdot\hat{v} & \hat{v}\cdot\hat{w} \\ \hat{w}\cdot\hat{u} & \hat{w}\cdot\hat{v} & \hat{w}\cdot\hat{w} \end{bmatrix} \underline{x}_U \tag{5.15}$$

Recall that the reference frame U is an orthonormal basis, and therefore:

$$\left(\mathbf{R}^{AU}\right)^T \underline{x}_A = \begin{bmatrix} \hat{u}\cdot\hat{u} & \hat{u}\cdot\hat{v} & \hat{u}\cdot\hat{w} \\ \hat{v}\cdot\hat{u} & \hat{v}\cdot\hat{v} & \hat{v}\cdot\hat{w} \\ \hat{w}\cdot\hat{u} & \hat{w}\cdot\hat{v} & \hat{w}\cdot\hat{w} \end{bmatrix} \underline{x}_U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \underline{x}_U \equiv \hat{\mathbf{I}}\,\underline{x}_U \tag{5.16}$$

$$\boxed{\left(\mathbf{R}^{AU}\right)^T \underline{x}_A = \underline{x}_U} \tag{5.17}$$

This result states that the product $\left(\mathbf{R}^{AU}\right)^T \mathbf{R}^{AU}$ yields the identity matrix. This would imply that $\left(\mathbf{R}^{AU}\right)^T$ undoes the transformation caused by $\mathbf{R}^{AU}$. This can be tested by multiplying equation 5.17 by $\mathbf{R}^{AU}$ and determining if that operation recovers equation 5.10.

$$\mathbf{R}^{AU}\left(\mathbf{R}^{AU}\right)^T \underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.18}$$

$$\begin{bmatrix} \hat{a}\cdot\hat{u} & \hat{a}\cdot\hat{v} & \hat{a}\cdot\hat{w} \\ \hat{b}\cdot\hat{u} & \hat{b}\cdot\hat{v} & \hat{b}\cdot\hat{w} \\ \hat{c}\cdot\hat{u} & \hat{c}\cdot\hat{v} & \hat{c}\cdot\hat{w} \end{bmatrix} \begin{bmatrix} \hat{a}\cdot\hat{u} & \hat{b}\cdot\hat{u} & \hat{c}\cdot\hat{u} \\ \hat{a}\cdot\hat{v} & \hat{b}\cdot\hat{v} & \hat{c}\cdot\hat{v} \\ \hat{a}\cdot\hat{w} & \hat{b}\cdot\hat{w} & \hat{c}\cdot\hat{w} \end{bmatrix} \underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.19}$$

Just like before, recalling equation 5.6 for shorthand notation:

$$\begin{bmatrix} u_a & v_a & w_a \\ u_b & v_b & w_b \\ u_c & v_c & w_c \end{bmatrix} \begin{bmatrix} u_a & u_b & u_c \\ v_a & v_b & v_c \\ w_a & w_b & w_c \end{bmatrix} \underline{x}_A = \begin{bmatrix} \hat{u} & \hat{v} & \hat{w} \end{bmatrix} \begin{bmatrix} \hat{u}^T \\ \hat{v}^T \\ \hat{w}^T \end{bmatrix} \underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.20}$$

$$\left(\hat{u}\hat{u}^T + \hat{v}\hat{v}^T + \hat{w}\hat{w}^T\right)\underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.21}$$

Expanding the left hand side:

$$\left(\begin{pmatrix} u_a \\ u_b \\ u_c \end{pmatrix}\begin{pmatrix} u_a & u_b & u_c \end{pmatrix} + \begin{pmatrix} v_a \\ v_b \\ v_c \end{pmatrix}\begin{pmatrix} v_a & v_b & v_c \end{pmatrix} + \begin{pmatrix} w_a \\ w_b \\ w_c \end{pmatrix}\begin{pmatrix} w_a & w_b & w_c \end{pmatrix}\right)\underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.22}$$

$$\left(\begin{bmatrix} u_a^2 & u_a u_b & u_a u_c \\ u_b u_a & u_b^2 & u_b u_c \\ u_c u_a & u_c u_b & u_c^2 \end{bmatrix} + \begin{bmatrix} v_a^2 & v_a v_b & v_a v_c \\ v_b v_a & v_b^2 & v_b v_c \\ v_c v_a & v_c v_b & v_c^2 \end{bmatrix} + \begin{bmatrix} w_a^2 & w_a w_b & w_a w_c \\ w_b w_a & w_b^2 & w_b w_c \\ w_c w_a & w_c w_b & w_c^2 \end{bmatrix}\right)\underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.23}$$

$$\begin{bmatrix} u_a^2+v_a^2+w_a^2 & u_a u_b + v_a v_b + w_a w_b & u_a u_c + v_a v_c + w_a w_c \\ u_b u_a + v_b v_a + w_b w_a & u_b^2+v_b^2+w_b^2 & u_b u_c + v_b v_c + w_b w_c \\ u_c u_a + v_c v_a + w_c w_a & u_c u_b + v_c v_b + w_c w_b & u_c^2+v_c^2+w_c^2 \end{bmatrix}\underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.24}$$

Which is recognizable as:

$$\begin{bmatrix} \hat{a}\cdot\hat{a} & \hat{a}\cdot\hat{b} & \hat{a}\cdot\hat{c} \\ \hat{b}\cdot\hat{a} & \hat{b}\cdot\hat{b} & \hat{b}\cdot\hat{c} \\ \hat{c}\cdot\hat{a} & \hat{c}\cdot\hat{b} & \hat{c}\cdot\hat{c} \end{bmatrix}\underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.25}$$

And due to the orthonormality of frame A:

$$\begin{bmatrix} \hat{a}\cdot\hat{a} & \hat{a}\cdot\hat{b} & \hat{a}\cdot\hat{c} \\ \hat{b}\cdot\hat{a} & \hat{b}\cdot\hat{b} & \hat{b}\cdot\hat{c} \\ \hat{c}\cdot\hat{a} & \hat{c}\cdot\hat{b} & \hat{c}\cdot\hat{c} \end{bmatrix}\underline{x}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\underline{x}_A = \hat{\mathbf{I}}\,\underline{x}_A = \mathbf{R}^{AU}\underline{x}_U \tag{5.26}$$

$$\boxed{\underline{x}_A = \mathbf{R}^{AU}\underline{x}_U} \tag{5.27}$$

Basis Vectors of A

To some, the step taken by equation 5.25 may seem far too convenient to be accurate. In order to put the reader's mind at ease, consider the basis vectors of frame A as linear combinations of the basis vectors of U.

$$\hat{a} = a_u\,\hat{u} + a_v\,\hat{v} + a_w\,\hat{w}$$
$$\hat{b} = b_u\,\hat{u} + b_v\,\hat{v} + b_w\,\hat{w}$$
$$\hat{c} = c_u\,\hat{u} + c_v\,\hat{v} + c_w\,\hat{w}$$

Utilizing the exact same process that was done to obtain equation 5.7:

$$\hat{a} = (\hat{a}\cdot\hat{u})\hat{u} + (\hat{a}\cdot\hat{v})\hat{v} + (\hat{a}\cdot\hat{w})\hat{w}$$
$$\hat{b} = (\hat{b}\cdot\hat{u})\hat{u} + (\hat{b}\cdot\hat{v})\hat{v} + (\hat{b}\cdot\hat{w})\hat{w}$$
$$\hat{c} = (\hat{c}\cdot\hat{u})\hat{u} + (\hat{c}\cdot\hat{v})\hat{v} + (\hat{c}\cdot\hat{w})\hat{w}$$

Which can be rewritten using the shorthand notation from equation 5.6.

$$\hat{a} = u_a\,\hat{u} + v_a\,\hat{v} + w_a\,\hat{w}$$
$$\hat{b} = u_b\,\hat{u} + v_b\,\hat{v} + w_b\,\hat{w}$$
$$\hat{c} = u_c\,\hat{u} + v_c\,\hat{v} + w_c\,\hat{w}$$

From here, the simplification of equation 5.24 into the form of dot products of the basis vectors of A is clear. There is another explanation that would have saved a lot of the algebra, however it would have relied on a "trust me" type of explanation. In linear algebra there is something called the completeness relation, which essentially states that a set of orthonormal basis vectors span their entire vector space. The completeness relation is written as the identity matrix being equal to the sum of the outer products of all basis vectors in the vector space. The U frame is a three dimensional space with three orthonormal basis vectors. So, by the completeness relation:

$$\sum_{i}^{N} \hat{e}_i \hat{e}_i^T = \hat{u}\hat{u}^T + \hat{v}\hat{v}^T + \hat{w}\hat{w}^T = \hat{\mathbf{I}}$$

Which is exactly the matrix coefficient to $\underline{x}_A$ in equation 5.21.

The results of equation 5.17 and equation 5.27 are powerful. Together, they show that:

$$\left(\mathbf{R}^{AU}\right)^T \mathbf{R}^{AU} = \mathbf{R}^{AU}\left(\mathbf{R}^{AU}\right)^T = \hat{\mathbf{I}} \tag{5.28}$$

Matrices that satisfy this condition are called orthogonal. An orthogonal matrix is such that the inverse of the matrix is equal to its own transpose. Because of the results of equations 5.17 and 5.27, it is concluded that any direction cosine matrix is an orthogonal matrix. To make the notation slightly more convenient, define $\mathbf{R}^{UA}$ such that:

$$\boxed{\mathbf{R}^{UA} = \left(\mathbf{R}^{AU}\right)^T = \left(\mathbf{R}^{AU}\right)^{-1}} \tag{5.29}$$

Consider also the case of vector $\underline{x}$ as expressed in terms of an entirely separate orthonormal reference frame $\mathcal{F}_V = (\hat{u}', \hat{v}', \hat{w}')$. Because A and V are both orthonormal reference frames, a direction cosine matrix exists that relates the coordinates of any vector expressed in these frames. The direction cosine matrix is such that:

$$\underline{x}_A = \mathbf{R}^{AV} \underline{x}_V \tag{5.30}$$

Since frame U is also orthonormal, it too has a direction cosine matrix that relates the coordinates of any vector expressed in these frames.

$$\underline{x}_U = \mathbf{R}^{UV} \underline{x}_V \tag{5.31}$$

Recall also that A has a direction cosine matrix with U.

$$\underline{x}_A = \mathbf{R}^{AU} \underline{x}_U = \mathbf{R}^{AU} \mathbf{R}^{UV} \underline{x}_V \tag{5.32}$$

Equating equation 5.30 and equation 5.32:

$$\underline{x}_A = \mathbf{R}^{AV} \underline{x}_V = \mathbf{R}^{AU} \mathbf{R}^{UV} \underline{x}_V \tag{5.33}$$

$$\mathbf{R}^{AV} = \mathbf{R}^{AU} \mathbf{R}^{UV} \tag{5.34}$$

To confirm whether or not this matrix is also a direction cosine matrix, the orthogonality condition must be verified:

$$\hat{\mathbf{I}} = \left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right)^T \left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right) \quad \& \quad \hat{\mathbf{I}} = \left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right) \left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right)^T \tag{5.35}$$

Since $\mathbf{R}^{AU}$ and $\mathbf{R}^{UV}$ are known direction cosine matrices:

$$\begin{aligned}
\left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right)^T \left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right) &= \left(\mathbf{R}^{UV}\right)^T \left(\mathbf{R}^{AU}\right)^T \mathbf{R}^{AU} \mathbf{R}^{UV} \\
&= \left(\mathbf{R}^{UV}\right)^T \hat{\mathbf{I}} \, \mathbf{R}^{UV} \\
&= \left(\mathbf{R}^{UV}\right)^T \mathbf{R}^{UV} \\
&= \hat{\mathbf{I}} \\
\left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right) \left(\mathbf{R}^{AU} \mathbf{R}^{UV}\right)^T &= \mathbf{R}^{AU} \mathbf{R}^{UV} \left(\mathbf{R}^{UV}\right)^T \left(\mathbf{R}^{AU}\right)^T \\
&= \mathbf{R}^{AU} \hat{\mathbf{I}} \left(\mathbf{R}^{AU}\right)^T \\
&= \mathbf{R}^{AU} \left(\mathbf{R}^{AU}\right)^T \\
&= \hat{\mathbf{I}}
\end{aligned} \tag{5.36}$$

The transformation matrix between A and V is therefore a direction cosine matrix. Since every direction cosine matrix is orthogonal, its inverse is equal to its transpose. Therefore, the product of two direction cosine matrices is itself a direction cosine matrix. The resulting transformation satisfies the same orthogonality condition as any direction cosine matrix.

$$\boxed{\begin{aligned} \mathbf{R}^{AV} &= \mathbf{R}^{AU} \mathbf{R}^{UV} \\ \mathbf{R}^{VA} &= \left(\mathbf{R}^{AV}\right)^T = \left(\mathbf{R}^{AV}\right)^{-1} \end{aligned}} \tag{5.37}$$

5.3 The Time Derivative of a Transformed Vector

If a vector $\underline{x}$ is expressed in an orthonormal non-inertial reference frame $\mathcal{F}_U = (\hat{u}, \hat{v}, \hat{w})$, its time derivative must be related to its derivative with respect to an inertial reference frame $\mathcal{F}_A = (\hat{a}, \hat{b}, \hat{c})$ when applying Newton's laws in their inertial form. This process will also yield an expression for the time rate of change of the direction cosine matrix. Consider vector $\underline{x}$, expressed in both reference frames and related through the direction cosine matrix.

$$\underline{x}_A = \mathbf{R}^{AU} \underline{x}_U \tag{5.27}$$

Taking a derivative with respect to time, using the chain rule, yields:

$$\frac{d}{dt}\left(\underline{x}_A\right) = \frac{d}{dt}\left(\mathbf{R}^{AU}\right)\underline{x}_U + \mathbf{R}^{AU}\frac{d}{dt}\left(\underline{x}_U\right) \tag{5.38}$$

Or, in more compact notation:

$$\dot{\underline{x}}_A = \dot{\mathbf{R}}^{AU}\underline{x}_U + \mathbf{R}^{AU}\dot{\underline{x}}_U \tag{5.39}$$

This isn't useful quite yet, as this equation has the unknown time derivative of the direction cosine matrix. The time derivative of the direction cosine matrix may be determined using the transport theorem for a time derivative of $\underline{x}$.

$$\left(\frac{d\underline{x}}{dt}\right)_A = \left(\frac{d\underline{x}}{dt}\right)_U + {}^A\underline{\omega}^U \times \underline{x} \tag{5.40}$$

Which, depending on the reference frame that $\underline{x}$ is expressed in, will have different forms. Beginning first by using the A frame representation:

$$\dot{\underline{x}}_A = \mathbf{R}^{AU}\dot{\underline{x}}_U + {}^A\underline{\omega}^U_A \times \underline{x}_A = \mathbf{R}^{AU}\dot{\underline{x}}_U + {}^A\underline{\omega}^U_A \times \left(\mathbf{R}^{AU}\underline{x}_U\right) \tag{5.41}$$

Where ${}^A\underline{\omega}^U_A$ is the angular velocity vector of the U frame with respect to the A frame, expressed in terms of the A frame coordinates. The cross product can be written in matrix form as:

$${}^A\underline{\omega}^U_A \times \underline{x}_A = \begin{vmatrix} \hat{a} & \hat{b} & \hat{c} \\ \omega_a & \omega_b & \omega_c \\ x_a & x_b & x_c \end{vmatrix} = \begin{pmatrix} 0 + \omega_b x_c - \omega_c x_b \\ \omega_c x_a + 0 - \omega_a x_c \\ -\omega_b x_a + \omega_a x_b + 0 \end{pmatrix} = \begin{bmatrix} 0 & -\omega_c & \omega_b \\ \omega_c & 0 & -\omega_a \\ -\omega_b & \omega_a & 0 \end{bmatrix} \begin{pmatrix} x_a \\ x_b \\ x_c \end{pmatrix} = \left[{}^A\omega^U_A\right]_\times \underline{x}_A \tag{5.42}$$

Equation 5.41 then becomes:

$$\dot{\underline{x}}_A = \mathbf{R}^{AU}\dot{\underline{x}}_U + \left[{}^A\omega^U_A\right]_\times \mathbf{R}^{AU}\underline{x}_U \tag{5.43}$$

Equating equations 5.39 and 5.43:

$$\boxed{\dot{\underline{x}}_A = \dot{\mathbf{R}}^{AU}\underline{x}_U + \mathbf{R}^{AU}\dot{\underline{x}}_U = \left[{}^A\omega^U_A\right]_\times \mathbf{R}^{AU}\underline{x}_U + \mathbf{R}^{AU}\dot{\underline{x}}_U} \tag{5.44}$$

Now examining the transport theorem in equation 5.40 and using the U frame representation, the time derivative of $\underline{x}$ is:

$$\dot{\underline{x}}_A = \mathbf{R}^{AU}\dot{\underline{x}}_U + \mathbf{R}^{AU}\left({}^A\underline{\omega}^U_U \times \underline{x}_U\right) \tag{5.45}$$

Where ${}^{A}\underline{\omega}_{U}^{U}$ is the angular velocity vector of the U frame with respect to the A frame, expressed in terms of the U frame coordinates. The cross product can be written in matrix form as:

$$ {}^{A}\underline{\omega}_{U}^{U} \times \underline{x}_U = \begin{vmatrix} \hat{u} & \hat{v} & \hat{w} \\ \omega_u & \omega_v & \omega_w \\ x_u & x_v & x_w \end{vmatrix} = \begin{pmatrix} 0 + \omega_v x_w - \omega_w x_v \\ \omega_w x_u + 0 - \omega_u x_w \\ -\omega_v x_u + \omega_u x_v + 0 \end{pmatrix} = \begin{bmatrix} 0 & -\omega_w & \omega_v \\ \omega_w & 0 & -\omega_u \\ -\omega_v & \omega_u & 0 \end{bmatrix} \begin{pmatrix} x_u \\ x_v \\ x_w \end{pmatrix} = \left[{}^{A}\omega_{U}^{U} \right]_{\times} \underline{x}_U \quad (5.46) $$

Equation 5.45 then becomes:

$$ \dot{\underline{x}}_A = \mathbf{R}^{AU} \dot{\underline{x}}_U + \mathbf{R}^{AU} \left[{}^{A}\omega_{U}^{U} \right]_{\times} \underline{x}_U \quad (5.47) $$

Equating equations 5.39 and 5.47:

$$ \boxed{ \dot{\underline{x}}_A = \dot{\mathbf{R}}^{AU} \underline{x}_U + \mathbf{R}^{AU} \dot{\underline{x}}_U = \mathbf{R}^{AU} \left[{}^{A}\omega_{U}^{U} \right]_{\times} \underline{x}_U + \mathbf{R}^{AU} \dot{\underline{x}}_U } \quad (5.48) $$

The form of equation 5.48 also exposes a neat property of the transport theorem and the time derivative of $\underline{x}$ as it is expressed in the A frame.

$$ \dot{\underline{x}}_A = \mathbf{R}^{AU} \left(\dot{\underline{x}}_U + \left[{}^{A}\omega_{U}^{U} \right]_{\times} \underline{x}_U \right) = \mathbf{R}^{AU} \left(\dot{\underline{x}}_U + {}^{A}\underline{\omega}_{U}^{U} \times \underline{x}_U \right) \quad (5.49) $$

Equation 5.49 demonstrates that the transport theorem is independent of the coordinate representation of $\underline{x}$. Whether $\underline{x}$ is expressed in the A frame or the U frame, the same physical result is obtained.

Using equation 5.44 and equation 5.48 examining terms, a matrix equation for the time rate of change of the direction cosine matrix has been obtained.

$$ \dot{\underline{x}}_A = \dot{\mathbf{R}}^{AU} \underline{x}_U + \mathbf{R}^{AU} \dot{\underline{x}}_U = \left[{}^{A}\omega_{A}^{U} \right]_{\times} \mathbf{R}^{AU} \underline{x}_U + \mathbf{R}^{AU} \dot{\underline{x}}_U = \mathbf{R}^{AU} \left[{}^{A}\omega_{U}^{U} \right]_{\times} \underline{x}_U + \mathbf{R}^{AU} \dot{\underline{x}}_U \quad (5.50) $$

Since $\underline{x}_U$ is the same arbitrary vector expressed in the U frame, the matrix that transforms it must be the equivilent between the three expressions. Therefore:

$$ \boxed{ \dot{\mathbf{R}}^{AU} = \left[{}^{A}\omega_{A}^{U} \right]_{\times} \mathbf{R}^{AU} = \mathbf{R}^{AU} \left[{}^{A}\omega_{U}^{U} \right]_{\times} } \quad (5.51) $$

Thus the time derivative of the direction cosine matrix is expressed by a matrix product between the angular veclocity of the U frame with respect to the A frame and the original direction cosine matrix. With equation 5.51, the time derivative of vector $\underline{x}$ may be expressed in terms of frame A if the original vector is known as measured in the U frame.

Equation 5.51 also happens to uncover the matrix similarity transformation.

$$ \left(\mathbf{R}^{AU} \right)^{T} \mathbf{R}^{AU} \left[{}^{A}\omega_{U}^{U} \right]_{\times} = \left(\mathbf{R}^{AU} \right)^{T} \left[{}^{A}\omega_{A}^{U} \right]_{\times} \mathbf{R}^{AU} \quad (5.52) $$

Using the fact that the transpose of a direction cosine matrix is its own inverse, the cross product matrix with respect to the U frame is written as:

$$ \boxed{ \left[{}^{A}\omega_{U}^{U} \right]_{\times} = \left(\mathbf{R}^{AU} \right)^{T} \left[{}^{A}\omega_{A}^{U} \right]_{\times} \mathbf{R}^{AU} } \quad (5.53) $$

This relation holds for all transformation matrices (not just for the cross product matrix) provided a direction cosine matrix exists between frames A and U. Note that both of the cross product matrices on either side of equation 5.53 encode the exact same physical process, with the similarity transformation only changing the basis for which either are expressed.

5.4 The Right Ascension/Declination Reference Frame

To be written.

5.5 The Body Centered Inertial Reference Frame

To be written.

5.6 The Body Centered, Body Fixed Reference Frame

To be written.

5.7 The Topocentric East, North, Up Reference Frame

To be written.

5.8 The Topocentric Azimuth/Elevation Reference Frame

To be written.

References

[1] Guinness World Records. *Largest artificial satellite*. `https://www.guinnessworldrecords.com/world-records/largest-artificial-satellite`. (Visited on 04/16/2026).